

Classical Mechanics Note by CHESTNUT

Yingqiu He

July 16, 2024

Preface

I got BSc degree from Zhejiang University of Technology, China, in 2024. My website is banligo.github.io. It's my honor if you would like to visit. Classical Mechanics contains both Newtonian mechanics and Analytical mechanics, the note goes along with a teacher who shared her teaching video in Bilibili, which is really good lesson for self-teaching learners, I recommend it greatly if you are Chinese. Reference books:

- Classical Mechanics Gailun, Li Shuming.
- Mechanics and Theoretical Mechanics, Qin Gan, Second Edition.
- Classical Mechanics, Herbert Goldstein, Third Edition.

Reference video: [Bilibili-Up: Shechanshan Theoretical Mechanics](#)

The work is done with the help of translator with Chinese to English. The origin of the figures in this document are all from other people, because it was too difficult and for brief it, I didn't cite them, I feel sorry about it. I also have other note related to Mechanics, please find in my hp later if you would like to. Thanks.

Some symbols here for your reference:

- **EX** means example or exercise
- s usually means the number of degree of freedom
- i usually means the parameter of i th particles
- $\mathcal{E}$ means the section there used Einstein Summation Convention

Welcome to communicate with me with anything if you would like to, my contact: he.yingqiu@hotmail.com. Thanks again for your visiting.

Yingqiu He ●
Zhejiang, hometown
2024.07.16

Contents

1	Newtonian Mechanics	1
1.1	Particle Kinetics	1
1.1.1	Coordinate System	1
1.2	Particle Dynamics	3
1.3	Moving Reference System	3
1.3.1	Translation Reference System	3
1.3.2	Rotation Reference System	3
1.3.3	Translation & Rotation Reference System	4
1.4	Mechanics of A Particle and A System of Particles	5
1.4.1	A Particle	5
1.4.2	A System of Particles	6
1.4.3	Center-of-Mass Frame	8
1.5	The Central Field	9
1.5.1	The Equations of Motion	9
1.5.2	The Differential Equation for the Orbit	9
1.5.3	Inverse-Square Law of Force	10
1.5.4	Scattering in a Central Force Field	11
1.5.5	The Two-Body Problem	12
2	Lagrangian Mechanics	13
2.1	Constraints	13
2.1.1	Constraints Classifications	13
2.1.2	Generalized Coordinates and Degrees of Freedom	13
2.2	D'Alembert's Principle and Lagrange's Equations	14
2.3	Variational Principles and Lagrange's Equations	16
2.3.1	Calculus of variations	16
2.3.2	Hamilton's Principle	18
2.4	Discussion – Properties and Statics	19
2.5	Motion Integral	21
2.5.1	Conservation Theorems	21
2.5.2	Symmetry Properties	22
2.5.3	Noether's Theorem &	23
2.5.4	Generalized Potential	23
2.5.5	Method of Lagrange Multipliers &	24
2.6	Oscillations	25
2.6.1	Small Vibrations in Single-Degree-of-Freedom Systems	25
2.6.2	Small Vibrations in Multi-Degree-of-Freedom Systems	26
2.6.3	Free Vibrations of a Linear Triatomic Molecule	28
3	Rigid Body Mechanics	30
3.1	Rigid Body Motion	30
3.2	Orthogonal Transformations &	32
3.2.1	Formal Properties of The Transformation Matrix	32
3.2.2	Tensors	33
3.3	The Kinetics of Rigid Body	34
3.3.1	The Euler Angels	34
3.3.2	Euler Kinetics Equations	34

3.4	The Principal Moment of Inertia Tensor	35
3.4.1	The Inertia Tensor and The Moment of Inertia	35
3.4.2	The Eigenvalues of The Inertia Tensor and The Principal Axis Transformation	37
3.4.3	Inertia Ellipsoid	38
3.5	Euler's Dynamics Equations	39
3.6	Torque-Free Motion of a Rigid Body	40
3.6.1	Asymmetric Euler Gyroscope	40
3.6.2	Symmetric Euler Gyroscope	41
4	Hamilton Mechanics ☞	43
4.1	The Canonical Equations of Hamilton	43
4.1.1	Legendre Transformations	43
4.1.2	The Hamilton Equations of Motion	43
4.1.3	Hamilton's Principle in Phase Space	45
4.2	Canonical Transformations	45
4.2.1	The Equations of Canonical Transformation	45
4.2.2	Poisson Bracket	47
4.2.3	Liouville's Theorem	48
4.3	Hamilton-Jacobi Theory	49
4.3.1	The Hamilton-Jacobi Equation For Hamilton's Principal Function	49
4.3.2	The Harmonic Oscillator Problem as an Example of The H-J Method	50

Chapter 1

Newtonian Mechanics

1.1 Particle Kinetics

- Motion of a particle: $\vec{r} = \vec{r}(t)$
- Displacement: $\Delta \vec{r} = \vec{r}(t + \Delta t) - \vec{r}(t)$
- Velocity: $\vec{v} = \frac{d\vec{r}}{dt} = \dot{\vec{r}}$
- Acceleration: $\vec{a} = \frac{dv}{dt} = \dot{\vec{v}} = \ddot{\vec{r}}$

1.1.1 Coordinate System

1. Cartesian coordinate (x, y, z)

$$\begin{aligned}\vec{r} &= x\vec{e}_x + y\vec{e}_y + z\vec{e}_z \\ \vec{v} &= \dot{x}\vec{e}_x + \dot{y}\vec{e}_y + \dot{z}\vec{e}_z \\ \vec{a} &= \ddot{x}\vec{e}_x + \ddot{y}\vec{e}_y + \ddot{z}\vec{e}_z\end{aligned}$$

2. Planar polar coordinate (r, θ)

$$x = r \cos \theta, y = r \sin \theta$$

- $\vec{e}_r = \vec{e}_r(\theta)$: keep θ constant, the infinitesimal displacement direction caused by an infinitesimal increase in r
- $\vec{e}_\theta = \vec{e}_\theta(\theta)$: keep r constant, the infinitesimal displacement direction caused by an infinitesimal increase in θ

$$d\vec{e}_r = \vec{e}_\theta d\theta, \quad d\vec{e}_\theta = -\vec{e}_r d\theta \quad \Rightarrow \quad \boxed{\frac{d\vec{e}_r}{d\theta} = \vec{e}_\theta, \quad \frac{d\vec{e}_\theta}{d\theta} = -\vec{e}_r}$$

$$\vec{r} = r\vec{e}_r$$

$$\vec{v} = \frac{d}{dt}(r\vec{e}_r) = \frac{dr}{dt}\vec{e}_r + r\frac{d\vec{e}_r}{dt} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta \quad \left(\begin{array}{l} v_r = \dot{r} \\ v_\theta = r\dot{\theta} \end{array} \right)$$

$$\vec{a} = \frac{d}{dt}(\dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta) = (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{e}_\theta \quad \left(\begin{array}{l} a_r = \ddot{r} - r\dot{\theta}^2 \\ a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) \end{array} \right)$$

3. Cylindrical coordinate (ρ, ϕ, z)

$$x = \rho \cos \phi, y = \rho \sin \phi, z = z$$

$$\begin{aligned}\vec{r} &= \rho\vec{e}_\rho + z\vec{e}_z \\ \vec{v} &= \dot{\rho}\vec{e}_\rho + \rho\dot{\phi}\vec{e}_\phi + \dot{z}\vec{e}_z \\ \vec{a} &= (\ddot{\rho} - \rho\dot{\phi}^2)\vec{e}_\rho + (\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi})\vec{e}_\phi + \ddot{z}\vec{e}_z\end{aligned}$$

4. Spherical coordinate (r, θ, ϕ)

$$x = r \sin \theta \cos \theta, y = r \sin \theta \sin \phi, z = r \cos \phi$$

$$\left. \begin{aligned} \frac{\partial \vec{e}_r}{\partial \theta} &= \vec{e}_\theta, & \frac{\partial \vec{e}_\theta}{\partial \theta} &= -\vec{e}_r, & \frac{\partial \vec{e}_\phi}{\partial \phi} &= -\vec{e}_\rho \\ \vec{e}_\rho &= \sin \theta \vec{e}_r + \cos \theta \vec{e}_\theta \\ \vec{e}_r &= \sin \theta \vec{e}_\rho + \cos \theta \vec{e}_z \\ \vec{e}_\theta &= \cos \theta \vec{e}_\rho - \sin \theta \vec{e}_z \end{aligned} \right\} \Rightarrow \boxed{\begin{aligned} \frac{\partial \vec{e}_r}{\partial \phi} &= \frac{\partial}{\partial \phi}(\vec{e}_\rho \sin \theta) = \vec{e}_\phi \sin \theta \\ \frac{\partial \vec{e}_\theta}{\partial \phi} &= \frac{\partial}{\partial \phi}(\vec{e}_\rho \cos \theta) = -\vec{e}_\phi \cos \theta \end{aligned}}$$

$$\vec{r} = r \vec{e}_r$$

$$\vec{v} = \frac{d}{dt}(r \vec{e}_r) = \dot{r} \vec{e}_r + r \cdot \frac{d \vec{e}_r}{dt} = \dot{r} \vec{e}_r + r(\dot{\theta} \vec{e}_\theta + \dot{\phi} \sin \theta \vec{e}_\phi)$$

$$\begin{aligned} \vec{a} &= \frac{d}{dt}(\dot{r} \vec{e}_r + r \dot{\theta} \vec{e}_\theta + r \dot{\phi} \sin \theta \vec{e}_\phi) \\ &= (\ddot{r} - r \dot{\theta}^2 - r \sin^2 \theta \dot{\phi}^2) \vec{e}_r + (r \ddot{\theta} + 2 \dot{r} \dot{\theta} - r \sin \theta \cos \theta \dot{\phi}^2) \vec{e}_\theta + (r \sin \theta \ddot{\phi} + 2 \sin \theta \dot{r} \dot{\phi} + 2 r \cos \theta \dot{\theta} \dot{\phi}) \vec{e}_\phi \end{aligned}$$

5. Orthogonal Curvilinear coordinate (u, v, w)

- Coordinate curve: one coordinate changing, and the other coordinates take the form of constants
- Unit vector: tangential direction of the coordinate curve

$$\vec{r} = \vec{r}(u, v, w) = x(u, v, w) \vec{e}_x + y(u, v, w) \vec{e}_y + z(u, v, w) \vec{e}_z$$

6. Natural coordinate

- Arc coordinate: s $|s| = \widehat{QP}$; $\dot{s} = v$
- Equation of Motion: $s = s(t)$
- Position vector: $\vec{r} = \vec{r}(s)$

(a) Tangent unit vector: $\vec{\tau} = \frac{d \vec{r}}{ds}$ (Direction: orbital direction; Magnitude: $|\vec{\tau}| = \frac{|d \vec{r}|}{ds} = 1$)

(b) Curvature vector: $\vec{N} = \frac{d \vec{\tau}}{ds} = N \vec{n} = \frac{\vec{n}}{\rho}$

- $\vec{n} \perp \vec{\tau}$, point to the concave side of the curve
- N : curvature; ρ : radius of curvature
- $\vec{n}$: Normal unit vector

(c) Binomial unit vector: $\vec{b} = \vec{\tau} \times \vec{n}$

- Winding vector: $\vec{k} = \frac{d \vec{b}}{ds} = \frac{d}{ds}(\vec{\tau} \times \vec{n}) = \frac{d \vec{\tau}}{ds} \times \vec{n} + \vec{\tau} \times \frac{d \vec{n}}{ds} = \vec{\tau} \times \frac{d \vec{n}}{ds}$
- $\vec{k} \perp \vec{\tau}$, $\vec{k} \perp \vec{b} \rightarrow \vec{k} = k \vec{n} = \frac{\vec{n}}{\sigma}$ k : winding; σ : radius of winding

* $\vec{\tau}$ and $\vec{n}$ form an osculating plane

* $\vec{n}$ and $\vec{b}$ form a normal plane

* $\vec{b}$ and $\vec{\tau}$ form a rectifying plane

$$\vec{v} = \frac{d \vec{r}}{dt} = \frac{d \vec{r}}{ds} \frac{ds}{dt} = \dot{s} \vec{\tau}$$

$$\vec{a} = \frac{d}{dt}(\dot{s} \vec{\tau}) = \ddot{s} \vec{\tau} + \frac{d}{d \vec{\tau} s} \frac{ds}{dt} = \ddot{s} \vec{\tau} + \frac{\dot{s}^2}{\rho} \vec{n} \quad \left(\begin{aligned} a_\tau &= \ddot{s} = \frac{dv}{dt} = v \frac{dv}{ds} \\ a_n &= \frac{\dot{s}^2}{\rho} = \frac{v^2}{\rho} \end{aligned} \right)$$

* $\vec{a}$ has no component on $\vec{b}$ direction

* if v is conserved, then $\ddot{s} = 0$, $\vec{a} \perp \vec{v}$

1.2 Particle Dynamics

Newton's second law and differential equation of motion:

$$\vec{F} = \vec{F}(\vec{r}, \dot{\vec{r}}, t), \quad \begin{cases} m\ddot{\vec{r}} = \vec{F} \\ \vec{r}(0), \dot{\vec{r}}(0) \end{cases}, \Rightarrow \vec{r} = \vec{r}(t)$$

or

$$\vec{F} = \vec{F}(\vec{r}, \vec{v}, t), \quad \begin{cases} m\ddot{\vec{r}} = \vec{F} \\ \dot{\vec{r}} = \vec{v} \end{cases}, \Rightarrow \begin{cases} \vec{r} = \vec{r}(t) \\ \vec{v} = \vec{v}(t) \end{cases}$$

Constraint motion: (active force $\vec{F}$ + constraint force $\vec{R}$)

$$\begin{cases} m\ddot{\vec{r}} = \vec{F}(\vec{r}, \dot{\vec{r}}, t) + \vec{R} \\ \text{constraint functions} \end{cases}$$

Ex 1. Projectile motion with air resistance $\vec{f} = -f(v)\vec{v}/v$, motion equation?
Choose natural coordinate, differential equation are

$$\begin{cases} ma_\tau = m \frac{dv}{ds} = -f(v) - mg \sin \theta \\ ma_n = m \frac{v^2}{\rho} = mg \cos \theta \end{cases} \quad \text{with} \quad \begin{cases} v = \frac{ds}{dt} \\ \rho = -\frac{ds}{d\theta} \quad (d\theta = |d\vec{\tau}|) \end{cases} \Rightarrow$$

$$\begin{aligned} mv \frac{dv}{ds} &= -f(v) - mg \sin \theta & \Rightarrow & \quad \frac{1}{v} \frac{dv}{d\theta} = \frac{f(v) + mg \sin \theta}{mg \sin \theta} & \Rightarrow & \quad v = v(\theta) & \Rightarrow & \quad \begin{cases} s = s(t) \\ \theta = \theta(t) \end{cases} \\ mv^2 \frac{d\theta}{ds} &= -mg \cos \theta \end{aligned}$$

1.3 Moving Reference System

Absolute space-time view: Observers in different frames of reference measure the same spatial distance and length of time. (High-speed motion $\rightarrow$ Relativity of space-time)

1.3.1 Translation Reference System

- S : inertial frame
- S' : non-inertial (Moving) frame
- $dt = dt'$
- $\vec{v}$: absolute velocity
- $\vec{v}'$: relatively velocity
- $\vec{v}_0$: convected velocity

$$\begin{aligned} \vec{r} &= \vec{r}' + \vec{r}_0 \\ \frac{d\vec{r}}{dt} &= \frac{d\vec{r}'}{dt} + \frac{d\vec{r}_0}{dt} \Rightarrow \vec{v} = \vec{v}' + \vec{v}_0; \vec{a} = \vec{a}' + \vec{a}_0 \\ \frac{d\vec{r}'}{dt} &= \frac{d}{dt}(x'\vec{e}_x + y'\vec{e}_y + z'\vec{e}_z) = \frac{dx'}{dt}\vec{e}_x + \frac{dy'}{dt}\vec{e}_y + \frac{dz'}{dt}\vec{e}_z \end{aligned}$$

From $\vec{F} = m\vec{a} = m\vec{a}' + m\vec{a}_0$, we have

$$\vec{F} - m\vec{a}_0 = m\vec{a}' \quad (\text{inertial force: } \vec{F} - m\vec{a}_0)$$

Ex 2. Wedge moves along the horizontal with acceleration of $\vec{A}$. Questions: The acceleration $\vec{a}'$ of the block with respect to the wedge; the pressure $\vec{R}$ of block on the wedge. (Fig: 1.1)

$$m\vec{g} + \vec{R} - m\vec{A} = m\vec{a}' \Rightarrow \begin{cases} mg \sin \alpha \pm mA \cos \alpha = ma' \\ R \pm mA \sin \alpha \mp mg \cos \alpha = 0 \end{cases} \Rightarrow \begin{cases} a' = g \sin \alpha \pm A \cos \alpha \\ R = mg \cos \alpha \mp mA \sin \alpha \end{cases}$$

1.3.2 Rotation Reference System

The origin point of S and S' coincide, considering normal vector $\vec{G}(=\vec{G}')$,

$$\vec{G} = G'_x \vec{e}'_x + G'_y \vec{e}'_y + G'_z \vec{e}'_z$$

The derivative of $\vec{G}$ for time in the S is

$$\frac{d\vec{G}}{dt} = \left(\frac{dG'_x}{dt} \vec{e}'_x + \frac{dG'_y}{dt} \vec{e}'_y + \frac{dG'_z}{dt} \vec{e}'_z \right) + \left(G'_x \frac{d\vec{e}'_x}{dt} + G'_y \frac{d\vec{e}'_y}{dt} + G'_z \frac{d\vec{e}'_z}{dt} \right)$$

$$\boxed{= \frac{d'\vec{G}}{dt} + \vec{\omega} \times \vec{G}} \Rightarrow \frac{d}{dt} = \frac{d'}{dt} + \vec{\omega} \times$$

- The derivative of $\vec{G}$ for time in the S' is $\frac{d'\vec{G}}{dt}$
- $\frac{d\vec{e}'_x}{dt} = \vec{\omega} \times \vec{e}'_x \rightarrow$ direction: $\frac{\vec{\omega}}{|\omega|} \times \vec{e}'_x$; magnitude: $\left| \frac{d\theta}{dt} \right| = \omega$

Considering $\vec{G} = \vec{r}$, then

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d'\vec{r}}{dt} + \vec{\omega} \times \vec{r} = \vec{v}' + \vec{\omega} \times \vec{r}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\vec{v}' + \vec{\omega} \times \vec{r}) = \frac{d\vec{v}'}{dt} + \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} = \left(\frac{d'\vec{v}'}{dt} + \vec{\omega} \times \vec{v}' \right) + \vec{\beta} \times \vec{r} + \vec{\omega} \times \left(\frac{d'\vec{r}}{dt} + \vec{\omega} \times \vec{r} \right)$$

$$= \vec{a}' + \vec{\beta} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) + 2\vec{\omega} \times \vec{v}'$$

$$m\vec{a}' = \vec{F} - m\vec{\beta} \times \vec{r} - m\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2m\vec{\omega} \times \vec{v}'$$

- $\vec{\omega} \times \vec{r}$: rotation convected velocity
- $\vec{\beta} \times \vec{r}$: rotation convected acceleration
- $\vec{\omega} \times (\vec{\omega} \times \vec{r})$: axial acceleration
- $\vec{\omega} \times \vec{v}'$: Coriolis acceleration
- $m\vec{\omega} \times (\vec{\omega} \times \vec{r})$: inertial centrifugal force
- $2m\vec{\omega} \times \vec{v}'$: Coriolis force

1.3.3 Translation & Rotation Reference System

$$\vec{r} = \vec{r}' + \vec{r}_0$$

$$\vec{v} = \vec{v}' + \vec{v}_0 + \vec{\omega} \times \vec{r}'$$

$$\vec{a} = \vec{a}' + \vec{a}_0 + \vec{\beta} \times \vec{r}' + \vec{\omega} \times (\vec{\omega} \times \vec{r}') + 2\vec{\omega} \times \vec{v}'$$

$$m\vec{a}' = \vec{F} - m\vec{a}_0 - m\vec{\beta} \times \vec{r}' - m\vec{\omega} \times (\vec{\omega} \times \vec{r}') - 2m\vec{\omega} \times \vec{v}'$$

Considering earth: (Fig: 1.2)

- Revolution: cancels out the gravitational pull of the sun and is not considered
- Rotation: at a constant speed ($|\vec{\beta}| = \text{const}$), causes inertial centrifugal force
- S' is ground reference frame, and translate with O' , rotate with z'

$$\vec{a}_0 = \frac{d^2\vec{R}}{dt^2} = \frac{d'^2\vec{R}}{dt'^2} + \vec{\beta} \times \vec{R} + \vec{\omega} \times (\vec{\omega} \times \vec{R}) + 2\vec{\omega} \times \frac{d'\vec{R}}{dt'} = \vec{\omega} \times (\vec{\omega} \times \vec{R})$$

$$\Rightarrow m\vec{a}' = \vec{F} - m\vec{\omega} \times (\vec{\omega} \times \vec{R}) - 2m\vec{\omega} \times \vec{v}'$$

Revise $\vec{g}$ as

$$m\vec{g}' = m\vec{g} - m\vec{\omega} \times (\vec{\omega} \times \vec{R}); \quad \vec{F}' = \vec{F} - m\vec{g}$$

get object motion equation at earth surface

$$\boxed{m\vec{a}' = \vec{F}' + m\vec{g}' - 2m\vec{\omega} \times \vec{v}'}$$

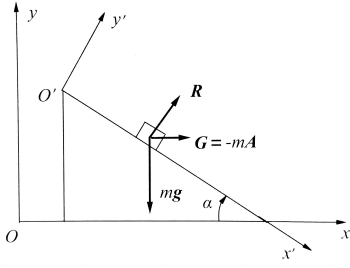


Fig 1.1: The movement of the smooth block with respect to the smooth wedge.

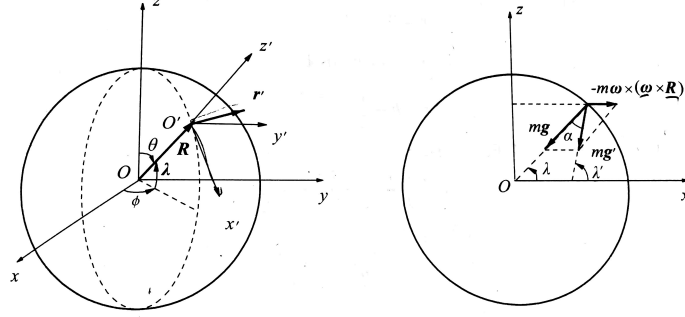


Fig 1.2: Effects of Earth's rotation and Inertial centrifugal force.

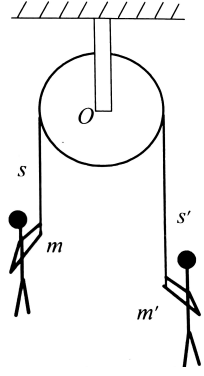


Fig 1.3: Climbers, pulley and rope.

EX 3. Free falling body offset to the east without air resistance (Fig: 1.2)

Assume the object is in latitude of λ , from height of h , then $\omega_x = -\omega \cos \lambda$, $\omega_y = 0$, $\omega_z = \omega \sin \lambda$

$$m\ddot{\vec{r}} = m\vec{g} - 2m\vec{\omega} \times \vec{v}$$

$$\ddot{x} = 2\omega\dot{y} \sin \lambda$$

$$\ddot{y} = -2\omega(\dot{x} \sin \lambda + \dot{z} \cos \lambda)$$

$$\ddot{z} = -g + 2\omega\dot{y} \cos \lambda$$

Approximation for ω :

- For zero-order approximation, it is free fall, we have

$$\ddot{x} = 0, \ddot{y} = 0, \ddot{z} = -g \quad \text{with} \quad x_0 = y_0 = 0, z_0 = h; \dot{x}_0 = \dot{y}_0 = \dot{z}_0 = 0$$

$$\Rightarrow x = y = 0, z = h - \frac{1}{2}gt^2$$

- For first-order approximation, we have ($\dot{x} \sim \mathcal{O}(\omega)$, $\dot{y} \sim \mathcal{O}(\omega)$, $\dot{z} \sim \mathcal{O}(\omega^0)$)

$$\begin{cases} \ddot{x} = 0 \\ \ddot{y} = -2\omega\dot{z} \cos \lambda \\ \ddot{z} = -g \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = \frac{1}{3}\omega g t^3 \cos \lambda \\ z = h - \frac{1}{2}gt^2 \end{cases}$$

Therefore, when object arrive at ground, $z = 0$, it will offset to the east with $y = \frac{1}{3}\sqrt{\frac{8h^3}{g}}\omega \cos \lambda$

1.4 Mechanics of A Particle and A System of Particles

1.4.1 A Particle

Momentum theorem

$$\vec{F} = m\vec{a} = m\frac{d\vec{v}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{d\vec{p}}{dt} \equiv \dot{\vec{p}} \quad (\vec{p} : \text{linear momentum})$$

$$\vec{F} = 0 \Rightarrow \vec{p} \text{ is conserved}$$

Angular momentum theorem

$$\vec{r} \times \vec{F} = \vec{r} \times \frac{\partial \vec{p}}{\partial t} = \frac{d}{dt}(\vec{r} \times \vec{p}) - \frac{dr}{dt} \times \vec{p} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \frac{d\vec{J}}{dt} \equiv \dot{\vec{J}}$$

$$\vec{M} = 0 \Rightarrow \vec{J} \text{ is conserved}$$

- $\vec{r}$: the radius vector from point O to the particles
- $\vec{M}$: torque (motion of force) about O
- $\vec{J}$: angular momentum of the particle about O

$$\vec{M} = \vec{r} \times \vec{F}, \quad \vec{J} = \vec{r} \times \vec{p}$$

Kinetic energy theorem

$$\vec{F} \cdot d\vec{r} = \frac{d\vec{p}}{dt} \cdot d\vec{r} = \frac{d(m\vec{v})}{dt} \cdot \vec{v} dt = d\left(\frac{1}{2}m\vec{v}^2\right)$$

Consider particle in going from point 1 to point 2, then

$$W_{12} = \int_1^2 \vec{F} \cdot d\vec{r} = T_2 - T_1 \quad T: \text{kinetic energy}$$

Potential energy

$$\oint \vec{F} \cdot d\vec{r} = 0 \Rightarrow \nabla \times \vec{F} = 0 \Rightarrow \vec{F} = -\nabla V(\vec{r}) \rightarrow$$

$$\vec{F} \cdot d\vec{r} = -dV \quad \text{or} \quad F_r = -\frac{\partial V}{\partial r} \rightarrow W_{12} = V_1 - V_2 \Rightarrow T_1 + T_2 = V_1 + V_2$$

- $\vec{F}$: conservative force

$$\vec{F} \text{ is conservative} \Rightarrow T + V \text{ is conserved}$$

Virial theorem

$$\begin{aligned} \vec{r} \cdot \vec{F} &= \vec{r} \cdot \frac{d\vec{p}}{dt} = \frac{d}{dt}(\vec{r} \cdot \vec{p}) - \frac{d\vec{r}}{dt} \cdot \vec{p} = \frac{d}{dt}(\vec{r} \cdot \vec{p}) - \vec{v} \cdot m\vec{v} \\ &\Rightarrow \vec{r} \cdot \vec{F} + 2T = \frac{d}{dt}(\vec{r} \cdot \vec{p}) \end{aligned}$$

"Limited motion", the time average over period τ is:

$$\frac{1}{\tau} \int_{t_0}^{t_0+\tau} (\vec{r} \cdot \vec{F} + 2T) dt = \frac{1}{\tau} \int_{t_0}^{t_0+\tau} \frac{d}{dt}(\vec{r} \cdot \vec{p}) dt = \frac{1}{\tau} \vec{r} \cdot \vec{p} \Big|_{t_0}^{t_0+\tau}$$

$$\text{when } \tau \rightarrow \infty, \overline{\vec{r} \cdot \vec{F} + 2T} = 0 \Rightarrow \overline{\vec{r} \cdot \vec{F}} = -2\bar{T}$$

Ex 4. V is the homogenous functions of $\vec{r}$, assume n order homogenous function: $f(cx, cy, cz) = c^n f(x, y, z)$
Derivative of c , and let $c \rightarrow 1$, then

$$\vec{r} \times \nabla f = nf \Rightarrow \vec{r} \cdot (-\vec{F}) = \vec{r} \cdot \nabla V = nV \Rightarrow 2\bar{T} = n\bar{V}$$

- Harmonic oscillator potential:

$$V = \frac{1}{2}m\omega^2 x^2 \xrightarrow{n=2} \bar{T} = \bar{V}$$

- Gravitational potential:

$$V = -\frac{\alpha}{r} \xrightarrow{n=-1} \bar{T} = -\frac{1}{2}\bar{V}$$

Integral of motion

$$\vec{F} = m\ddot{\vec{r}} \Leftrightarrow \dot{\vec{r}} = \vec{v}, m\dot{\vec{v}} = \vec{F}(\vec{r}, \vec{v}, t)$$

Six integral constants $C_i = C_i(\vec{r}, \vec{v}, t)$, $i = 1, \dots, 6$

$$\begin{cases} \vec{r} = \vec{r}(t, C_1, \dots, C_6) \\ \vec{v} = \vec{v}(t, C_1, \dots, C_6) \end{cases}$$

If $C_i = C_i(\vec{r}, \vec{v})$, then there is one conservative quantity and 5 integral constant at most.

1.4.2 A System of Particles

- $\vec{F}_i$: external force on i th particle
- $\vec{f}_{ij}$: internal force on the i th due to the j th particle, and

$$\vec{f}_{ij} = -\vec{f}_{ji}, \quad \vec{f}_{ii} = 0$$

1. Momentum theorem

$$\frac{d\vec{p}_i}{dt} = \vec{F}_i + \sum_j \vec{f}_{ij} \rightarrow \sum_i \frac{d\vec{p}_i}{dt} = \sum_i \vec{F}_i + \sum_{ij} \vec{f}_{ij}$$

$$\text{and } \sum_i \sum_j \vec{f}_{ij} = \sum_{ij} \vec{f}_{ij} = \sum_{ji} \vec{f}_{ji} = \sum_{ij} \vec{f}_{ji}, \text{ therefore } \sum_i \frac{d\vec{p}_i}{dt} = \sum_i \vec{F}_i + \frac{1}{2} \sum_{ij} (\vec{f}_{ij} + \vec{f}_{ji}) = \sum_i \vec{F}_i \Rightarrow$$

$$\boxed{\frac{d\vec{P}}{dt} = \vec{F}}$$

◇ *Conservation Theorem for the Linear Momentum: If the total external force is zero, the total linear momentum is conserved.*

2. Angular momentum theorem

$$\begin{aligned} \frac{d\vec{J}_i}{dt} &= \vec{r}_i \times \left(\vec{F}_i + \sum_j \vec{f}_{ij} \right) \rightarrow \frac{d}{dt} \sum_i \vec{J}_i = \sum_i \vec{r}_i \times \vec{F}_i + \sum_{ij} \vec{r}_i \times \vec{f}_{ij} \\ &= \sum_i \vec{r}_i \times \vec{F}_i + \frac{1}{2} \left(\sum_{ij} \vec{r}_i \times \vec{f}_{ij} + \sum_{ji} \vec{r}_i \times \vec{f}_{ji} \right) \end{aligned}$$

$$\text{and } \vec{f}_{ij} = \vec{f}_{ij}(\vec{r}_{ij}) \frac{\vec{r}_{ij}}{|\vec{r}_{ij}|}, \vec{r}_{ij} \equiv \vec{r}_i - \vec{r}_j, \text{ therefore } \frac{d}{dt} \sum_i \vec{J}_i = \sum_i \vec{r}_i \times \vec{F}_i \Rightarrow$$

$$\boxed{\frac{d\vec{J}}{dt} = \vec{M}}$$

◇ *Conservation Theorem for Total Angular Momentum: $\vec{J}$ is constant in time if the applied (external) torque is zero.*

3. Kinetic energy theorem

$$\begin{aligned} dT_i &= \left(\vec{F}_i + \sum_j \vec{F}_{ij} \right) d\vec{r}_i \rightarrow d \sum_i T_i = \sum_i \vec{F}_i d\vec{r}_i + \sum_{ij} \vec{F}_{ij} d\vec{r}_i \\ &= \sum_i \vec{F}_i d\vec{r}_i + \frac{1}{2} \left(\sum_{ij} \vec{f}_{ij} d\vec{r}_i + \sum_{ji} \vec{f}_{ji} d\vec{r}_j \right) \\ &= \sum_i \vec{F}_i d\vec{r}_i + \frac{1}{2} \sum_{ij} \vec{f}_{ij} d\vec{r}_{ij} \end{aligned}$$

* for rigid body, $\vec{r}_{ij} \perp d\vec{r}_{ij}$, and the internal force is zero

$$\begin{aligned} \vec{F}_i &= -\frac{\partial V_i(\vec{r}_i)}{\partial \vec{r}_i}, \quad \vec{F}_{ij} = -\frac{\partial V_{ij}(\vec{r}_{ij})}{\partial \vec{r}_{ij}} \rightarrow \\ dT &= -\sum_i \frac{\partial V_i(\vec{r}_i)}{\partial \vec{r}_i} d\vec{r}_i - \frac{1}{2} \sum_{ij} \frac{\partial V_{ij}}{\partial \vec{r}_{ij}} d\vec{r}_{ij} = -\sum_i dV_i - \frac{1}{2} \sum_{ij} dV_{ij} = -dV \end{aligned}$$

Therefore,

$$d(T + V) = 0 \Rightarrow T + V \text{ is conserved}$$

Ex 5. If the climbers climb up at a uniform acceleration and arrive at same time, how long it takes? (Fig. 1.3)

Assume the pulley radius is R , and $m \rightarrow v, a$; $m' \rightarrow v', a'$

$$\frac{d\vec{J}}{dt} = \vec{M} \Rightarrow \frac{d}{dt} (-mvR + mv'R) = mgR - m'gR \Rightarrow ma - ma' = (m' - m)g$$

$$\text{and } a = \frac{2s}{t^2}, a' = \frac{2s'}{t'^2}, \text{ then } \frac{1}{t^2} (2ms - 2m's') = (m' - m)g \Rightarrow t = \sqrt{\frac{2(ms - m's')}{(m - m')g}}$$

1.4.3 Center-of-Mass Frame

Centroid motion theorem:

$$m_i \ddot{\vec{r}} = \vec{F}_i + \sum_j \vec{f}_{ij}, \quad m = \sum_i m_i, \quad \vec{r}_c = \frac{1}{m} \sum_i m_i \vec{r}_i$$

$$\Rightarrow m \ddot{\vec{r}}_c = \vec{F}$$

In the inertial reference system of S ,

$$\vec{r}_i = \vec{r}_c + \vec{r}'_i$$

$$\frac{1}{m} \sum_i m_i \vec{r}'_i = \frac{1}{m} \sum_i (\vec{r}_i - \vec{r}_c) = \vec{r}_c - \vec{r}_c = 0 \quad \Rightarrow \quad \frac{1}{m} \sum_i m_i \vec{v}'_i = 0$$

Then, $\vec{J}$ and T can be decomposed to

$$\begin{aligned} \vec{J} &= \sum_i \vec{r}_i \times m_i \vec{v}_i = \sum_i (\vec{r}_c + \vec{r}'_i) \times m_i (\vec{v}_c + \vec{v}'_i) \\ &= \left(\sum_i m_i \right) \vec{r}_c \times \vec{v}_c + \cancel{\vec{r}_c \times \sum_i m_i \vec{v}'_i} + \cancel{\sum_i m_i \vec{r}'_i \times \vec{v}_c} + \sum_i m_i \vec{r}'_i \times \vec{v}'_i \\ &= \vec{r}_c \times m \vec{v}_c + \sum_i m_i \vec{r}'_i \times \vec{v}'_i = \vec{J}_c + \vec{J}' \\ T &= \sum_i \frac{1}{2} m_i \vec{v}_i^2 = \sum_i \frac{1}{2} (\vec{v}_c + \vec{v}'_i)^2 = \sum_i \frac{1}{2} m_i (\vec{v}_c^2 + \vec{v}'_i^2 + 2 \vec{v}_c \cdot \vec{v}'_i) \\ &= \frac{1}{2} \left(\sum_i m_i \right) \vec{v}_c^2 + \sum_i \frac{1}{2} m_i \vec{v}'_i^2 + \cancel{\vec{v}_c \cdot \sum_i m_i \vec{v}'_i} \\ &= \frac{1}{2} m v_c^2 + \sum_i \frac{1}{2} m_i v_i'^2 = T_c + T'_c \end{aligned}$$

König Theorem:

$$T = T_c + T'_c$$

1. Momentum theorem

$$\sum_i m_i \ddot{\vec{r}}'_i = \sum_i \vec{F}_{ij} + \sum_{ij} \vec{f}_{ij} - \left(\sum_i m_i \ddot{\vec{r}}_c \right) = 0 \quad \Rightarrow \quad \sum_i m_i \vec{v}'_i = \text{const}$$

* $-m_i \ddot{\vec{r}}_c$: inertial force

2. Angular momentum theorem

$$\begin{aligned} \frac{d}{dt} \sum_i \vec{r}'_i \times m_i \dot{\vec{r}}'_i &= \sum_i \vec{r}'_i \times \vec{F}_i + \sum_{ij} \vec{r}'_i \times \vec{f}_{ij} + \cancel{\dot{\vec{r}}_c \times \sum_i m_i \vec{r}'_i} \\ &= \sum_i \vec{r}_i \times \vec{F}_i + \cancel{\frac{1}{2} \sum_{ij} \vec{r}'_{ij} \times \vec{f}_{ij}} = \sum_i \vec{r}_i \times \vec{F}_i \quad \Rightarrow \quad \frac{\partial \vec{J}'}{\partial t} = \vec{M}' \end{aligned}$$

3. Kinetic energy theorem

$$\begin{aligned} \sum_i m_i \ddot{\vec{r}} \cdot d\vec{r}'_i &= \sum_i \vec{F}_i d\vec{r}'_i + \sum_{ij} \vec{f}_{ij} d\vec{r}'_i + \cancel{(-\ddot{\vec{r}}) \sum_i m_i d\vec{r}'_i} \\ dT' &= d \left(\sum_i \frac{1}{2} m v_i'^2 \right) = \sum_i \vec{F}_i d\vec{r}'_i + \frac{1}{2} \sum_{ij} \vec{f}_{ij} d\vec{r}'_{ij} \end{aligned}$$

Ex 6. Find the velocity of the ring $\vec{V}$ and the worm $\vec{v}$ relative to the ground. (Fig: 1.4)

C : the mass center ; $\vec{V}'$: velocity of the contact point with the worm on the ring relative to O

$$\begin{cases} m\vec{v} + M\vec{V} = 0 \\ \vec{r} \times m\vec{v} + (\vec{R} \times M\vec{V} + \vec{\rho} \times M\vec{V}') = 0 \end{cases} \quad \begin{cases} \vec{v} = \vec{V} + \vec{V}' + \vec{u} \\ M\vec{R} + m\vec{r} = 0 \\ \vec{r} - \vec{R} = \vec{\rho} \end{cases} \Rightarrow \vec{\rho} \times \left(\frac{Mn}{M+m} \vec{v} - \frac{Mm}{M+m} \vec{V} + M\vec{V}' \right) = 0$$

$$\vec{v} - \vec{V} + \frac{M+m}{m} \vec{V}' = 0 \Rightarrow \begin{cases} \vec{v} = \frac{M}{M+2m} \vec{u} \\ \vec{V} = \vec{V}' = -\frac{m}{M+2m} \vec{u} \end{cases}$$

1.5 The Central Field

1.5.1 The Equations of Motion

The central force:

$$\vec{F} = F(\vec{r}) \vec{e}_r, \quad \vec{e}_r = \frac{\vec{r}}{r}$$

Properties:

1. Angular Momentum is conserved

Proof. $\frac{d\vec{J}}{dt} = \vec{r} \times \vec{F} = \vec{r} \times F \frac{\vec{r}}{r} = 0$ □

2. Motion in the plane

Proof. $\vec{r} \cdot \vec{J} = \vec{r} \cdot (\vec{r} \times m\vec{v}) = 0$ □

3. The area swept by the sagittal diameter per unit time is constant (Kepler's third law)

Proof. $\frac{d\vec{S}}{dt} = \frac{1}{2} \vec{r} \times \vec{v} = \frac{\vec{J}}{2m} \equiv \frac{1}{2} \vec{h}, \quad \vec{h}$: velocity torque □

4. If $\vec{F}(\vec{r}) = \vec{F}(r)$, then the central force is conservative force, and kinetic energy is conserved ¹

Proof. $\nabla \times \vec{F} = \nabla \times \left(F(r) \frac{\vec{r}}{r} \right) = \left(\nabla \times \frac{F(r)}{r} \right) \times \vec{r} + \frac{F(r)}{r} \times \nabla \vec{r} = \frac{d}{dt} \left(\frac{F(r)}{r} \right) \vec{e}_r \times \vec{r} = 0$ □

1.5.2 The Differential Equation for the Orbit

Motion differential equation in polar coordinate:

$$\begin{aligned} \vec{v} &= \dot{r} \vec{e}_r + r\dot{\theta} \vec{e}_\theta \Rightarrow F(r) = m(\ddot{r} - r\dot{\theta}^2) \\ 0 &= m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = m \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta}) \Rightarrow mr^2 \dot{\theta} = J \quad (\text{const}) \end{aligned}$$

and $\frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + V(r) = E \quad (\text{const})$, then

$$\frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r) = E, \quad \text{where} \quad V_{\text{eff}} = \frac{J^2}{2mr^2} + V(r)$$

- V_{eff} : effective potential, use V_{eff} curve to judge the orbit motion

Ex 7. Considering $V(r) = -\frac{\alpha}{r^n} \quad (\alpha, n > 0)$, then $V_{\text{eff}}(r) = \frac{J^2}{2mr^2} - \frac{\alpha}{r^n}$

- $0 < n < 2$: when $E = E_1$, motion area $r > r_1$; $E = E_2 \rightarrow (r_2, r_3)$; $E = E_3 \rightarrow r = r_4$, and the circle orbit is stable
- $n > 2$: $E = E_1 \rightarrow r = r_1$, and unstable; $E = E_3 \rightarrow r < r_2 \wedge r > r_3$; $E = E_3 \rightarrow r < r_4$
- $n = 2, J > \alpha$: $E = E_1 \rightarrow r > r_1$
- $n = 2, J < \alpha$: $E = E_1 \rightarrow$ all area; $E = E_2 \rightarrow r < r_2$

¹ $\nabla \times (\psi \vec{a}) = \nabla \psi \times \vec{a} + \psi \nabla \times \vec{a}$; $\nabla \times \vec{r} = 0$; $\nabla \psi = \vec{e}_r \frac{\partial \psi}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial \psi}{\partial \theta} + \vec{e}_\phi \frac{1}{r \sin \theta} \frac{\partial \psi}{\partial \phi}$

Let $u = \frac{1}{r}$, $F(r) = F(\frac{1}{u}) \equiv F(u)$,

$$\begin{aligned}\dot{\theta} &= \frac{J}{mr^2} = \frac{J}{m}u^2 = hu^2 \\ \dot{r} &= \frac{dr}{du} \frac{du}{d\theta} \frac{d\theta}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} \cdot hu^2 = -h \frac{du}{d\theta} \\ \ddot{r} &= \frac{d}{d\theta} \left(-h \frac{du}{d\theta} \right) \frac{d\theta}{dt} = -h^2 u^2 \frac{d^2u}{d\theta^2}\end{aligned}$$

substitute them into motion differential equations, then we got Binet Equation:

$$\boxed{\frac{d^2u}{d\theta^2} + u = \frac{f(u)}{mh^2u^2}}$$

1.5.3 Inverse-Square Law of Force

$$F = \frac{\alpha}{r^2}, \quad E = \frac{\alpha}{r} + \frac{p^2}{2m}$$

Excepted E and $\vec{J}$ as four conservative quantity, there is "Laplace-Rurge-Lenz vector (eccentricity vector)" which is conserved

$$\vec{e} = \vec{e}_r + \frac{\vec{p} \times \vec{J}}{m\alpha} \quad \text{with} \quad e = \sqrt{1 + \frac{2J^2E}{m\alpha^2}}$$

but they are dependent and satisfy two scale equation:

$$\vec{e} \cdot \vec{J} = 0, \quad e^2 = 1 + \frac{2J^2E}{m\alpha^2}$$

Proof. ² From $\frac{d\vec{p}}{dt} = \vec{F} = F\vec{e}_r$, we have

$$\vec{F} \times \vec{J} = F \frac{\vec{r}}{r} \times (\vec{r} \times m\dot{\vec{r}}) = \frac{mF}{r} \vec{r} \times (\vec{r} \times \dot{\vec{r}}) = \frac{mF}{r} [\vec{r}(\vec{r} \cdot \dot{\vec{r}}) - \dot{\vec{r}}r^2] = mF(\dot{\vec{r}}\vec{r} - r\dot{\vec{r}}) = -mFr^2 \frac{d}{dt} \left(\frac{\vec{r}}{r} \right) = -m\alpha \dot{\vec{e}}_r$$

and $\frac{d\vec{p}}{dt} \times \vec{J} = \frac{d}{dt}(\vec{p} \times \vec{J})$, therefore $\frac{d}{dt}(\vec{p} \times \vec{J} + m\alpha \vec{e}_r) = 0 \Rightarrow \vec{e}$ is conserved □

Proof. ³

$$e^2 = \vec{e}^2 = \vec{e}_r^2 + \frac{(\vec{p} \times \vec{J})^2}{m^2\alpha^2} + \frac{2}{m\alpha} \vec{e}_r \cdot (\vec{p} \times \vec{J}) = 1 + \frac{p^2J^2}{m^2\alpha^2} + \frac{2}{m\alpha} \vec{J} \cdot (\vec{e}_r \times \vec{p}) = 1 + \frac{2J^2}{m\alpha^2} \left(\frac{\alpha}{r} + \frac{p^2}{2m} \right) = 1 + \frac{2J^2E}{m\alpha^2}$$

$$\star \vec{J} \cdot (\vec{e}_r \times \vec{p}) = \frac{1}{r} \vec{J} \cdot (\vec{r} \times \vec{p}) = \frac{J^2}{r}$$

□

Oribit equation of particle in inverse-square field

Let $\vec{e}$ as polar axis, the origin is the focal point of the conic curve

$$\vec{e}_r \cdot \vec{e} = e \cos \theta = \vec{e}_r \cdot \left(\vec{e}_r + \frac{\vec{p} \times \vec{J}}{m\alpha} \right) = 1 + \frac{1}{m\alpha r} (\vec{r} \times \vec{J}) \cdot \vec{J} = 1 + \frac{J^2}{m\alpha r}$$

$$r = \frac{q}{1 - e \cos \theta} \quad \left(q = -\frac{J^2}{m\alpha} \right)$$

- $e < 1$: ellipse, $E < 0, \alpha < 0$ (gravatational force)
- $e = 1$: parabola, $E = 0, \alpha < 0$ (gravatational force)
- $e > 1$: hyperbola, $E > 0, \alpha < 0$ (gravatational force); $E > 0, \alpha > 0$ (repulsive force)

² $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$; $\vec{r} \cdot \dot{\vec{r}} = \frac{1}{2} \frac{d}{dt} (r^2) = r\dot{r}$
³ $\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$

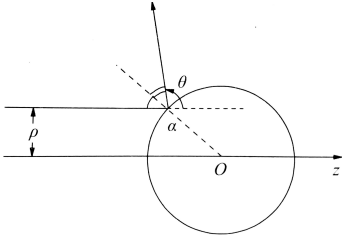


Fig 1.7: Particles against fixed balls of classic elastic collision

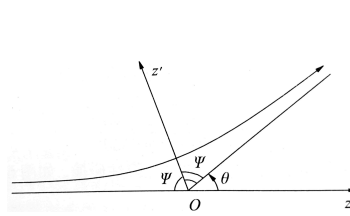


Fig 1.8: Rutherford Scattering

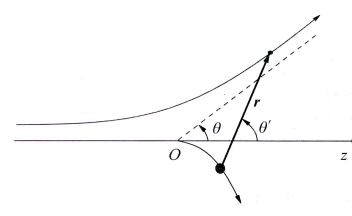


Fig 1.9: Scattering between two particles observed in the experimental system.

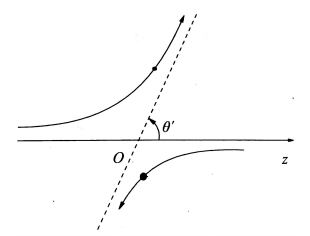


Fig 1.10: Observed in a centroid system.

1.5.5 The Two-Body Problem

Assume particles m_1 and m_2 do not affected by external forces, , then

$$\begin{cases} m_1 \ddot{\vec{r}}_1 = \vec{f}_{12} \\ m_2 \ddot{\vec{r}}_2 = \vec{f}_{21} = -\vec{f}_{12} \end{cases} \quad \begin{cases} \vec{r}_c = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \\ \vec{r}_{12} = \vec{r}_1 - \vec{r}_2 \end{cases} \quad \begin{array}{l} \text{(centroid coordinates)} \\ \text{(relative coordinate)} \end{array}$$

Therefore,

$$\text{centroid motion: } m \ddot{\vec{r}}_c = 0$$

$$\text{relative motion: } \ddot{\vec{r}}_1 - \ddot{\vec{r}}_2 = \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \vec{f}_{12} = \frac{1}{m_{12}} \vec{f}_{12}$$

$$\bullet \frac{1}{m_{12}} = \frac{1}{m_1} + \frac{1}{m_2}; m_{12} \text{ called reduced mass}$$

The motion equation of relative coordinate:

$$m_{12} \ddot{\vec{r}}_{12} = \vec{f}_{12}$$

In the m_2 translation reference frame(non-inertial), the motion of m_1 is like the motion of a mass m_{12} in an inertial frame. The two-body problem becomes a monomer problem.

$$\begin{aligned} \vec{r}'_1 &= \vec{r}_1 - \vec{r}_c = \frac{m_2}{m_1 + m_2} \vec{r}_{12} \\ \vec{r}_1 &= \vec{r}_c + \frac{m_2}{m_1 + m_2} \vec{r}_{12} \end{aligned}$$

$$\begin{aligned} \vec{r}'_2 &= \vec{r}_2 - \vec{r}_c = -\frac{m_1}{m_1 + m_2} \vec{r}_{12} \\ \vec{r}_2 &= \vec{r}_c - \frac{m_1}{m_1 + m_2} \vec{r}_{12} \end{aligned}$$

Total kinetic energy (inertial system):

$$T = \frac{1}{2} m_1 \dot{\vec{r}}_1^2 + \frac{1}{2} m_2 \dot{\vec{r}}_2^2 = \frac{1}{2} m v_c^2 + \frac{1}{2} m_{12} v_{12}^2 = T_c + T'$$

Centriod system kinetic energy:

$$T' = \frac{1}{2} m_1 \dot{\vec{r}}_1'^2 + \frac{1}{2} m_2 \dot{\vec{r}}_2'^2 = \frac{1}{2} m_{12} v_{12}^2$$

Look at Fig 1.9 and Fig 1.10, the experimental system is inertial system

- θ' : scattering angel in monomer body problem = centroid system (theoretical)
- θ : scattering in experimental system (experimental)

Chapter 2

Lagrangian Mechanics

2.1 Constraints

2.1.1 Constraints Classifications

Consider N particles

1. if integrable?

- Holonomic constraints:

$$f(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, t) = 0$$
$$\text{or } f(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, \dot{\vec{r}}_1, \dot{\vec{r}}_2, \dots, \dot{\vec{r}}_N, t) = 0 \quad (\text{Differential Constraints})$$

For integrable differential constraints, we have

$$\dot{\vec{r}}_1 - \dot{\vec{r}}_2 = \vec{c} \quad \rightarrow \quad \vec{r}_1 - \vec{r}_2 = \vec{c} \cdot t$$

- Nonholonomic constraints:

Ex 8. Vertical disk rolling on a horizontal xy plane. Describe the motion with (x, y, θ, ϕ) , x, y of the center, and angle of rotation ϕ about the axis of the disk, and an angle θ between the axis of the disk and x axis. (Fig: 2.1)

As a result of the constraint the velocity $\vec{v}$ of the center has a magnitude proportional to $\dot{\phi}$,

$$v = a\dot{\phi} \quad (a: \text{radius})$$

combining with $\dot{x} = v \sin \theta$, $\dot{y} = -v \cos \theta$, we have two differential equations of constraint:

$$dx - a \sin \theta d\phi = 0; \quad dy + a \cos \theta d\phi = 0 \quad \Rightarrow \quad \sin \theta dx - \cos \theta dy = 0 \quad (\text{non-integrable}^a)$$

^aIntegrability condition: $\vec{F} \cdot (\nabla \times \vec{F}) = 0$

2. if contain the time?

- Rhenomous constraints: contian the time as an explicit variable
- Scleronomous constraints: not dependent on time

$$f(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, \dot{\vec{r}}_1, \dot{\vec{r}}_2, \dots, \dot{\vec{r}}_N) = 0$$

3. if equation form?

- Unilateral constraints: inequalities
- Double constraints: equations

2.1.2 Generalized Coordinates and Degrees of Freedom

A system of N particles, $3N$ coordinates, k constraints, $s = 3N - k$ independent coordinates.

By introduction of new, s independent variables $q_1, q_2, \dots, q_s$ in terms of $\vec{r}_i$

$$\vec{r}_i = \vec{r}_i(q_1, \dots, q_s, t) \quad i = 1, \dots, 3N$$

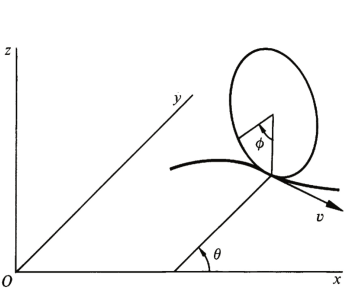


Fig 2.1: Vertical disk rolling on a horizontal plane.

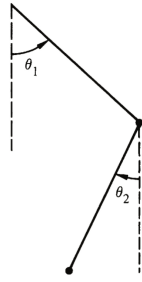


Fig 2.2: Double pendulum.

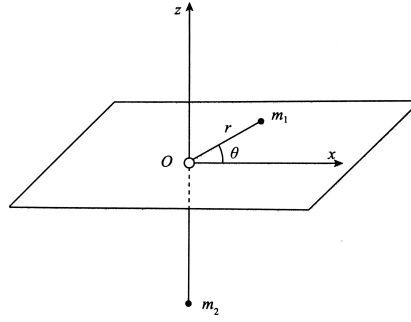


Fig 2.3: A light rope with a length of l passes through the small hole.

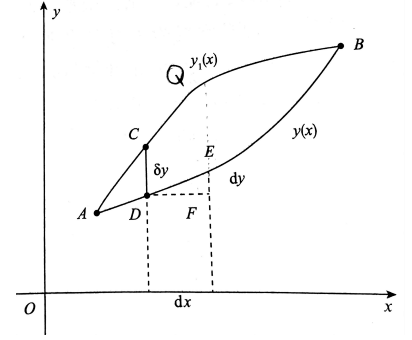


Fig 2.4: Comparison of variational versus differential.

- s : degrees of freedom
- $q_1, q_2, \dots, q_s$: generalized coordinates
- $\dot{q}_\alpha$: generalized velocity $\alpha = 1, \dots, s$
- Configuration space:

$$(\vec{r}_1, \dots, \vec{r}_N) \Leftrightarrow (q_1, \dots, q_s)$$

Ex 9. Double pendulum. (Fig: 2.2)

Degree of freedom is $s = 2$, the generalized coordinates are θ_1, θ_2 .

Furthermore, if there are l nonholonomic constraints, then degrees of freedom of dependent velocity is $3N - k - l$.

2.2 D'Alembert's Principle and Lagrange's Equations

1. Virtual Displacement

A virtual (infinitesimal) displacement of a system refers to a change in the configuration of the system as the result of any arbitrary infinitesimal change of the coordinates $\delta \vec{r}$, consistent with the forces and constraints imposed on the system at the given instant t .

$$\delta \vec{r}_i = \sum_{\alpha=1}^s \frac{\partial \vec{r}_i}{\partial q_\alpha} \delta q_\alpha$$

- instant happened
- satisfy constraints
- hypothetical (may not real, may not satisfy laws of physics, may have infinite variety)
- " δ ": viriational symble
 - when used for spatial coordinate, then the laws are same with " d ", and can swap with " d "
 - when used for time, $\delta t = 0$

2. Virtual work

The imaginary work of the force in the virtual displacement.

$$\delta W = \sum_i (\vec{F}_i + \vec{r}_i) \cdot \delta \vec{r}_i$$

- $\vec{F}_i$: applied force
- $\vec{r}_i$: constraint force

3. Ideal constraints

$$\sum_i \vec{r}_i \cdot \delta \vec{r}_i = 0$$

★ We are always discussing: unilateral & holonomic & ideal constraints.

4. D'Alembert principle

$$\vec{F}_i + \vec{r}_i = m_i \ddot{\vec{r}}_i \rightarrow \sum_{i=1}^N (\vec{F}_i + \vec{r}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0 \xrightarrow{\text{ideal constraints}} \boxed{\sum_{i=1}^N (\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0}$$

- * The total of applied force and inertial force is zero.
- * Equivalent to Newton's second law.

Ex 10. The motion function? (Fig. 2.3)

Degrees of freedom: (r, θ)

$$\begin{aligned} \begin{cases} \vec{r}_1 = r \vec{e}_r \\ \vec{r}_2 = -(l-r) \vec{e}_z \end{cases} &\Rightarrow \begin{cases} \ddot{\vec{r}}_1 = (\ddot{r} - r\dot{\theta}) \vec{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \vec{e}_\theta \\ \ddot{\vec{r}}_2 = \ddot{r} \vec{e}_z \end{cases} \Rightarrow \\ \{-m_1 g \vec{e}_z - m_1[(\ddot{r} - r\dot{\theta}) \vec{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \vec{e}_\theta]\delta(r \vec{e}_r) + \{-m_2 g \vec{e}_z - m_2(\ddot{r} \vec{e}_z)\delta(r \vec{e}_z) = 0, &\begin{cases} \delta(r \vec{e}_r) = \vec{e}_r \delta r + r \vec{e}_\theta \delta \theta \\ \delta(r \vec{e}_z) = \vec{e}_z \delta r \end{cases} \\ \Rightarrow [(m_1 + m_2)\ddot{r} - m_1 r \dot{\theta} + m_2 g] \delta r + [m_1 r (r\ddot{\theta} + 2\dot{r}\dot{\theta})] \delta \theta = 0 &\Rightarrow \begin{cases} (m_1 + m_2)\ddot{r} - m_1 r \dot{\theta} + m_2 g = 0 \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \end{cases} \end{aligned}$$

5. Static problem

d'Alembert Principle of Virtual Work:

$$\sum_{i=1}^N \vec{F}_i \cdot \delta \vec{r}_i = 0 \quad \text{The virtual work of applied force is zero} \Leftrightarrow \text{The system is in equilibrium}$$

- If conservative system,

$$0 = \sum_i \delta \vec{r}_i = - \sum_i \frac{\partial V}{\partial \vec{r}_i} \delta \vec{r}_i = \delta V \Rightarrow \delta V = 0$$

- Generally,

$$\begin{aligned} 0 = \sum_i \delta \vec{r}_i &= \sum_i \vec{F}_i \left(\sum_\alpha \frac{\partial \vec{r}_i}{\partial q_\alpha} \delta q_\alpha \right) = \sum_\alpha \left(\sum_i \vec{F}_i \frac{\partial \vec{r}_i}{\partial q_\alpha} \right) \delta q_\alpha \\ \text{Generalized force: } Q_\alpha &= \sum_i \vec{F}_i \frac{\partial \vec{r}_i}{\partial q_\alpha} \Rightarrow Q_\alpha = 0 \end{aligned}$$

Lagrange Relation:

$$\dot{\vec{r}} = \sum_\alpha \frac{\partial \vec{r}_i}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial \vec{r}_i}{\partial t} \Rightarrow \boxed{\frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha} = \frac{\partial \vec{r}}{\partial q_\alpha}}$$

Re-write D'Alembert principle and generalized force, we have

$$\begin{aligned} 0 &= \sum_i (\vec{F}_i - m_i \ddot{\vec{r}}_i) \sum_\alpha \frac{\partial \vec{r}_i}{\partial q_\alpha} \delta q_\alpha = \sum_\alpha \left[\sum_i (\vec{F}_i - m_i \ddot{\vec{r}}_i) \frac{\partial \vec{r}_i}{\partial q_\alpha} \right] \delta q_\alpha \\ Q_\alpha &= \sum_i \left[m_i \frac{d\dot{\vec{r}}_i}{dt} \right] \frac{\partial \vec{r}_i}{\partial q_\alpha} = \frac{d}{dt} \left[\sum_i m_i \dot{\vec{r}}_i \frac{\partial \vec{r}_i}{\partial q_\alpha} \right] - \sum_i m_i \dot{\vec{r}}_i \frac{d}{dt} \left[\frac{\partial \vec{r}_i}{\partial q_\alpha} \right] = \frac{d}{dt} \left[\sum_i m_i \dot{\vec{r}}_i \frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha} \right] - \sum_i m_i \dot{\vec{r}}_i \frac{d}{dt} \left[\frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha} \right] \\ &= \frac{d}{dt} \left[\sum_i \frac{\partial (\frac{1}{2} m \dot{\vec{r}}^2)}{\partial q_\alpha} \right] - \sum_i m_i \frac{\partial (\frac{1}{2} m \dot{\vec{r}}^2)}{\partial q_\alpha} \Rightarrow \end{aligned}$$

$$\text{Lagrange's Equation: } \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_\alpha} \right) - \frac{\partial T}{\partial q_\alpha} = Q_\alpha$$

- Conservative system: $\vec{F}_i = -\nabla_i V \Rightarrow Q_\alpha = - \sum_i \frac{\partial V}{\partial \vec{r}_i} \cdot \frac{\partial \vec{r}_i}{\partial q_\alpha} = - \frac{\partial V}{\partial q_\alpha}$

if $V = V(q_\alpha)$, we can find $\frac{d}{dt} \left(\frac{\partial V}{\partial \dot{q}_\alpha} \right) - \frac{\partial V}{\partial q_\alpha} = Q_\alpha$. Then $\frac{d}{dt} \left(\frac{\partial (T - V)}{\partial \dot{q}_\alpha} \right) - \frac{\partial (T - V)}{\partial q_\alpha} = 0$

Let Lagrange function $L \equiv T - V = L(q, \dot{q}, t)$, we have

Euler-Lagrange's equation in conservative system:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = 0$$

- Both conservative force and non-conservative force, the equation is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = Q_\alpha$$

Ex 11. Using Lagrange's Equation to solve Fig: 2.3

$$L = T - V = \frac{1}{2}m_1(\dot{r} + r^2\dot{\theta}^2) + \frac{1}{2}m_2\dot{r}^2 - (-m_2g(l - r))$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = 0 \Rightarrow (m_1 + m_2)\ddot{r} - (m_1r\dot{\theta}^2 - m_2g) = 0$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0 \Rightarrow m_1r^2\dot{\theta} = \text{const}$$

2.3 Viriational Principles and Lagrange's Equations

2.3.1 Calculus of variations

Varied paths of the function $y(x)$ in the one-dimensional extremum problem. (Fig: 2.4)

1. Functional

$$J[y(x)]$$

- Depends on the overall form of the argument $y(x)$
- $f(y(x))$ is compound function
- Extremum of viriational $\rightarrow$ Calculus of variations

2. Variations of $y(x)$ and $J[y(x)]$

$$\delta y = y_1(x) - y(x)$$

$$\delta J = J[y_1(x)] - J[y(x)] = J[y(x) + \delta y] - J[y(x)]$$

- The difference between the two functions when x is fixed
- Isochronous variation or fixed endpoint variation: $\delta x = 0$

3. Algorithms of variations

$$(a) \delta(J_1 + J_2) = \delta J_1 + \delta J_2$$

$$(b) \delta(J_1 J_2) = J_2 \delta J_1 + J_1 \delta J_2$$

(c) The order of isochronous variation and differential and integration of argument can be changed

- $\delta dy = d\delta y$

Proof.

$$D \rightarrow C \rightarrow G: \quad y(x) \rightarrow y(x) + \delta y \rightarrow y(x) + \delta y + d(y(x) + \delta y)$$

$$D \rightarrow E \rightarrow G: \quad y(x) \rightarrow y(x) + dy \rightarrow y(x) + dy + \delta(y(x) + dy) \Rightarrow$$

□

- $\delta \frac{d}{dx} y = \frac{d}{dx} \delta y$

Proof.

$$\delta \frac{d}{dx} y \triangleq \frac{d}{dx} (y + \delta y) - \frac{d}{dx} y = \frac{d}{dx} \delta y$$

□

$$\bullet \delta \int_a^b f(y(x))dx = \int_a^b \delta f(y(x))dx$$

Proof.

$$\delta \int_a^b f(y(x))dx \triangleq \int_a^b f(y(x) + \delta y)dx - \int_a^b f(y(x))dx = \int_a^b [f(y(x) + \delta y) - f(y(x))]dx = \int_a^b \delta f(y(x))dx$$

□

4. The condition for the functional to take the extremum in $y_0(x)$

$$(J[y(x)])_{\max} = J(y_0(x)) \Leftrightarrow \delta J \Big|_{y_0(x)} = 0$$

Proof. α : an infinitesimal parameter; $\eta(x)$: any function

$$y(x) = y_0(x) + \delta y(x) \rightarrow y(x, \alpha) = y(x, 0) + \alpha \eta(x), \quad y(x, 0) = y_0(x), \quad \alpha \eta(x) = \delta y(x)$$

Because J is taken as an extreme value at $\alpha = 0$, then

$$\left. \frac{\partial J[y(x, 0) + \alpha \eta(x)]}{\partial \alpha} \right|_{\alpha=0} = 0$$

$$\begin{aligned} \delta J \Big|_{y_0(x)} &= J[y_0(x) + \delta y(x)] - J[y_0(x)] = J[y(x, 0) + \alpha \eta(x)] - J[y(x, 0)] \\ &= \left. \frac{\partial J[y(x, 0) + \alpha \eta(x)]}{\partial \alpha} \right|_{\alpha=0} \cdot \alpha \quad (\text{Taylor expanding}) \\ &= 0 \end{aligned}$$

□

5. Euler's Equation

(a) Functional in one-dimensional

$$J = \int_{x_1}^{x_2} f(y(x), \dot{y}(x), x)dx$$

Consider the condition for J to take extremum, then ($\delta x = 0$)

$$\delta J = \int_{x_1}^{x_2} \delta f(y(x), \dot{y}(x), x)dx = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial \dot{y}} \delta \dot{y} \right) dx$$

Focus the right side of the addition, we have

$$\begin{aligned} \int_{x_1}^{x_2} \frac{\partial f}{\partial \dot{y}} \delta \left(\frac{d}{dx} y \right) dx &= \int_{x_1}^{x_2} \frac{\partial f}{\partial \dot{y}} \left(\frac{d}{dx} \delta y \right) dx = \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial f}{\partial \dot{y}} \delta y \right) dx - \int_{x_1}^{x_2} \left(\frac{d}{dx} \frac{\partial f}{\partial \dot{y}} \right) \delta y dx \\ &= \left. \frac{\partial f}{\partial \dot{y}} \delta y \right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \left(\frac{d}{dx} \frac{\partial f}{\partial \dot{y}} \right) \delta y dx \Rightarrow \delta J = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial \dot{y}} \right) \delta y dx \end{aligned}$$

δJ is true for any δy , therefore we got Euler differential equation:

$$\frac{d}{dx} \left(\frac{\partial f}{\partial \dot{y}} \right) - \frac{\partial f}{\partial y} = 0$$

(b) Multi-argument functional

$$J = \int_{x_1}^{x_2} f(y_1(x), \dots, y_s(x), \dot{y}_1(x), \dots, \dot{y}_s(x), x)dx$$

When $\delta J = 0$, because $\delta y_1, \dots, \delta y_s$ are dependent with each other, we got

Euler-Lagrange Differential Equations:

$$\frac{d}{dx} \left(\frac{\partial f}{\partial \dot{y}_\alpha} \right) - \frac{\partial f}{\partial y_\alpha} = 0 \quad \alpha = 1, \dots, s$$

2.3.2 Hamilton's Principle

◇ *Hamilton Principle: The motion of the system from time t_1 to time t_2 is such that the line integral.*

For the integral in Hamilton's principle,

$$S = \int_{t_1}^{t_2} L(q_1(t), \dots, \dot{q}_s(t), t) dt = \int_{t_1}^{t_2} L dt$$

with the transformation

$$\begin{aligned} x &\rightarrow t \\ y_\alpha(x) &\rightarrow q_\alpha(x) \\ f(y, \dot{y}, t) &\rightarrow L(q, \dot{q}, t) \\ J &\rightarrow S \end{aligned}$$

from $\delta s = 0$, then we show that Euler-Lagrange's equations follow from Hamilton's principle – for monogenic system with holonomic constraints:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = 0$$

For non-conservative systems,

$$\int_{t_1}^{t_2} \left(\delta L + \sum_{\alpha=1}^s Q_\alpha \delta q_\alpha \right) dt = 0 \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = Q_\alpha$$

Ex 12. The motion equations of the particle in gravitational field

$$\begin{aligned} L = T - V &= \frac{1}{2} m (\ddot{x} + \ddot{y} + \ddot{z}) - mgz & \dot{x}\delta\dot{x} &= \dot{x} \frac{d}{dt}(\delta x) = \frac{d}{dt}(x\delta x) - \ddot{x}\delta x = -\ddot{x}\delta x \\ \Rightarrow 0 = \delta S &= m \int_{t_1}^{t_2} (\dot{x}\delta\dot{x} + \dot{y}\delta\dot{y} + \dot{z}\delta\dot{z} - mg\delta z) dt = m \int_{t_1}^{t_2} (-\ddot{x}\delta x - \ddot{y}\delta y - (\ddot{z} + g)\delta z) dt \\ \Rightarrow \ddot{x} &= 0, \quad \ddot{y} = 0, \quad \ddot{z} = -g \end{aligned}$$

Simple examples of the applicaiton:

1. Shorted distance between two points in a plane $s[y(x)]$

$$\begin{aligned} s &= \int_1^2 ds = \int_1^2 \sqrt{dx^2 + dy^2} = \int_{x_1}^{x_2} \sqrt{1 + \dot{y}^2} dx \Rightarrow f(y, \dot{y}, x) = \sqrt{1 + \dot{y}^2} \\ \frac{\partial f}{\partial y} &= 0; \quad \frac{\partial f}{\partial \dot{y}} = \frac{\dot{y}}{\sqrt{1 + \dot{y}^2}} \Rightarrow \frac{d}{dx} \left(\frac{\dot{y}}{\sqrt{1 + \dot{y}^2}} \right) = 0 \Rightarrow \dot{y} = \pm \sqrt{\frac{c_1^2}{1 - c_1^2}} = k \\ &\Rightarrow \text{A straight line: } y = kx + b \end{aligned}$$

2. Shorted distance between two points in a sphere $s[\phi(\theta)]$

$$\begin{aligned} s &= \int_1^2 ds = r \int_1^2 \sqrt{d\theta^2 + \sin^2 \theta d\phi^2} = r \int_1^2 \sqrt{1 + \sin^2 \theta \dot{\phi}^2} d\theta \Rightarrow f(\phi, \dot{\phi}, \theta) = \sqrt{1 + \sin^2 \theta \dot{\phi}^2} \\ \frac{\partial f}{\partial \phi} &= 0; \quad \frac{\partial f}{\partial \dot{\phi}} = \frac{\dot{\phi} \sin^2 \theta}{\sqrt{1 + \sin^2 \theta \dot{\phi}^2}} = \text{const} = c_1 \rightarrow c_3 \sin(c_2 - \phi) = \cot \theta \\ &\Rightarrow \text{A transversal that cross spherical center plane and sphere: } Ay - Bz = z \end{aligned}$$

3. Catenary $V[y(x)]$

A flexible chain is loosely hung between two fixed points, let linear density as ρ , and we have $\delta V = 0$ ¹

$$V = \int_1^2 (\rho ds) gy = \rho g \int_{x_1}^{x_2} y \sqrt{1 + \dot{y}^2} dx \Rightarrow f(y, \dot{y}, x) = y \sqrt{1 + \dot{y}^2}$$

$$\frac{\partial f}{\partial y} = \sqrt{1 + \dot{y}^2}; \quad \frac{\partial f}{\partial \dot{y}} = \frac{y \dot{y}}{\sqrt{1 + \dot{y}^2}} \rightarrow y \ddot{y} - (q + \dot{y}^2) = 0, \quad \ddot{y} = \frac{1}{2\dot{y}} \frac{d(\dot{y}^2)}{dt} = \frac{1}{2} \frac{d(\dot{y}^2)}{dy} \Rightarrow$$

$$\frac{2dy}{y} = \frac{d\dot{y}}{1 + \dot{y}^2} \Rightarrow y^2 = c_1^2(1 + \dot{y}^2) \Rightarrow y_1 = c_1 \cosh \frac{x + c_2}{c_1}$$

Then, we can solve c_1 inversely with length of chain: $l = \int_1^2 ds = \int_{-a}^a \sqrt{1 + \dot{y}^2} dx = 2c_1 \sinh \frac{a}{c_1}$

4. The brachistochrone problem $t[y(x)]$

Find the curve joining two points, along which a particle falling from rest under the influence of gravity travels from the higher to the lower in the least time.

$$t_{12} = \int_1^2 \frac{ds}{v} = \int_1^2 \frac{\sqrt{1 + \dot{y}^2}}{\sqrt{2gy}} dx \Rightarrow f(y, \dot{y}, t) = \sqrt{\frac{1 + \dot{y}^2}{2gy}}$$

$$\rightarrow (1 + \dot{y})y = a \rightarrow \begin{cases} x = a\left(\theta - \frac{\sin 2\theta}{2}\right) \\ y = a \sin^2 \theta \end{cases} \rightarrow t_{12} = \sqrt{\frac{2a}{g}} \theta$$

It is sketched in Fig. 2.5 for the first cycle ($0 \leq x \leq 2\pi a$) and the beginning of the second cycle.

2.4 Discussion – Properties and Statics

$$\delta s = 0 \rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = 0, \quad \text{where } s = \int_{t_1}^{t_2} L dt \quad \text{and} \quad L = T - V$$

1. Additiveness:

$$T = T_1 + T_2$$

$$V = V_1 + V_2$$

$$L = (T_1 + T_2) - (V_1 + V_2) = L_1 + L_2$$

2. Non-uniqueness: The Lagrange's equation doesn't change by

- $L \rightarrow L + c$
- $L \rightarrow c' L$
- $L \rightarrow L' = L + \dot{f}(q_\alpha, t)$

Proof.

$$s \rightarrow s' = \int_{t_1}^{t_2} (L + \dot{f}(q_\alpha, t)) dt = s + f \Big|_{t_1}^{t_2} = s + f(q_\alpha^{(2)}, t_2) - f(q_\alpha^{(1)}, t_1)$$

$$\delta q_\alpha^{(1,2)} = 0, \quad \delta f = 0 \Rightarrow \delta s = \delta s'$$

□

Ex 13. A simple pendulum with forced oscillation $x_0 = a \cos \omega t$, L ? (Fig: 2.6)

$$\text{dof}=1(\theta) \quad x = a \cos \omega t + l \sin \theta, \quad y = \rho \cos \theta \Rightarrow \dot{x} = -a\omega \sin \omega t + \rho \dot{\theta} \cos \theta, \quad \dot{y} = -\rho \dot{\theta} \sin \theta$$

$$V = -mgl \cos \theta, \quad T = \frac{1}{2} m (a^2 \omega^2 \sin^2 \omega t - 2a\omega \sin \omega t \cos \theta + l^2 \dot{\theta}^2) \Rightarrow$$

$$L = T - V = \frac{1}{2} m a^2 \omega^2 \sin^2 \omega t - \frac{d}{dt} (m a \omega l \sin \omega t \sin \theta) + m a \omega^2 l \cos \omega t \sin \theta + \frac{1}{2} m l^2 \dot{\theta}^2 + mgl \cos \theta$$

$$\sin^2 \omega t = \frac{1}{2} (1 - \cos 2\omega t) = \frac{1}{2} \frac{d}{dt} (\sin 2\omega t) \frac{1}{4\omega} \Rightarrow L \rightarrow L = m a \omega^2 l \cos \omega t \sin \theta + \frac{1}{2} m l^2 \dot{\theta}^2 + mgl \cos \theta$$

¹ $\cosh x^2 - \sinh x^2 = 1, (\cosh x)' = \sinh x$

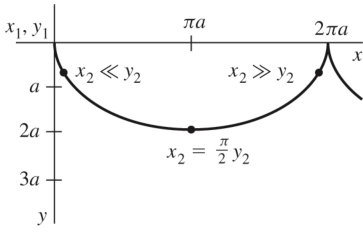


Fig 2.5: Rotary wheel line.

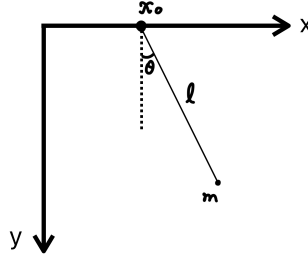


Fig 2.6: The simple pendulum.

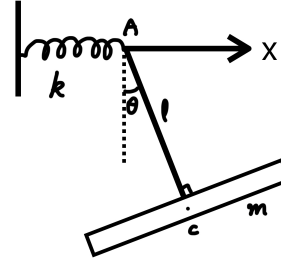


Fig 2.7: Wall clock.

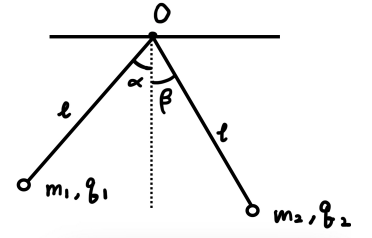


Fig 2.8: Two charges.

3. L is scale in generalized coordinate transformation

Consider rotation reference system, (x, y) coordinate is inertial coordinate, and (X, Y) coordinate rotate with ωt on point O . Then $L' = L = \frac{1}{2}m\dot{x}^2 + \dot{y}^2$

$$\begin{cases} x = X \cos \omega t - Y \sin \omega t \\ y = X \sin \omega t + Y \cos \omega t \end{cases} \Rightarrow L' = \frac{1}{2}m(\dot{X}^2 + \dot{Y}^2) + \frac{1}{2}m\omega^2(X^2 + Y^2) + m\omega(X\dot{Y} - Y\dot{X})$$

$$\begin{cases} \frac{d}{dt} \frac{\partial L'}{\partial \dot{X}} - \frac{\partial L'}{\partial X} = 0 \\ \frac{d}{dt} \frac{\partial L'}{\partial \dot{Y}} - \frac{\partial L'}{\partial Y} = 0 \end{cases} \Rightarrow \begin{cases} \ddot{X} - 2\omega\dot{Y} - \omega^2 X = 0 \\ \ddot{Y} + 2\omega\dot{X} - \omega^2 Y = 0 \end{cases} \quad \text{or} \quad \begin{cases} m a_X = m\omega^2 X + 2m\omega\dot{Y} \\ m a_Y = m\omega^2 Y - 2m\omega\dot{X} \end{cases}$$

We can also know left is centrifugal force, and right is Coriolis force.

Ex 14. The wall clock is a simple pendulum, and the right connect with a $k = \frac{mg}{l}$ string, L ? (Fig: 2.7)
dof=2 (x, θ) , choose center-of-mass frame, $T = T_c + T'$ and $x_c = x + l \sin \theta$, $y = l \cos \theta$, then

$$\begin{aligned} T_c &= \frac{m}{2}(\dot{x}_c^2 + \dot{y}_c^2), \quad T' = \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2} \cdot \frac{1}{3}ml^2 \cdot \dot{\theta}^2, \quad V = -mgl \cos \theta + \frac{1}{2}kx^2 \\ L &= T - V = T_c + T' - V = \frac{1}{2}m\left(\dot{x}^2 + \frac{4}{3}l^2\dot{\theta}^2 + 2l\dot{x} \cos \theta\right) + mgl \cos \theta - \frac{1}{2}\frac{mg}{l}x^2 \end{aligned}$$

Statics

Equilibrium configuration problems and constraint force problem.

1. Virtual work principle: $\sum_i \vec{F}_i \cdot \delta \vec{r}_i = 0$
2. $\delta V = 0 \quad \leftarrow \quad \frac{\partial V}{\partial q_\alpha} = 0 \quad \leftarrow \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} - \frac{\partial L}{\partial q_\alpha} = 0$
3. $Q_\alpha = 0 \quad \leftarrow \quad T = 0 \quad \leftarrow \quad \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_\alpha} - \frac{\partial T}{\partial q_\alpha} = Q_\alpha$

Ex 15. When system is in equilibrium, $\alpha + \beta = \frac{\pi}{2}$, m_2 ? (Fig: 2.8)

$$\delta V = \frac{\partial V}{\partial \alpha} \delta \alpha + \frac{\partial V}{\partial \beta} \delta \beta = 0, \text{ and let } k = \frac{q_1 q_2}{4\pi \epsilon_0}$$

$$\begin{aligned} V &= -m_1 gl \cos \alpha - m_2 gl \cos \beta + \frac{k}{\sqrt{l^2 + l^2 - 2l^2 \cos(\alpha + \beta)}} = -m_1 gl \cos \alpha - m_2 gl \cos \beta + \frac{k}{2l \sin \frac{\alpha + \beta}{2}} \\ \frac{\partial V}{\partial \alpha} &= 0 = m_1 gl \sin \alpha - \frac{k \cos \frac{\alpha + \beta}{2}}{4l \sin^2 \frac{\alpha + \beta}{2}} \xrightarrow{\alpha + \beta = \pi/2} m_1 gl \sin \alpha - \frac{k}{2\sqrt{2}l} = 0 \\ \frac{\partial V}{\partial \beta} &= 0 = m_2 gl \sin \beta - \frac{k \cos \frac{\alpha + \beta}{2}}{4l \sin^2 \frac{\alpha + \beta}{2}} \xrightarrow{\alpha + \beta = \pi/2} m_2 gl \cos \alpha - \frac{k}{2\sqrt{2}l} = 0 \\ \Rightarrow m_2 &= \left(\frac{2g^2 l^4}{k^2} - \frac{1}{m_2^2} \right)^{-\frac{1}{2}} \end{aligned}$$

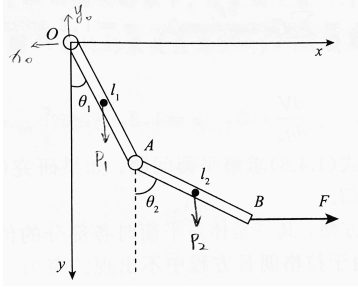


Fig 2.9: Double pendulum with applied force.

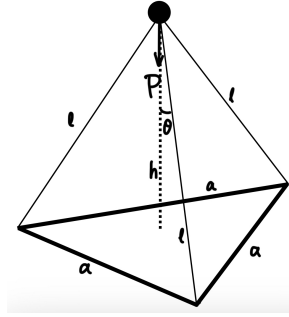


Fig 2.10: Use a rubber band to prop up a small ball held up by a light bar.

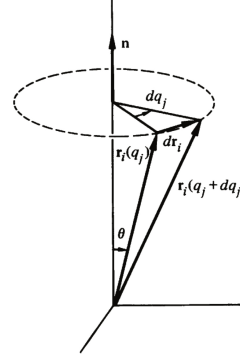


Fig 2.11: Change of a position vector under rotation of the system.

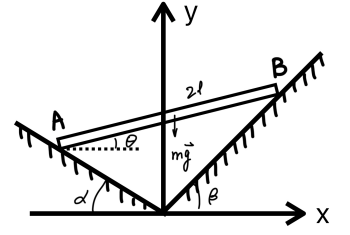


Fig 2.12: The rod is stationary on an inclined plane.

Ex 16. Double pendulum with applied force, P_1 and P_2 are gravitational force. When in equilibrium, θ_1 ? θ_2 ? and the constraint force of R_x, R_y at point O? (Fig: 2.9)

$$Q_\alpha = \sum_i \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_\alpha} = 0, \text{dof} = 2(\theta_1, \theta_2), \quad y_1 = \frac{l_2}{2} \cos \theta_1, \quad y_2 = l_1 \cos \theta_1 + \frac{l_2}{2} \cos \theta_2, \quad x_3 = l_1 \sin \theta_1 + l_2 \sin \theta_2$$

$$\begin{cases} Q_1 = P_1 \frac{\partial y_1}{\partial \theta_1} + P_2 \frac{\partial y_2}{\partial \theta_1} + F \frac{\partial x_3}{\partial \theta_1} = 0 \\ Q_2 = P_1 \frac{\partial y_1}{\partial \theta_2} + P_2 \frac{\partial y_2}{\partial \theta_2} + F \frac{\partial x_3}{\partial \theta_2} = 0 \end{cases} \Rightarrow \begin{cases} \tan \theta_1 = \frac{2F}{P_1 + 2P_2} \\ \tan \theta_2 = \frac{P_2}{P_1} \end{cases}$$

seem constraint force as applied force, then

$$\text{dof}=4 (\theta_1, \theta_2, x_0, y_0), \quad y_1 = \frac{l_2}{2} \cos \theta_1 + y_0, \quad y_2 = l_1 \cos \theta_1 + \frac{l_2}{2} \cos \theta_2 + y_0, \quad x_3 = l_1 \sin \theta_1 + l_2 \sin \theta_2 + x_0$$

$$\begin{cases} Q_1 = P_1 \frac{\partial y_1}{\partial \theta_1} + P_2 \frac{\partial y_2}{\partial \theta_1} + F \frac{\partial x_3}{\partial \theta_1} + \vec{R} \cdot \frac{\partial \vec{r}_O}{\partial \theta_1} = 0 \\ Q_2 = P_1 \frac{\partial y_1}{\partial \theta_2} + P_2 \frac{\partial y_2}{\partial \theta_2} + F \frac{\partial x_3}{\partial \theta_2} + \vec{R} \cdot \frac{\partial \vec{r}_O}{\partial \theta_2} = 0 \\ Q_3 = P_1 \frac{\partial y_1}{\partial x_0} + P_2 \frac{\partial y_2}{\partial x_0} + F \frac{\partial x_3}{\partial x_0} + \vec{R} \cdot \frac{\partial \vec{r}_O}{\partial x_0} = 0 \\ Q_4 = P_1 \frac{\partial y_1}{\partial y_0} + P_2 \frac{\partial y_2}{\partial y_0} + F \frac{\partial x_3}{\partial y_0} + \vec{R} \cdot \frac{\partial \vec{r}_O}{\partial y_0} = 0 \end{cases} \Rightarrow \begin{cases} F + R_x = 0 \\ P_1 + P_2 + R_y = 0 \end{cases} \Rightarrow \begin{cases} R_x = -F \\ R_y = -P_1 - P_2 \end{cases}$$

Ex 17. System is in equilibrium, the tension of rubber band? (Fig: 2.10)

dof=1, θ , the virtual work of applied force is zero, so

$$\begin{aligned} \sum_i \vec{F}_i \cdot \delta \vec{r}_i &= 0 \quad \rightarrow \quad \delta W = -P \delta h - T \delta a = 0 \\ h &= l \sin \theta, \quad a = 3\sqrt{3}l \sin \theta \quad \rightarrow \quad -(Pl \sin \theta - 3\sqrt{3}lT \cos \theta) \delta \theta = 0 \quad \Rightarrow \quad T = \frac{P}{3\sqrt{3}} \tan \theta \end{aligned}$$

2.5 Motion Integral

2.5.1 Conservation Theorems

Motion Differentiation and The Conservation of Momentum

A system of s degrees of freedom $\rightarrow s$ second order differential equations $\rightarrow 2s$ first order differential equations $\rightarrow 2s$ constants of integration: "integrals of motion"

Define generalized momentum (canonical momentum / conjugate momentum):

$$p_\alpha \equiv \frac{\partial L}{\partial \dot{q}_\alpha} \quad ; \quad \frac{dp_\alpha}{dt} = \frac{\partial L}{\partial q_\alpha}$$

Furthermore, if L is not an explicit function of generalized coordinate q_α , which means that $p_\alpha = \text{const}$. Here,

- q_α is called "cyclic coordinate"
- p_α is called "cyclic integral"

◇ The generalized momentum conjugate to a cyclic coordinate is conserved.

For example, in Cartesian coordinate $L = T = \frac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \rightarrow p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$, so x, y, z is cyclic coordinates and $m\dot{x}, m\dot{y}, m\dot{z}$ are conserved.

Energy Function and The Conservation of Energy

$$\begin{aligned} \frac{dL}{dt} &= \sum_{\alpha} \frac{\partial L}{\partial q_{\alpha}} \frac{dq_{\alpha}}{dt} + \sum_{\alpha} \frac{\partial L}{\partial \dot{q}_{\alpha}} \frac{d\dot{q}_{\alpha}}{dt} + \frac{\partial L}{\partial t} = \sum_{\alpha} \frac{\partial p_{\alpha}}{\partial t} \dot{q}_{\alpha} + \sum_{\alpha} p_{\alpha} \frac{d\dot{q}_{\alpha}}{dt} + \frac{\partial L}{\partial t} = \frac{d}{dt} \left(\sum_{\alpha} p_{\alpha} \dot{q}_{\alpha} \right) + \frac{\partial L}{\partial t} \\ \Rightarrow \frac{d}{dt} \left(\sum_{\alpha} p_{\alpha} \dot{q}_{\alpha} - L \right) &= -\frac{\partial L}{\partial t} \Rightarrow \frac{dH}{dt} = -\frac{\partial L}{\partial t} \quad \text{where} \end{aligned}$$

Hamilton function:

$$H(q_{\alpha}, \dot{q}_{\alpha}, t) = \sum_{\alpha} p_{\alpha} \dot{q}_{\alpha} - L$$

If L is not an explicit function of time t , then H is conserved. Here, H is generalized energy integral. The physics meaning of H : (Considering $V = V(q_{\alpha}, t)$ and $\vec{r}_i = \vec{r}_i(q_{\alpha}, t)$)

$$\begin{aligned} T &= \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i^2 = \frac{1}{2} \sum_i m_i \left(\sum_{\alpha} \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \dot{q}_{\alpha} + \frac{\partial \vec{r}_i}{\partial t} \right)^2 \\ &= \frac{1}{2} \sum_i m_i \left(\sum_{\alpha} \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \dot{q}_{\alpha} \right) \left(\sum_{\beta} \frac{\partial \vec{r}_i}{\partial q_{\beta}} \dot{q}_{\beta} \right) + \sum_i m_i \left(\sum_{\alpha} \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \dot{q}_{\alpha} \right) \left(\frac{\partial \vec{r}_i}{\partial t} \right) + \frac{1}{2} \sum_i m_i \left(\frac{\partial \vec{r}_i}{\partial t} \right)^2 \\ &= \frac{1}{2} \sum_{\alpha\beta} \left(m_i \sum_i \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \frac{\partial \vec{r}_i}{\partial q_{\beta}} \right) \dot{q}_{\alpha} \dot{q}_{\beta} + \sum_{\alpha} \left(\sum_i m_i \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \frac{\partial \vec{r}_i}{\partial t} \right) \dot{q}_{\alpha} + \frac{1}{2} \sum_i m_i \left(\frac{\partial \vec{r}_i}{\partial t} \right)^2 \\ &= T_2 + T_1 + T_0 \end{aligned}$$

- $T_2(q, \dot{q})$ is a quadtric function of the $\dot{q}$'s
- $T_1(q, \dot{q})$ is linear in $\dot{q}$'s
- $T_0(q)$ is a function of $\dot{q}$'s only

Euler's theorem states that if f is a homogeneous function of degree n in the variables x_i , then

$$\sum_i x_i \frac{\partial f}{\partial x_i} = n f \Rightarrow \sum_{\alpha} \left(\frac{\partial T_2}{\partial \dot{q}_{\alpha}} + \frac{\partial T_1}{\partial \dot{q}_{\alpha}} + \frac{\partial T_0}{\partial \dot{q}_{\alpha}} \right) \dot{q}_{\alpha} = 2T_2 + T_1 + 0$$

Therefore,

$$H = \sum_{\alpha} p_{\alpha} \dot{q}_{\alpha} - L = \sum_{\alpha} \frac{\partial T}{\partial \dot{q}_{\alpha}} \dot{q}_{\alpha} - L = 2T_2 + T_1 - (T - V) = T_2 - T_0 + V$$

For scleronomous constraint, $\vec{r}_i = \vec{r}_i(q_{\alpha})$, then $H = T_2 + V$, by $T = T_2$, and $H = T + V$ is mechanical energy.

Ex 18. Consider in the horizontal plane, the spring rotates around the origin at a constant speed. H ? Constraints are $\theta = \omega t, \omega = \text{const}$. Let generalized coordinate is r .

$$\begin{aligned} T &= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) = \frac{1}{2} m (\dot{r}^2 + r^2 \omega^2) = T_2 + T_0, \quad V = \frac{1}{2} k (r - r_0)^2 \Rightarrow \\ H &= T_2 - T_0 + V = \frac{1}{2} m (\dot{r}^2 - r^2 \omega^2) + \frac{1}{2} k (r - r_0)^2 = \text{const} \end{aligned}$$

2.5.2 Symmetry Properties

* Symmetry is about physics laws.

1. L is not explicit of $q_{\alpha} = x_{\alpha} \Leftrightarrow$ "Spatial Translation Invariance". Then

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_{\alpha}} = 0 \Rightarrow p_{\alpha} = \text{const}$$

Table 2.1: Comparison between Newtonian and Lagrangian Mechanics

(Dynamics)		(Symmetry)	
Newtonian Mechanics	Conserved Quantity	Lagrangian Mechanics	
Applied force is zero	→ Linear Momentum	←	Spatial Homogeneous
Applied torque is zero	→ Angular Momentum	←	Spatial Isotropic
Conserved force	→ Kinetic Energy	←	Time Homogeneous

◇ The generalized momentum conjugate to a cyclic coordinate is conserved

Spatial Translation Invariance / Spatially Homogeneous → Momentum is conserved

2. L is not explicit of angel coordinate $q_\alpha \Leftrightarrow$ "Spatial Rotation Invariance". Then (Fig: 2.11)

$$\frac{\partial \vec{r}_i}{\partial q_\alpha} = \lim_{dq_\alpha \rightarrow 0} \frac{\vec{r}_i(q_\alpha + dq_\alpha) - \vec{r}_i(q_\alpha)}{dq_\alpha} = \frac{dq_\alpha}{dq_\alpha} \vec{n} \times \vec{r}_i = \vec{n} \times \vec{r}_i$$

$$p_\alpha = \frac{\partial T}{\partial \dot{q}_\alpha} = \sum_i m_i \dot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial \dot{q}_\alpha} = \sum_i m_i \vec{v}_i \cdot (\vec{n} \times \vec{r}_i) = \vec{n} \cdot \sum_i (\vec{r}_i \times m \dot{\vec{r}}_i) = \vec{n} \cdot \vec{J}$$

Spatial Rotation Invariance / Spatial Isotropic → Angular momentum is conserved

3. L is not explicit of $t \Leftrightarrow$ "Time Translation Invariance". Then

$$H = T + V = E = \text{const}$$

Time Translation Invariance / Time Homogeneous → Kinetic energy is conserved

2.5.3 Noether's Theorem ¶

From $q_\alpha \rightarrow q_\alpha + \epsilon \Delta q_\alpha$,

$$\epsilon \Delta L = \frac{\partial L}{\partial q_\alpha} (\epsilon \Delta q_\alpha) + \frac{\partial L}{\partial \dot{q}_\alpha} \frac{d}{dt} (\epsilon \Delta q_\alpha) = \epsilon \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \Delta q_\alpha \right) + \epsilon \left(\frac{\partial L}{\partial q_\alpha} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_\alpha} \right) \Delta q_\alpha$$

and $L \rightarrow L + \epsilon \Delta L = L + \epsilon \frac{d}{dt} f$, we have

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \Delta q_\alpha - f \right) = 0$$

Therefore, the conservation quantity:

$$Q = \frac{\partial L}{\partial \dot{q}_\alpha} \Delta q_\alpha - f$$

Symmetry → Conservation laws

1. Translation of space: $x \rightarrow x' + \epsilon, \Delta q_\alpha = 1; f = 0 \Rightarrow Q = \frac{\partial L}{\partial \dot{x}} = p_x = \text{const}$

2. Rotation of space: $\phi \rightarrow \phi' + \epsilon, \Delta \phi = 1; f = 0 \Rightarrow Q = \frac{\partial L}{\partial \dot{\phi}} = p_\phi = \text{const}$

3. Translation of time: $t \rightarrow t' + \epsilon, q_\alpha \rightarrow q_\alpha(1 + \epsilon) = q_\alpha(t) + \epsilon \dot{q}_\alpha, \Delta q_\alpha = \dot{q}_\alpha$;
 $L \rightarrow L(t + \epsilon) = L(t) + \epsilon \frac{d}{dt} L, f = L \Rightarrow Q = \frac{\partial L}{\partial \dot{q}_\alpha} - L = p_\alpha \dot{q}_\alpha - L = H = \text{const}$

2.5.4 Generalized Potential

If there exists $U = U(q_\alpha, \dot{q}_\alpha, t)$, let

$$Q_\alpha = -\frac{\partial U}{\partial q_\alpha} + \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_\alpha} \right)$$

then U is called generalized potential or velocity-dependent potential. And for $L \equiv T - U$, the Lagrange's equation in conservative system is still true.

Ex 19. Calculate the generalized potential of charged particles in electromagnetic field: ^{ab}

Change the form of $\vec{E}, \vec{B}$ to potential form, $\vec{B} = \nabla \times \vec{A}$ and $\nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t} \right) = 0 \rightarrow \vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$, then

$$\begin{aligned}\vec{F} &= e(\vec{E} + \dot{\vec{r}} \times \vec{B}) = e\left(-\nabla\phi - \frac{\partial \vec{A}}{\partial t} + \dot{\vec{r}} \times (\nabla \times \vec{A})\right) = e\left(-\nabla\phi - \frac{\partial \vec{A}}{\partial t} + \nabla(\vec{A} \cdot \dot{\vec{r}}) - (\dot{\vec{r}} \cdot \nabla)\vec{A}\right) \\ &= e\left(-\nabla\phi - \frac{d}{dt}\vec{A} + \nabla(\vec{A} \cdot \dot{\vec{r}})\right) \quad \leftarrow \quad \frac{\partial \vec{A}}{\partial t} + (\dot{\vec{r}} \cdot \nabla)\vec{A} = \frac{d\vec{A}}{dt} \\ &= e\left(-\nabla(\phi - \vec{A} \cdot \dot{\vec{r}}) + \frac{d}{dt} \frac{\partial}{\partial \dot{\vec{r}}}(\phi - \vec{A} \cdot \dot{\vec{r}})\right) \Rightarrow U = e(\phi - \vec{A} \cdot \dot{\vec{r}})\end{aligned}$$

Furthermore, the Lagrange's equation in conservative system of $L = \frac{1}{2} \vec{v}^2 - e\phi + e\vec{A} \cdot \vec{v}$ is motion equations of the charged particle in electromagnetic field $\vec{F} = e(\vec{E} + \dot{\vec{r}} \times \vec{B}) = m\ddot{\vec{r}}$. And we have generalized momentum $\vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} + e\vec{A}$

^aMaxwell equations: $\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$; $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$; $\nabla \times \vec{B} - \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \vec{j}$; $\nabla \cdot \vec{B} = 0$

^b $\nabla(\vec{A} \cdot \dot{\vec{r}}) = (\dot{\vec{r}} \cdot \nabla)\vec{A} + \vec{B} \times (\nabla \times \vec{A}) + (\vec{A} \cdot \nabla)\dot{\vec{r}} + \vec{A} \times (\nabla \times \dot{\vec{r}})$; $\nabla \cdot \dot{\vec{r}} = 0, \nabla \times \dot{\vec{r}} = 0$

2.5.5 Method of Lagrange Multipliers $\mathfrak{E}$

Using $3N$ independent coordinates u_m ($m = 1, \dots, N$) to represent Lagrange's equation, and satisfy k constraints

$$f_j(u_i, t) = 0 \quad (j = 1, \dots, k)$$

Introduce k multipliers λ_j , and

$$\begin{aligned}L' &= L + \lambda_j f_j \\ S' &= \int_{t_1}^{t_2} L' dt = \int_{t_1}^{t_2} (L + \lambda_j f_j) dt\end{aligned}$$

Then after variation of λ_j and u_m , we have

$$\begin{aligned}\delta S' &= \int_{t_1}^{t_2} f_j \delta \lambda_j dt \rightarrow \text{Constraints: } f_j = 0 \\ \delta S' &= \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial u_m} - \frac{d}{dt} \frac{\partial L}{\partial \dot{u}_m} + \lambda_j \frac{\partial f_j}{\partial u_m} \right) \delta u_m dt\end{aligned}$$

Take the appropriate value of λ_j for $\delta S' = 0$, since u_m are independent, we have

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{u}_m} \right) - \frac{\partial L}{\partial u_m} = \lambda_j \frac{\partial f_j}{\partial u_m}$$

- which is Lagrange's equations using independent coordinates
- k independent values and $3N - k$ dependent values for δu_m
- For statics, $T = 0$, we got the virtual work principle using independent coordinates

$$Q_m + \lambda_j \frac{\partial f_j}{\partial u_m} = 0$$

- Q_m : generalized force (applied force)
- $R_m = \lambda_j \frac{\partial f_j}{\partial u_m}$: generalized constraint force

Ex 20. The angel of θ when system in equilibrium and the constraint force $\vec{R}$ of points A and B. (Fig: 2.12)
Choose independent coordinates: $u_m = (x_A, y_A, \theta)$. The constraints are

$$\begin{cases} f_1 = x_A \tan \alpha + y_A = 0 \\ f_2 = (x_A + 2l \cos \theta) \tan \beta - (y_A + 2l \sin \theta) = 0 \end{cases} \Rightarrow \begin{cases} \lambda_1 \frac{\partial f_1}{\partial x_A} + \lambda_2 \frac{\partial f_2}{\partial x_A} = \lambda_1 \tan \alpha + \lambda_2 \tan \beta \\ \lambda_1 \frac{\partial f_1}{\partial y_A} + \lambda_2 \frac{\partial f_2}{\partial y_A} = \lambda_1 - \lambda_2 \\ \lambda_1 \frac{\partial f_1}{\partial \theta} + \lambda_2 \frac{\partial f_2}{\partial \theta} = 2\lambda_2 l(-\sin \theta \tan \beta - \cos \theta) \end{cases}$$

$$Q_m = \vec{F}_i \frac{\partial \vec{r}_i}{\partial u_m} = -mg \frac{\partial}{\partial u_m} (y_A + l \sin \theta)$$

$$\Rightarrow Q_x = -mg \frac{\partial}{\partial x_A} (y_A + l \sin \theta) = 0, \quad Q_y = -mg \frac{\partial}{\partial y_A} (y_A + l \sin \theta) = -mg, \quad Q_\theta = -mg \frac{\partial}{\partial \theta} (y_A + l \sin \theta) = -mgl \cos \theta$$

Then, we got Lagrange multipliers

$$\begin{cases} \lambda_1 \tan \alpha + \lambda_2 \tan \beta = 0 \\ \lambda_1 - \lambda_2 = -mg \\ 2\lambda_2 l(-\sin \theta \tan \beta - \cos \theta) = -mg \cos \theta \end{cases} \Rightarrow \begin{cases} \tan \theta = \frac{1}{2}(\cot \alpha - \cot \beta) \\ \lambda_1 = \frac{\tan \beta}{\tan \alpha + \tan \beta} mg \\ \lambda_2 = -\frac{\tan \alpha}{\tan \alpha + \tan \beta} mg \end{cases}$$

The constraint force of A:

$$\vec{R}_A = \lambda_1 \frac{\partial f_1}{\partial x_A} \vec{e}_x + \lambda_1 \frac{\partial f_1}{\partial y_A} \vec{e}_y = \frac{\sin \beta}{\sin(\alpha + \beta)} mg \vec{n}, \quad \vec{n} = \sin \alpha \vec{e}_x + \cos \alpha \vec{e}_y$$

The constraint force of B:

$$\vec{R}_B = \lambda_2 \frac{\partial f_2}{\partial x_A} \vec{e}_x + \lambda_2 \frac{\partial f_2}{\partial y_A} \vec{e}_y = \frac{\sin \alpha}{\sin(\alpha + \beta)} mg \vec{n}', \quad \vec{n}' = \sin \beta \vec{e}_x + \cos \beta \vec{e}_y$$

2.6 Oscillations

We consider conservative systems and generalized coordinates do not involve the time explicitly. The system is said to be in equilibrium when the generalized forces acting on the system vanish:

$$Q_i = -\left(\frac{\partial V}{\partial q_i}\right)_0 = 0$$

Equilibrium position: (Fig: 2.13)

- Stable: if a small disturbance of the system from equilibrium results only a small bounded motion about the rest position.
- Unstable: if an infinitesimal disturbance eventually produces unbounded motion.

2.6.1 Small Vibrations in Single-Degree-of-Freedom Systems

Considering free vibrations, and in equilibrium position, the potential energy is zero, expanding the potential energy about q_0 , we obtain

$$V(q) = V(q_0) + \left(\frac{dV}{dq}\right)_{q_0} (q - q_0) + \frac{1}{2} \left(\frac{d^2V}{dq^2}\right)_{q_0} (q - q_0)^2 + \dots$$

Because q_0 is equilibrium position, therefore $(dV/dq)_{q_0}$ is zero, and let $q - q_0 = x$ is infinitesimal, we can get the potential of harmonic oscillation

$$V = \frac{1}{2} kx^2, \quad k = \left(\frac{d^2V}{dq^2}\right)_{q_0}$$

The equation of motion is

$$T = \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i^2 = \frac{1}{2} \sum_i m_i \left(\frac{\partial \vec{r}_i}{\partial q} \dot{q}\right)^2 = \frac{1}{2} m_i \left(\frac{\partial \vec{r}_i}{\partial q} \frac{\partial \vec{r}_i}{\partial q}\right) \dot{q}^2 = \frac{1}{2} a(q) \dot{q}^2 \approx \frac{1}{2} a(q_0) \dot{q}^2 = \frac{1}{2} m \dot{x}^2$$

$$L = T - V = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} kx^2 \xrightarrow{\text{Lagrange eq}} m\ddot{x} + kx = 0 \quad (\text{linear simple harmonic vibration})$$

$$\Rightarrow x = A \cos(\omega t + \phi) \quad \text{or} \quad x = A \sin(\omega t + \phi) \quad \text{or} \quad x = Ae^{i(\omega t + \phi)}$$

2.6.2 Small Vibrations in Multi-Degree-of-Freedom Systems

The potential energy is zero at the equilibrium configuration of the system, $q_{01}, q_{02}, \dots, q_{0s}$. The deviations of the generalized coordinates from equilibrium will be denoted by η_α :

$$q_\alpha = q_{0\alpha} + \eta_\alpha$$

Expanding the potential energy about $q_{0\alpha}$ in origin ($q_\alpha = \eta_\alpha$), we obtain

$$\begin{aligned} V(q_1, \dots, q_s) &= V(q_{01}, \dots, q_{0s}) + \sum_{\alpha} \left(\frac{\partial V}{\partial q_{\alpha}} \right)_0 q_{\alpha} + \frac{1}{2} \sum_{\alpha\beta} \left(\frac{\partial^2 V}{\partial q_{\alpha} \partial q_{\beta}} \right)_0 q_{\alpha} q_{\beta} + \dots \\ &= \frac{1}{2} \sum_{\alpha\beta} k_{\alpha\beta} q_{\alpha} q_{\beta} = \frac{1}{2} \mathbf{q}^T \mathbf{k} \mathbf{q} \end{aligned}$$

- $k_{\alpha\beta} = k_{\beta\alpha} = \sum_{\alpha\beta} \left(\frac{\partial^2 V}{\partial q_{\alpha} \partial q_{\beta}} \right)_0$

- $[k_{\alpha\beta}] \equiv \mathbf{k}$ is symmetric positive definite matrix; $[q_{\alpha}] = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_s \end{bmatrix} \equiv \mathbf{q}$

Same with vibration in single degree of system, we have

$$\begin{aligned} T &= \frac{1}{2} \sum_i m_i \dot{\vec{r}}_i^2 = \frac{1}{2} \sum_i m_i \left(\sum_{\alpha} \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \dot{q}_{\alpha} \right) \left(\sum_{\beta} \frac{\partial \vec{r}_i}{\partial q_{\beta}} \dot{q}_{\beta} \right) = \frac{1}{2} \sum_{\alpha\beta} \left(\sum_i \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \frac{\partial \vec{r}_i}{\partial q_{\beta}} m_i \right) \dot{q}_{\alpha} \dot{q}_{\beta} \\ &= \frac{1}{2} \sum_{\alpha\beta} a_{\alpha\beta}(q) \dot{q}_{\alpha} \dot{q}_{\beta} \approx \frac{1}{2} \sum_{\alpha\beta} a_{\alpha\beta}(0) \dot{q}_{\alpha} \dot{q}_{\beta} = \frac{1}{2} \sum_{\alpha\beta} m_{\alpha\beta} \dot{q}_{\alpha} \dot{q}_{\beta} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} \end{aligned}$$

- $a_{\alpha\beta} = a_{\beta\alpha} = \sum_i \frac{\partial \vec{r}_i}{\partial q_{\alpha}} \frac{\partial \vec{r}_i}{\partial q_{\beta}} m_i$, $m_{\alpha\beta} = a_{\alpha\beta}(0)$

- $[m_{\alpha\beta}] \equiv \mathbf{M}$ is a symmetric positive definite matrix, $[\dot{q}_{\alpha}] \equiv \dot{\mathbf{q}}$

Then we got Lagrangian and the motion of equations:

$$\begin{aligned} L &= T - V = \frac{1}{2} \sum_{\alpha\beta} (m_{\alpha\beta} \dot{q}_{\alpha} \dot{q}_{\beta} - k_{\alpha\beta} q_{\alpha} q_{\beta}) \\ \frac{\partial L}{\partial \dot{q}_{\alpha}} &= \sum_{\beta} m_{\alpha\beta} \dot{q}_{\beta}, \quad \frac{\partial L}{\partial q_{\alpha}} = \sum_{\beta} -k_{\alpha\beta} q_{\beta} \Rightarrow \sum_{\beta} (m_{\alpha\beta} \ddot{q}_{\beta} + k_{\alpha\beta} q_{\beta}) = 0 \end{aligned}$$

which is equivalent to

$$\boxed{\mathbf{M} \ddot{\mathbf{q}} + \mathbf{k} \mathbf{q} = 0} \quad \text{where} \quad \mathbf{M} = \begin{bmatrix} m_{11} & \cdots & m_{1s} \\ \vdots & \ddots & \vdots \\ m_{s1} & \cdots & m_{ss} \end{bmatrix}, \mathbf{k} = \begin{bmatrix} k_{11} & & \\ & \ddots & \\ & & k_{ss} \end{bmatrix}, \mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_s \end{bmatrix}$$

When $\omega \neq 0$, let solution is

$$\mathbf{q} = \mathbf{A}^{(\omega)} \cos(\omega t + \phi) = \begin{bmatrix} A_{1\omega} \\ \vdots \\ A_{s\omega} \end{bmatrix} \cos(\omega t + \phi)$$

- $\mathbf{A}^{(0)}$ is a column matrix

- $A_{\alpha\omega}$: the subscription α means generalized coordinate and ω is angular frequency

Take it into the differential equation, we got the eigen function:

$$\boxed{(-\omega^2 \mathbf{M} + \mathbf{k}) \mathbf{A}^{(\omega)} = 0} \quad \text{or} \quad \begin{bmatrix} -\omega^2 m_{11} + k_{11} & -\omega^2 m_{12} + k_{12} & \cdots & -\omega^2 m_{1s} + k_{1s} \\ -\omega^2 m_{21} + k_{21} & -\omega^2 m_{22} + k_{22} & \cdots & -\omega^2 m_{2s} + k_{2s} \\ \vdots & \vdots & \ddots & \vdots \\ -\omega^2 m_{s1} + k_{s1} & -\omega^2 m_{s2} + k_{s2} & \cdots & -\omega^2 m_{ss} + k_{ss} \end{bmatrix} \begin{bmatrix} A_{1\omega} \\ A_{2\omega} \\ \vdots \\ A_{s\omega} \end{bmatrix} = 0$$

They constitute s linear homogeneous equations about ω^2 , and consequently can have a nontrivial solution only if $\det(-\omega^2 \mathbf{M} + \mathbf{k}) = 0$, therefore

$$\begin{aligned} \mathbf{A}^{(\omega)\top} (-\omega^2 \mathbf{M} + \mathbf{k}) \mathbf{A}^{(\omega)} = 0 & \rightarrow -\omega^2 \mathbf{A}^{(\omega)\top} \mathbf{M} \mathbf{A}^{(\omega)} + \mathbf{A}^{(\omega)\top} \mathbf{k} \mathbf{A}^{(\omega)} = 0 \\ \Rightarrow \omega^2 = \frac{\mathbf{A}^{(\omega)\top} \mathbf{k} \mathbf{A}^{(\omega)}}{\mathbf{A}^{(\omega)\top} \mathbf{M} \mathbf{A}^{(\omega)}} > 0 & \text{ (positive definite matrix)} \end{aligned}$$

For every ω^2 there is $\pm\omega$, and for each of them there is only phase difference, therefore we only take $+\omega = \omega_l$ one. For each of the ω_l , from eigen function we can get eigen vector $\mathbf{A}^{(l)}$, and one of the special solutions is

$$\mathbf{q}^{(l)} = \mathbf{A}^{(l)} \cos(\omega_l t + \phi_l)$$

General solution:

$$\mathbf{q} = \sum_l C_l \mathbf{q}^{(l)} = \sum_l C_l \mathbf{A}^{(l)} \cos(\omega_l t + \phi_l)$$

- C_l and ϕ_l is determined by initial conditions
- Remember here only take s non-negative $\omega_1, \dots, \omega_s$ which are called normal frequency

Some discussion of $\mathbf{A}^{(l)}$:

- The equations are homogeneous $\mathbf{A}^{(l)}$ cannot be completely determined, since a holistic factor is missing
- For different $\mathbf{A}^{(l)}$ of ω_l , they are orthogonal, which means $\langle \mathbf{A}^{(l)}, \mathbf{A}^{(m)} \rangle = 0 \quad (l \neq m)$
 - Inner product: $\langle \mathbf{A}^{(l)}, \mathbf{A}^{(m)} \rangle = \mathbf{A}^{(l)\top} \mathbf{M} \mathbf{A}^{(m)}$, where $\mathbf{M}$ is metric matrix
 - Length: $|\mathbf{A}^{(l)}| = \sqrt{\langle \mathbf{A}^{(l)}, \mathbf{A}^{(l)} \rangle}$
 - Unit vector: $\mathbf{a}^{(l)} = \frac{\mathbf{A}^{(l)}}{|\mathbf{A}^{(l)}|}$

Proof.

$$\begin{aligned} \mathbf{A}^{(l)\top} (-\omega_m^2 \mathbf{M} + \mathbf{k}) \mathbf{A}^{(m)} = 0 & \rightarrow -\omega_m^2 \mathbf{A}^{(l)\top} \mathbf{M} \mathbf{A}^{(m)} + \mathbf{A}^{(l)\top} \mathbf{k} \mathbf{A}^{(m)} = 0 \rightarrow \\ \mathbf{A}^{(m)\top} (-\omega_l^2 \mathbf{M} + \mathbf{k}) \mathbf{A}^{(l)} = 0 & \rightarrow -\omega_l^2 \mathbf{A}^{(m)\top} \mathbf{M} \mathbf{A}^{(l)} + \mathbf{A}^{(m)\top} \mathbf{k} \mathbf{A}^{(l)} = 0 \\ \xrightarrow{\text{transposed}} & -\omega_l^2 \mathbf{A}^{(l)\top} \mathbf{M} \mathbf{A}^{(m)} + \mathbf{A}^{(l)\top} \mathbf{k} \mathbf{A}^{(m)} = 0 \rightarrow \\ \Rightarrow (\omega_m^2 - \omega_l^2) \mathbf{A}^{(l)\top} \mathbf{M} \mathbf{A}^{(m)} = 0 & \Rightarrow \langle \mathbf{A}^{(l)}, \mathbf{A}^{(m)} \rangle = 0 \quad \text{if } \omega_m \neq \omega_l \end{aligned}$$

□

- If ω have multiple root, using Schmidt orthogonalization

Or the general solution: (Method 2)

$$\begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_s \end{bmatrix} = \begin{bmatrix} C_1 A_{11} \cos(\omega_1 t \phi_1) + C_2 A_{12} \cos(\omega_2 t \phi_2) + \cdots + C_s A_{1s} \cos(\omega_s t \phi_s) \\ C_1 A_{21} \cos(\omega_1 t \phi_1) + C_2 A_{22} \cos(\omega_2 t \phi_2) + \cdots + C_s A_{2s} \cos(\omega_s t \phi_s) \\ \vdots \\ C_1 A_{s1} \cos(\omega_1 t \phi_1) + C_2 A_{s2} \cos(\omega_2 t \phi_2) + \cdots + C_s A_{ss} \cos(\omega_s t \phi_s) \end{bmatrix}$$

Introduce normal coordinates $\xi_l = C_l \cos(\omega_l t + \phi_l)$, then

$$\mathbf{q} = \sum_l \mathbf{A}^{(l)} \xi_l = \begin{bmatrix} \mathbf{A}^{(1)} & \cdots & \mathbf{A}^{(s)} \end{bmatrix} \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_s \end{bmatrix} \equiv \mathbf{A} \boldsymbol{\xi} \Rightarrow \mathbf{q} = \mathbf{A} \boldsymbol{\xi}, \quad \boldsymbol{\xi} = \mathbf{A}^{-1} \mathbf{q}$$

$\mathbf{A}$ ($s \times s$) diagonalizes $\mathbf{M}$ and $\mathbf{k}$ at same time, which means we have

$$\mathbf{A}^T \mathbf{M} \mathbf{A} = \boldsymbol{\mu} \quad \mathbf{A}^T \mathbf{k} \mathbf{A} = \boldsymbol{\lambda} \quad (\mu_l \omega_l^2 = \lambda_l)$$

Proof. From $\langle \mathbf{A}^{(l)}, \mathbf{A}^{(m)} \rangle = \mathbf{A}^{(l)T} \mathbf{M} \mathbf{A}^{(m)} = \delta_{lm} \mu_l$, then

$$\begin{bmatrix} \mathbf{A}^{(1)T} \\ \mathbf{A}^{(2)T} \\ \vdots \\ \mathbf{A}^{(s)T} \end{bmatrix} \mathbf{M} \begin{bmatrix} \mathbf{A}^{(1)} & \mathbf{A}^{(2)} & \dots & \mathbf{A}^{(s)} \end{bmatrix} = \begin{bmatrix} \mathbf{A}^{(1)T} \mathbf{M} \mathbf{A}^{(1)} & \mathbf{A}^{(1)T} \mathbf{M} \mathbf{A}^{(2)} & \dots & \mathbf{A}^{(1)T} \mathbf{M} \mathbf{A}^{(s)} \\ \mathbf{A}^{(2)T} \mathbf{M} \mathbf{A}^{(1)} & \mathbf{A}^{(2)T} \mathbf{M} \mathbf{A}^{(2)} & \dots & \mathbf{A}^{(2)T} \mathbf{M} \mathbf{A}^{(s)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}^{(s)T} \mathbf{M} \mathbf{A}^{(1)} & \mathbf{A}^{(s)T} \mathbf{M} \mathbf{A}^{(2)} & \dots & \mathbf{A}^{(s)T} \mathbf{M} \mathbf{A}^{(s)} \end{bmatrix} = \begin{bmatrix} \mu_1 & & & \\ & \mu_2 & & \\ & & \ddots & \\ & & & \mu_s \end{bmatrix}$$

□

Proof. From $(-\omega_l^2 \mathbf{M} + \mathbf{k}) \mathbf{A}^{(l)} = 0 \Rightarrow \mathbf{k} \mathbf{A}^{(l)} = \omega_l^2 \mathbf{M} \mathbf{A}^{(l)}$, then

$$\begin{aligned} \mathbf{k} \begin{bmatrix} \mathbf{A}^{(1)} & \mathbf{A}^{(2)} & \dots & \mathbf{A}^{(s)} \end{bmatrix} &= \mathbf{M} \begin{bmatrix} \omega_1^2 \mathbf{A}^{(1)} & \omega_2^2 \mathbf{A}^{(2)} & \dots & \omega_s^2 \mathbf{A}^{(s)} \end{bmatrix} \\ &= \mathbf{M} \begin{bmatrix} \mathbf{A}^{(1)} & \mathbf{A}^{(2)} & \dots & \mathbf{A}^{(s)} \end{bmatrix} \begin{bmatrix} \omega_1^2 & & & \\ & \omega_2^2 & & \\ & & \ddots & \\ & & & \omega_s^2 \end{bmatrix} \equiv \mathbf{M} \mathbf{A} \boldsymbol{\Omega} \end{aligned}$$

Therefore, $\mathbf{k} \mathbf{A} = \mathbf{M} \mathbf{A} \boldsymbol{\Omega} \Rightarrow \mathbf{A}^T \mathbf{k} \mathbf{A} = \mathbf{A}^T \mathbf{M} \mathbf{A} \boldsymbol{\Omega} \Rightarrow$

$$\boldsymbol{\lambda} = \boldsymbol{\mu} \boldsymbol{\Omega}$$

□

Lagrange's equations using normal coordinates:

$$\begin{aligned} T &= \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} = \frac{1}{2} \dot{\boldsymbol{\xi}}^T \mathbf{A}^T \mathbf{M} \mathbf{A} \dot{\boldsymbol{\xi}} = \frac{1}{2} \dot{\boldsymbol{\xi}}^T \boldsymbol{\mu} \dot{\boldsymbol{\xi}} = \frac{1}{2} \sum_l \mu_l \dot{\xi}_l^2 \\ V &= \frac{1}{2} \mathbf{q}^T \mathbf{k} \mathbf{q} = \frac{1}{2} \boldsymbol{\xi}^T \mathbf{A}^T \mathbf{k} \mathbf{A} \boldsymbol{\xi} = \frac{1}{2} \boldsymbol{\xi}^T \boldsymbol{\lambda} \boldsymbol{\xi} = \frac{1}{2} \sum_l \lambda_l \xi_l^2 \\ L = T - V &= \frac{1}{2} \sum_l \mu_l \dot{\xi}_l^2 - \frac{1}{2} \sum_l \lambda_l \xi_l^2, \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{\xi}_l} - \frac{\partial L}{\partial \xi_l} = 0, \quad \frac{\partial}{\partial \xi_l} \left(\sum_m \mu_m \dot{\xi}_m^2 \right) = 2\mu_l \dot{\xi}_l \Rightarrow \\ \ddot{\xi} + \omega_l^2 \xi_l &= 0 \rightarrow \xi_l = C_l \cos(\omega_l t + \phi) \end{aligned}$$

2.6.3 Free Vibrations of a Linear Triatomic Molecule

Summary: (Notice: There is equilibrium position.)

- Method 1

1. Get $T = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}}$ and $V = \frac{1}{2} \mathbf{q}^T \mathbf{k} \mathbf{q}$ in second order, then we know $\mathbf{M}$ and $\mathbf{k}$
2. Write eigen function $(-\omega^2 \mathbf{M} + \mathbf{k}) \mathbf{A}^{(\omega)} = 0$
3. Solve eigen value function $\det(-\omega^2 \mathbf{M} + \mathbf{k}) = 0$, get $\omega_1, \dots, \omega_s$
4. Substitute it into eigen function and normalize, get eigen vector $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(s)}$
5. Write both special solution and general solution

- Method 2

- Find $\mathbf{M}$ and $\mathbf{k}$
- Seek $\mathbf{A}$ that let $\mathbf{A}^T \mathbf{M} \mathbf{A} = \boldsymbol{\mu}$, $\mathbf{A}^T \mathbf{k} \mathbf{A} = \boldsymbol{\lambda}$ at same time
- Let $\mathbf{q} = \mathbf{A} \boldsymbol{\xi}$, then $T = \frac{1}{2} \sum_l \mu_l \dot{\xi}_l^2$, $V = \frac{1}{2} \sum_l \lambda_l \xi_l^2$
- $\ddot{\xi} + \omega_l^2 \xi_l = 0 \rightarrow \xi_l = C_l \cos(\omega_l t + \phi) \quad \left(\omega_l^2 = \frac{\lambda_l}{\mu_l} \right)$, and get general solution

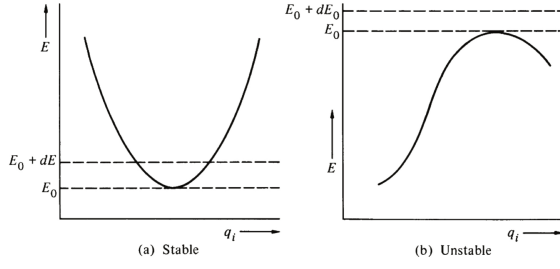


Fig 2.13: Shape of the potential energy curve at equilibrium.

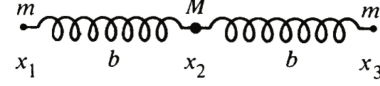


Fig 2.14: Model of a linear symmetrical triatomic molecule.

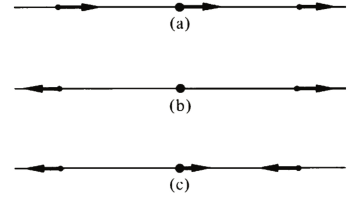


Fig 2.15: Longitudinal normal modes of the linear symmetric triatomic molecule.

Ex 21. In equilibrium configuration of the molecule, two atoms are symmetrically located on each side of an atom, consider only vibration along the line, and the interatomic potential will be approximated by two springs of force constant k joining the atoms. (Fig: 2.14)

dof= (x_1, x_2, x_3) ; equilibrium position: $(x_1 = 0, x_2 = b, x_3 = b)$. Here we define generalized coordinates as

$$q_1 = x_1, \quad q_2 = x_2 - b, \quad q_3 = x_3 - 2b \quad \Rightarrow \quad \text{equilibrium position: } q_1 = q_2 = q_3 = 0$$

Then we can get $\mathbf{M}$ and $\mathbf{k}$:

$$T = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}M\dot{x}_2^2 + \frac{1}{2}m\dot{x}_3^2 = \frac{1}{2}m\dot{q}_1^2 + \frac{1}{2}M\dot{q}_2^2 + \frac{1}{2}m\dot{q}_3^2$$

$$V = \frac{1}{2}k(x_2 - x_1 - b)^2 + \frac{1}{2}k(x_3 - x_2 - b)^2 = \frac{1}{2}k(q_1^2 + 2q_2^2 + q_3^2 - 2q_1q_2 - 2q_2q_3)$$

$$\Rightarrow \quad \mathbf{M} = \begin{bmatrix} m & & \\ & M & \\ & & m \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{bmatrix}$$

Using Method 1, take them into $(-\omega^2\mathbf{M} + \mathbf{k})\mathbf{A}^{(\omega)} = 0$ and since $\det(-\omega^2\mathbf{M} + \mathbf{k}) = 0$, we got the eigen values:

$$\begin{vmatrix} -\omega^2 m + k & -k & 0 \\ -k & -\omega^2 M + 2k & -k \\ 0 & -k & -\omega^2 m + k \end{vmatrix} = \omega(-m\omega^2 + k)(mM\omega^2 - 2km - kM) = 0 \Rightarrow$$

$$\omega_1 = 0, \quad \omega_2 = \sqrt{\frac{k}{m}}, \quad \omega_3 = \sqrt{\frac{k}{m}\left(1 + \frac{2m}{M}\right)}$$

1. $\omega_1 = 0$

$$\begin{bmatrix} k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{bmatrix} \begin{bmatrix} A_{11} \\ A_{21} \\ A_{31} \end{bmatrix} = 0 \rightarrow A_{11} = A_{21} = A_{31}; \quad \mathbf{A}^{(1)\top} \mathbf{M} \mathbf{A}^{(1)} = 1 \rightarrow A_{11} = A_{21} = A_{31} = \frac{1}{\sqrt{2m + M}}$$

$$\Rightarrow \mathbf{q}^{(1)} = \frac{1}{\sqrt{2m + M}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} (t + C) \quad (* \text{ Fig 2.15 (a) an overall movement of three atoms with the same displacement})$$

The corresponding normal coordinate is

$$\ddot{\xi} = 0 \Rightarrow \xi_1 = C_1(t + C)$$

2. $\omega_2 = \sqrt{\frac{k}{m}}$

$$\begin{bmatrix} 0 & -k & 0 \\ 0 & -\frac{M}{m}k + 2k & -k \\ 0 & -k & 0 \end{bmatrix} \begin{bmatrix} A_{12} \\ A_{22} \\ A_{32} \end{bmatrix} = 0 \rightarrow A_{22} = 0, A_{32} = -A_{12}; \quad m(A_{12}^2 + A_{22}^2) = 1 \rightarrow A_{12} = -A_{32} = \frac{1}{\sqrt{2m}}$$

$$\Rightarrow \mathbf{q}^{(2)} = \frac{1}{\sqrt{2m}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cos\left(\sqrt{\frac{k}{m}}t + \phi_2\right) \quad (* \text{ Fig 2.15 (b) the center does not move, the two sides move in the opposite direction})$$

The corresponding normal coordinate is

$$\ddot{\xi}_2 + \frac{k}{m}\xi_2 = 0 \Rightarrow \xi_2 = C_2 \cos\left(\sqrt{\frac{k}{m}}t + \phi_2\right)$$

3. $\omega_3 = \sqrt{\frac{k}{m}\left(1 + \frac{2m}{M}\right)}$

$$A_{13} = A_{33} = -\frac{M}{2m}A_{23}; \quad \mathbf{q}^{(3)} = \begin{bmatrix} \frac{1}{\sqrt{2m(1 + \frac{2m}{M})}} \\ -\frac{M}{2m} \\ \frac{1}{\sqrt{2m(1 + \frac{2m}{M})}} \end{bmatrix} \cos\left(\sqrt{\frac{k}{m}\left(1 + \frac{2m}{M}\right)}t + \phi_3\right)$$

(* Fig 2.15 (c) the two sides move in the same direction, the center in opposite direction, but the mass point isn't change)

The general solution is

$$\mathbf{q} = C_1\mathbf{q}^{(1)} + C_2\mathbf{q}^{(2)} + C_3\mathbf{q}^{(3)}$$

Chapter 3

Rigid Body Mechanics

3.1 Rigid Body Motion

"A rigid body was defined previously as a system of mass points subject to the holonomic constraints that the distance between all pairs of points remain constant throughout the motion."

Degrees of Freedom of Rigid Body

- For N particles system, the constraints: $c_N^2 = \frac{1}{2}N(N-1) \xrightarrow{n \rightarrow \infty} \sim N^2 \gg N$
- Confirm the three points on the rigid body that are not collinear, and the rigid body is completely determined
- Six independent coordinate
 - three external coordinates to specify the position of some reference point in the body
 - * fix a point as base point in the rigid body, got three Cartesian coordinates
 - three more to specify how the body is oriented with respect to the external coordinate
 - * Take any line on the base point as the axis, two are independent direction cosine, and one more is the angle at which the rigid body is turned relative to the axis
- Classifications of rigid body motion
 - translation: dof=3
 - move on a surface: dof=2+1(rotation of the plane)
 - fixed-point rotation (interested): dof=3
 - fixed-axis rotation: dof=1
- * Finite rotation is not vector

Proof. $\mathbf{x} \rightarrow \mathbf{A}_1 \mathbf{x} \rightarrow \mathbf{A}_2 \mathbf{A}_1 \mathbf{x}; \quad \mathbf{x} \rightarrow \mathbf{A}_2 \mathbf{x} \rightarrow \mathbf{A}_1 \mathbf{A}_2 \mathbf{x},$

since $\mathbf{A}_2 \mathbf{A}_1 \neq \mathbf{A}_1 \mathbf{A}_2$, therefore the finite rotation composition don't satisfy commutative law. □

- * Infinitesimal rotation is vector

Proof. $\mathbf{A}_1 = \mathbb{1} + \epsilon_1, \quad \mathbf{A}_2 = \mathbb{1} + \epsilon_2$, and we take first order approximation, then

$2\mathbf{A}_1 = \mathbb{1} + \epsilon_2 + \epsilon_1, \quad \mathbb{1}\mathbf{A}_2 = \mathbb{1} + \epsilon_1 + \epsilon_2$, satisfy the commutative law. □

Euler's Theorem on the Motion of a Rigid Body

◇ *The general displacement of a rigid body with one point fixed is a rotation about some point.*

Proof. (Fig: 3.1)

Isosceles sphere: $\widehat{AP} = \widehat{A'P}$; $\widehat{BP} = \widehat{B'P}$; Rigid body: $\widehat{AB} = \widehat{A'B'}$.

Therefore, sphere $\triangle ABP \cong \triangle A'B'P$, and $\angle APB = \angle A'P'A'$, $\angle APA' = \angle BPB'$, which means when A is turned to A' with OP as the axis, B goes to B' at the same time. The positional displacement of the rigid body with respect to O is equivalent to a fixed axis rotation about the axis.

Deduction: Motion of rigid body = Translation of the base point + Rotation around the base point □

Velocity

Fixed-point motion (Fig: 3.2)

$$\text{Displacement: } \Delta \vec{r} = \vec{n} \times \vec{r} \Delta \varphi = (\vec{n} \Delta \varphi) \times \vec{r} = \Delta \vec{\varphi} \times \vec{r}$$

$$\text{Angular Velocity: } \vec{\omega} = \frac{d\vec{\varphi}}{dt} = \frac{d\varphi}{dt} \vec{n} = \omega \vec{n}$$

$$\text{Angular Acceleration: } \vec{\beta} = \frac{d\vec{\omega}}{dt} = \frac{d\omega}{dt} \vec{n} = \dot{\omega} \vec{n}$$

$$\text{Linear Velocity: } \vec{v} = \frac{d\vec{r}}{dt} = \frac{d\vec{\varphi}}{dt} \times \vec{r} = \vec{\omega} \times \vec{r}$$

$$\text{Linear Acceleration: } \vec{a} = \frac{d\vec{v}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} = \vec{\beta} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

- $\Delta \vec{\varphi}$: infinitesimal angle displacement vector
- $\vec{n} \varphi$: finite angular displacement is not a vector, since it don't satisfy commutative law.
- Angular velocity is vector, which can commute.
- The direction of $\vec{\beta}$ and $\vec{\omega}$ are not same usually.
- For the points on the axis, linear velocity is vanished.

General motion:

$$\vec{v} = \vec{v}_A + \vec{\omega} \times \vec{r}'$$

$$\vec{a} = \vec{v}_A + \vec{\beta} \times \vec{r}' + \vec{\omega} \times (\vec{\omega} \times \vec{r}')$$

- P : base point
- $\vec{A}'$: the position vector relative to base point
- Absoluteness of angular velocity: Angular velocity is independent of base point selection.

Proof. Choose base point of A and B , we have: $\vec{v} = \vec{v}_A + \vec{\omega}_A \times \vec{AP}$; $\vec{v} = \vec{v}_B + \vec{\omega}_B \times \vec{BP}$

And we have $\vec{v}_B = \vec{v}_A + \vec{\omega}_A \times \vec{AB}$, take it into, then

$$0 = \vec{\omega}_A \times (\vec{AP} - \vec{AB}) - \vec{\omega}_B \times \vec{BP} = (\vec{\omega}_A - \vec{\omega}_B) \times \vec{BP}$$

which means, for any P , we have $\vec{\omega}_A = \vec{\omega}_B$

□

Instantaneous Axis of Rotation

The time rate of change of the vector $\vec{G}$ as the two observers is then obtained by

$$\left(\frac{d\vec{G}}{dt} \right)_{\text{space}} = \left(\frac{d\vec{G}}{dt} \right)_{\text{body}} + \vec{\omega} \times \vec{G}$$

- $\vec{\omega}$ lies along the axis of the infinitesimal rotation occurring between t and $t + dt$
 - direction: instantaneous axis of rotation
 - magnitude: instantaneous rate of rotation of the body
- The instantaneous velocity of a straight line that passes a fixed point and is parallel to the $\vec{\omega}$ direction is zero
- Real-time variation, acceleration can be non-zero

Let fixed point as origin, build space coordinates (inertial coordinate) and body-fixed coordinate (Fig: 3.3)

- Herpolhode cone: As it rotates with a fixed point, the instantaneous axis of rotation traces out a cone plane in space coordinate with the fixed point as the vertex
- Polhode cone: As it rotates with a fixed point, the instantaneous axis of rotation traces out a cone plane in body-fixed coordinate with the fixed point as the vertex

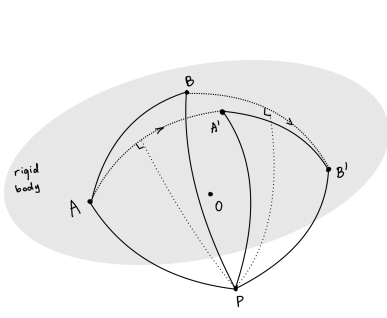


Fig 3.1: Rotation around the O point.

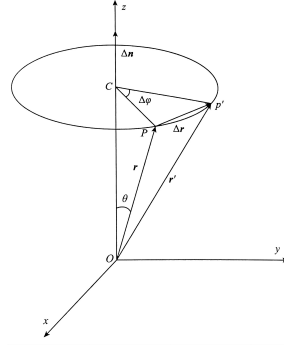


Fig 3.2: Derivation of the rotation formula.

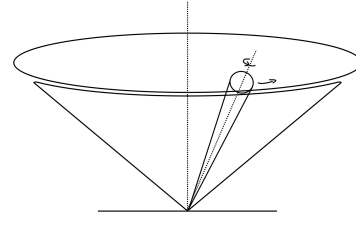


Fig 3.3: Polhode cone and herpolhode cone when the rigid body is in a fixed point motion.

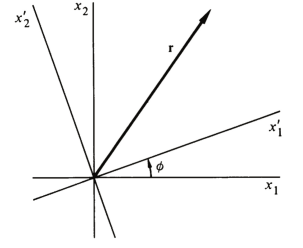


Fig 3.4: Rotation of the coordinate axes, as equivalent to two-dimensional orthogonal transformation.

3.2 Orthogonal Transformations

Any linear transformation, $x'_i = a_{ij}x_j$ ($i = 1, 2, 3$), that has the properties required by an $a_{ij}a_{ik} = \delta_{jk}$ (orthogonality condition) is called an orthogonal transformation.

A two-dimensional transformation from one Cartesian coordinate system to another corresponds to a rotation of the axes in the plane (Fig: 3.4)

$$\begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- (x_1, x_2, x_3) is space coordinate; (x'_1, x'_2, x'_3) is body-fixed coordinate
- Two view:
 - Passive view: In different coordinates to look a same vector, the components are different
 - Initiative view: the vector is rotated with space coordinate on rigid body, therefore

$$x_1 = x'_1 \cos \phi - x'_2 \sin \phi; \quad x_2 = x'_1 \sin \phi + x'_2 \cos \phi$$

Matrix of transformation (operator):

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

A operating on the vector $\mathbf{r}$, changing it to a different vector $\mathbf{r}'$

$$\mathbf{r}' = \mathbf{A}\mathbf{r}$$

- Or

$$\mathbf{r} = \mathbf{A}^T \mathbf{r}' = \mathbf{A}^{-1} \mathbf{r}'$$

- $\mathbf{A}$ depicts a coordinate transformation from space to body-fixed
- $\mathbf{A}^T$ also a rotation operation

3.2.1 Formal Properties of The Transformation Matrix

Consider 3×3 matrix

$$1. \quad \mathbf{A}^T \mathbf{A} = \mathbf{A} \mathbf{A}^T = \mathbb{1} \quad \rightarrow \quad \text{Orthogonal matrix}$$

Proof.

$$\mathbf{r}'^T \mathbf{r}' = (\mathbf{A}\mathbf{r})^T (\mathbf{A}\mathbf{r}) = \mathbf{r}^T (\mathbf{A}^T \mathbf{A}) \mathbf{r} \xrightarrow[\text{change magnitude}]{\text{Rotation didn't}} \mathbf{r}^T \mathbf{r} \Rightarrow \mathbf{A}^T \mathbf{A} = \mathbb{1}$$

Therefore, $\mathbf{A}$ and $\mathbf{A}^T$ are both symmetric matrix, which have six dependent equations, a_{ij} only has three dependent components $\square$

2. $\det \mathbf{A} = 1 \rightarrow$ Special orthogonal matrix (Rotation matrix)

Proof.

$$\mathbf{A}^T \mathbf{A} = \mathbb{1} \rightarrow \det(\mathbf{A}^T \mathbf{A}) = \det(\mathbf{A}) \det(\mathbf{A}^T) = (\det(\mathbf{A}))^2 = 1 \Rightarrow \det(\mathbf{A}) = \pm 1$$

The identity transformation is a special case of continuous rotation and corresponds to $\mathbb{1}$ which determinant is 1. It is impossible for the determinant of continuous rotation to mutate to -1. $\square$

3. There are must one of the eigenvalues are 1 for $\mathbf{A}$

Proof.

$$(\mathbf{A} - \mathbb{1})\mathbf{A}^T = \mathbb{1} - \mathbf{A}^T$$

$$\text{Left: } \det[(\mathbf{A} - \mathbb{1})\mathbf{A}^T] = \det(\mathbf{A} - \mathbb{1}) \det(\mathbf{A}^T) = \det(\mathbf{A} - \mathbb{1})$$

$$\text{Right: } \det(\mathbb{1} - \mathbf{A}^T)^T = \det(\mathbb{1} - \mathbf{A}) = -\det(\mathbf{A} - \mathbb{1})$$

$$\text{Since } (\mathbf{A} - \lambda \mathbb{1})\mathbf{x} = 0 \text{ and } \det(\mathbf{A} - \lambda \mathbb{1}) = 0 \Rightarrow \lambda = 1$$

$\square$

Then, we have

$$\mathbf{A}\mathbf{x} = \mathbf{x}$$

- $\mathbf{x}$: the direction of rotation axis
- Matrix representation of Euler's theorem
- $\mathbf{A} : 3 \times 3$, $\mathbf{A}^T \mathbf{A} = \mathbb{1}$, $\det(\mathbf{A}) = 1$ made a group called spatial rotation group

3.2.2 Tensors

In a Cartesian three-dimensional space, a tensor $\vec{\vec{T}}$ of the N th rank may be defined for our purposes as a quantity having 3^N components $T_{\mu\nu\lambda}$ (with N indices) that transform under an orthogonal transformation of coordinates, $\mathbf{A}$, according to the following scheme:

$$T'_{\alpha\beta\gamma\dots} = \underbrace{a_{\alpha\mu}a_{\beta\nu}a_{\gamma\lambda}\dots}_N T_{\mu\nu\lambda\dots}$$

- $N = 0$: T , scale, one component, $T' = T$
- $N = 1$: T_α , vector, three components, $T'_\alpha = a_{\alpha\mu}T_\mu$, $\vec{T} = T_\alpha \vec{e}_\alpha$
- $N = 2$: $T_{\alpha\beta}$, second-rank, nine components, $T'_{\alpha\beta} = a_{\alpha\mu}a_{\beta\nu}T_{\mu\nu}$, $\vec{\vec{T}} = T_{\alpha\beta} \vec{e}_\alpha \vec{e}_\beta$, i.e. Electromagnetic tensor:

$$F_{\mu\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & B_y & B_x & 0 \end{bmatrix} \quad \text{*is a tensor in Lorentz transformation}$$

- $N = 3$: $T_{\alpha\beta\gamma}$, third-rank, twenty-seven components, i.e. Levi-Civita tensor: $\epsilon_{\alpha\beta\gamma}$

Ex 22. Second-rank unit tensor is a tensor:

$$\vec{\vec{T}} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \vec{\vec{1}} = \delta_{\alpha\beta} \vec{e}_\alpha \vec{e}_\beta$$

Proof.

$$a_{\alpha\mu}a_{\beta\nu}\delta_{\mu\nu} = a_{\alpha\mu}a_{\beta\mu} = \mathbf{A}_{\alpha\mu}\mathbf{A}_{\mu\beta}^T = (\mathbf{A}\mathbf{A}^T)_{\alpha\beta} = \delta_{\alpha\beta} = \delta'_{\alpha\beta}$$

$\square$

3.3 The Kinetics of Rigid Body

3.3.1 The Euler Angels

- * $\mathbf{A}$ only has three dependent elements of matrix
- * $Ox'y'z'$: body-fixed axes; $Oxyz$: space-fixed axes

(ϕ, θ, ψ) is a set of Euler Angels (Fig: 3.5)

- $\phi \in [0, 2\pi)$: angle of precession (causes a change in rotation around the z -axis)
- $\theta \in [0, \pi)$: angle of nutation (causes a change in included angle of $z'Oz$)
- $\psi \in [0, \pi)$: angle of rotation (does not cause a change in z' -axis orientation)

Rotation matrix of plane $x - y$, plane $y - z$, plane $x - z$:

$$\mathbf{A}_\phi = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{A}_\theta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}, \quad \mathbf{A}_\psi = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{r}' = \mathbf{A}_\psi(\mathbf{A}_\theta(\mathbf{A}_\phi \mathbf{r})) = (\mathbf{A}_\psi \mathbf{A}_\theta \mathbf{A}_\phi) \mathbf{r} = \mathbf{A} \mathbf{r}$$

$$\mathbf{A} = \mathbf{A}_\psi \mathbf{A}_\theta \mathbf{A}_\phi = \begin{bmatrix} \cos \psi \cos \phi - \cos \theta \sin \phi \sin \psi & \cos \psi \sin \phi + \cos \theta \cos \phi \sin \psi & \sin \psi \sin \theta \\ -\sin \psi \cos \phi - \cos \theta \sin \phi \cos \psi & -\sin \psi \sin \phi + \cos \theta \cos \phi \cos \psi & \cos \psi \sin \theta \\ \sin \theta \sin \phi & -\sin \theta \cos \phi & \cos \theta \end{bmatrix}$$

Consider from body-fixed to space-fixed, then

$$\mathbf{A}^{-1} = \mathbf{A}^\tau = \mathbf{A}_\phi \mathbf{A}_\theta \mathbf{A}_\psi$$

There are $3 \times 2 \times 2 = 12$ kinds of Euler's angles definition.

3.3.2 Euler Kinetics Equations

$$\vec{\omega} = \dot{\phi} \vec{e}_z + \dot{\theta} \vec{e}_N + \dot{\psi} \vec{e}_{z'}$$

Decompose $\vec{\omega}$ in the body-fixed reference,

$$\begin{aligned} \vec{e}_z &= \cos \theta \vec{e}_{z'} + \sin \theta \cos \psi \vec{e}_{y'} + \sin \theta \sin \psi \vec{e}_{x'} \\ \vec{e}_N &= \cos \psi \vec{e}_{x'} - \sin \psi \vec{e}_{y'} \\ \Rightarrow \vec{\omega} &= (\dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi) \vec{e}_{x'} + (\dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi) \vec{e}_{y'} + (\dot{\phi} \cos \theta + \dot{\psi}) \vec{e}_{z'} \end{aligned}$$

Euler Motion Equations:

$$\begin{aligned} \omega_{x'} &= \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi \\ \omega_{y'} &= \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \\ \omega_{z'} &= \dot{\phi} \cos \theta + \dot{\psi} \end{aligned}$$

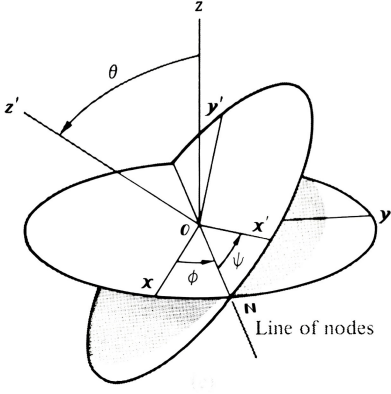


Fig 3.5: Euler Angles.

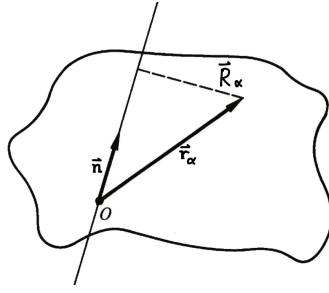


Fig 3.6: The definition of the moment of inertia.

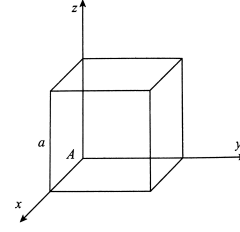


Fig 3.7: A homogeneous cube.

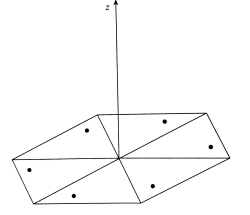


Fig 3.8: Six rotations are symmetrical about the axis.

3.4 The Principal Moment of Inertia Tensor

3.4.1 The Inertia Tensor and The Moment of Inertia

Rigid body seems as particles $m_1, m_2, \dots, m_N$, the rotation angular momentum around a fixed point is ¹²

$$\begin{aligned}\vec{J} &= \sum_{\alpha=1}^N m_{\alpha} \vec{r}_{\alpha} \times \vec{v}_{\alpha} = \sum_{\alpha=1}^N m_{\alpha} \vec{r}_{\alpha} \times (\vec{\omega} \times \vec{r}_{\alpha}) = \sum_{\alpha=1}^N m_{\alpha} (\vec{\omega} r_{\alpha}^2 - \vec{r}_{\alpha} (\vec{r}_{\alpha} \cdot \vec{\omega})) \\ &= \sum_{\alpha=1}^N m_{\alpha} (r_{\alpha}^2 \vec{1} - \vec{r}_{\alpha} \vec{r}_{\alpha}) \cdot \vec{\omega} = \vec{I} \cdot \vec{\omega}\end{aligned}$$

Angular momentum - Rotation around a fixed point:

$$\vec{J} = \vec{I} \cdot \vec{\omega}$$

Inertia Tensor:

$$\vec{I} = \sum_{\alpha} m_{\alpha} (r_{\alpha}^2 \vec{1} - \vec{r}_{\alpha} \vec{r}_{\alpha})$$

- Components: $I_{ij} = \sum_{\alpha} m_{\alpha} (r_{\alpha}^2 \delta_{ij} - r_{\alpha,i} r_{\alpha,j})$, $r_{\alpha}^2 = x_{\alpha}^2 + y_{\alpha}^2 + z_{\alpha}^2$

$$\text{or } I_{ij} = \int_V \rho(\vec{r}) (r_{\alpha}^2 \delta_{ij} - r_{\alpha,i} r_{\alpha,j}) dV, \quad \rho(r) : \text{density of continuous distribution}$$

- Matrix form:

$$\mathbf{I} = \begin{bmatrix} \sum_{\alpha} m_{\alpha} (y_{\alpha}^2 + z_{\alpha}^2) & -\sum_{\alpha} m_{\alpha} x_{\alpha} y_{\alpha} & -\sum_{\alpha} m_{\alpha} x_{\alpha} z_{\alpha} \\ -\sum_{\alpha} m_{\alpha} x_{\alpha} y_{\alpha} & \sum_{\alpha} m_{\alpha} (x_{\alpha}^2 + z_{\alpha}^2) & -\sum_{\alpha} m_{\alpha} y_{\alpha} z_{\alpha} \\ -\sum_{\alpha} m_{\alpha} x_{\alpha} z_{\alpha} & -\sum_{\alpha} m_{\alpha} y_{\alpha} z_{\alpha} & \sum_{\alpha} m_{\alpha} (x_{\alpha}^2 + y_{\alpha}^2) \end{bmatrix}$$

Kinetics energy:

$$\begin{aligned}T_{\text{rotation}} &= \frac{1}{2} \sum_{\alpha} m_{\alpha} \vec{v}_{\alpha}^2 = \frac{1}{2} \sum_{\alpha} m_{\alpha} \vec{v}_{\alpha} \cdot (\vec{\omega} \times \vec{r}_{\alpha}) = \frac{1}{2} \sum_{\alpha} \vec{\omega} \cdot (\vec{r}_{\alpha} \times m_{\alpha} \vec{v}_{\alpha}) \\ &= \frac{1}{2} \vec{\omega} \cdot \vec{J} = \frac{1}{2} \vec{\omega} \cdot \vec{I} \cdot \vec{\omega} = \frac{1}{2} \omega^2 \vec{n} \cdot \vec{I} \cdot \vec{n} = \frac{1}{2} I \omega^2\end{aligned}$$

- $\vec{n}$: a unit vector in the direction of $\vec{\omega}$ (axis), so that

$$\vec{n} = \frac{\vec{\omega}}{\omega}$$

¹ $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$

² Dyad $\mathfrak{E}$: $\vec{r} \vec{r} = (r_i \vec{e}_i)(r_j \vec{e}_j) = r_i r_j \vec{e}_i \vec{e}_j$ (symmetric, special 2-order tensor). i.e. $i, j = (x, y, z)$ $\vec{r} \vec{r} = \begin{bmatrix} x^2 & xy & xz \\ xy & y^2 & yz \\ xz & yz & z^2 \end{bmatrix}$

- Components: $T = \frac{1}{2} \sum_{i,j} \omega_i I_{ij} \omega_j$
- Matrix form: $\mathbf{T} = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I} \boldsymbol{\omega}$

where I is a scale, defined by (Fig: 3.6)

Moment of Inertia about the axis of rotation:

$$I = \vec{n} \cdot \vec{\vec{I}} \cdot \vec{n} \quad \text{or} \quad I = \mathbf{n}^T \mathbf{I} \mathbf{n}$$

- Customary definition:

$$I = \vec{n} \cdot \left[\sum_{\alpha} m_{\alpha} \left(r_{\alpha}^2 \vec{\vec{1}} - \vec{r}_{\alpha} \vec{r}_{\alpha} \right) \right] \cdot \vec{n} = \sum_{\alpha} m_{\alpha} \left[r_{\alpha}^2 - (\vec{r}_{\alpha} \cdot \vec{n})^2 \right] = \sum_{\alpha} m_{\alpha} (\vec{r}_{\alpha} \times \vec{n}) \cdot (\vec{r}_{\alpha} \times \vec{n}) = \sum_{\alpha} m_{\alpha} \vec{R}_{\alpha}^2$$

Parallel Axis Theorem

$$\vec{\vec{I}} = \vec{\vec{I}}_c + \vec{\vec{I}}'$$

- $\vec{\vec{I}}$: rigid body relative to fixed point
- $\vec{\vec{I}}_c$: center of mass relative to fixed point
- $\vec{\vec{I}}'$: rigid body relative to center of mass
- * König's theorem: $T = T_c + T'$

Proof. $\vec{r}_{\alpha} = \vec{r}_c + \vec{r}'_{\alpha}$, $\sum_{\alpha} m_{\alpha} \vec{r}'_{\alpha} = 0$, notice that the direction vector of center of mass in barycentric coordinates is 0

$$\vec{\vec{I}} = \sum_{\alpha} m_{\alpha} \left[(\vec{r}_c + \vec{r}'_{\alpha})^2 \vec{\vec{1}} - (\vec{r}_c + \vec{r}'_{\alpha})(\vec{r}_c + \vec{r}'_{\alpha}) \right] = \left(\sum_{\alpha} m_{\alpha} \right) \left(r_c^2 \vec{\vec{1}} - \vec{r}_c \vec{r}_c \right) + \sum_{\alpha} m_{\alpha} \left[r_{\alpha}'^2 \vec{\vec{1}} - \vec{r}'_{\alpha} \vec{r}'_{\alpha} \right] = \vec{\vec{I}}_c + \vec{\vec{I}}'$$

□

The Inertia Tensor Changed in Space-fixed Axes Rotation

The rotation matrix is $\mathbf{A}$, for $\vec{r} \vec{r}^T$ and $T = (1/2) \boldsymbol{\omega}^T \mathbf{I} \boldsymbol{\omega}$, after rotation we have

$$r_i r_j \xrightarrow{\text{rotate}} r'_i r'_j = \sum_{mn} (a_{im} r_m) (a_{jn} r_n) = \sum_{mn} a_{im} a_{jn} r_m r_n$$

$$T \xrightarrow{\text{rotate}} T' = \boldsymbol{\omega}'^T \mathbf{I}' \boldsymbol{\omega}' = \boldsymbol{\omega}^T \mathbf{A}^T \mathbf{I} \mathbf{A} \boldsymbol{\omega} \quad \Rightarrow \quad \mathbf{I} = \mathbf{A}^T \mathbf{I}' \mathbf{A}, \mathbf{I}' = \mathbf{A} \mathbf{I} \mathbf{A}^T$$

- Components: $I'_{ij} = \sum_{mn} a_{im} I_{mn} a_{jn} = \sum_{mn} a_{im} a_{jn} I_{mn}$
- also proofed that $\vec{r} \vec{r}^T$ and $\mathbf{I}$ are 2nd-order tensor
- In space reference: every component of $\vec{\vec{I}}$ is a function of t ; in body-fixed: components of $\vec{\vec{I}}$ is constant

Ex 23. A homogeneous cube of mass m , side a . The inertia tensor? (Fig: 3.7)

1. when origin to be at one corner;

$$\left. \begin{aligned} I_{11} &= \int (y^2 + z^2) dm = \int_0^a \int_0^a \int_0^a \frac{m}{a^3} (y^2 + z^2) dx dy dz = \frac{2}{3} ma^2 = I_{22} = I_{33} \\ I_{12} &= - \int xy dm = - \int_0^a \int_0^a \int_0^a \frac{m}{a^3} xy dx dy dz = -\frac{1}{4} ma^2 = I_{13} = I_{23} \end{aligned} \right\} \quad \vec{\vec{I}} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{2}{3} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{2}{3} \end{bmatrix} ma^2$$

2. when the origin is random. Use $\vec{\vec{I}} = \vec{\vec{I}}_c + \vec{\vec{I}}'$

$$\vec{\vec{I}}_c = m(r_c^2 \vec{\vec{1}} - \vec{r}_c \vec{r}_c) \quad \rightarrow \quad I_{c,ij} = m(r_c^2 \delta_{ij} - r_{c,i} r_{c,j}) = m \left(\frac{3}{4} a^2 \delta_{ij} - \frac{a^2}{4} \right) \quad \Rightarrow \quad \vec{\vec{I}}' = \begin{bmatrix} \frac{1}{6} & & \\ & \frac{1}{6} & \\ & & \frac{1}{6} \end{bmatrix} ma^2$$

It embodies the high symmetry of the rigid body.

3.4.2 The Eigenvalues of The Inertia Tensor and The Principal Axis Transformation

³Real Symmetric Matrix $\xrightarrow{\text{similarity transformation}}$ Diagonal Matrix

$$\mathbf{I} = \begin{bmatrix} I_{11} & I_{12} & I_{13} \\ I_{21} & I_{22} & I_{23} \\ I_{31} & I_{32} & I_{33} \end{bmatrix} \rightarrow \mathbf{E}^T \mathbf{I} \mathbf{E} = \begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix}$$

- $\mathbf{E}$ performs a similarity transformation
- I_1, I_2, I_3 : principal moment of inertia

Therefore, the question is what's the eigenvalues of $\mathbf{I}$. From $\mathbf{I}\mathbf{x} = \lambda\mathbf{x} \Rightarrow$

$$\begin{bmatrix} I_{11} - \lambda & I_{12} & I_{13} \\ I_{21} & I_{22} - \lambda & I_{23} \\ I_{31} & I_{32} & I_{33} - \lambda \end{bmatrix} \begin{bmatrix} x_{1\lambda} \\ x_{2\lambda} \\ x_{3\lambda} \end{bmatrix} = 0$$

And $\det(\mathbf{I} - \lambda\mathbf{1}) = 0 \rightarrow \lambda = I_1, I_2, I_3$ then $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are eigen vector (normalization) for each λ .
Situations:

- Three single roots: $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are orthogonal with each other
- Double roots and one single root: if $I_1 = I_2$, take any two vector that perpendicular to each other, another is $\mathbf{x}_3$
- Triple roots: take any three vector that perpendicular to each other

Properties of $\mathbf{E}$

- $\mathbf{E} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3]$, $\mathbf{x}_i^T \mathbf{x}_j = \delta_{ij}$ (ortho-nomal).

Proof.

$$\mathbf{E}^T \mathbf{E} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \mathbf{x}_3^T \end{bmatrix} [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3] = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$$

Diagonalize $\mathbf{I}$, and eigen function is $\mathbf{x}_i^T (\mathbf{I}\mathbf{x}_j) = \lambda_j (\mathbf{x}_i^T \mathbf{x}_j) = \lambda_j \delta_{ij}$

$$\mathbf{E}^T \mathbf{I} \mathbf{E} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \mathbf{x}_3^T \end{bmatrix} \mathbf{I} [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3] = \begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix}$$

□

- $\mathbf{E}\mathbf{e}_1 = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{x}_1$, $\mathbf{E}\mathbf{e}_2 = \mathbf{x}_2$, $\mathbf{E}\mathbf{e}_3 = \mathbf{x}_3$.

Let $\mathbf{E}^T = \mathbf{A}$, then $\mathbf{e}_i = \mathbf{A}\mathbf{x}_i$, $i = 1, 2, 3$, which means in the rotation transformation matrix $\mathbf{A}$, $\mathbf{x}_i$ change to $\mathbf{e}_i$ in the coordinate transformation. Or The eigenvector becomes the coordinate axis in the new coordinate system.

$$\mathbf{I} \rightarrow \mathbf{I}' = \mathbf{A}\mathbf{I}\mathbf{A}^T = \mathbf{E}^T \mathbf{I} \mathbf{E} = \begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix}$$

- The new coordinate system is called the principal axis coordinate system
- The coordinate axis is the principal axis
- In the principal axis coordinate system, we have

$$\begin{aligned} \vec{J} &= \vec{I} \cdot \vec{\omega} = I_1 \omega_1 \vec{e}_1 + I_2 \omega_2 \vec{e}_2 + I_3 \omega_3 \vec{e}_3 \\ T &= \frac{1}{2} \vec{\omega} \cdot \vec{I} \cdot \vec{\omega} = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) \end{aligned}$$

³Similarity: $\mathbf{A} = \mathbf{P}^{-1}\mathbf{B}\mathbf{P}$; congruent: $\mathbf{A} = \mathbf{P}^T\mathbf{B}\mathbf{P}$. If $\mathbf{P}$ orthogonal, then similarity=congruent.

Properties of $\mathbf{I}$

1. The principal axes of the center of inertia: set the mass point as origin
2. • The axis of symmetry and the principal axis of rotation of homogeneous rigid body are the principal axes of the center of inertia

Proof. If the rigid body is about z -axis symmetric, then $(x_\alpha, y_\alpha, z_\alpha)$ and $(-x_\alpha, -y_\alpha, z_\alpha)$ appear in pairs. We need to proof $\mathbf{Ie}_z = \lambda_3 \mathbf{e}_z$, since $\sum_\alpha m_\alpha x_\alpha z_\alpha = 0$ and $\sum_\alpha m_\alpha y_\alpha z_\alpha = 0$, therefore

$$\mathbf{Ie}_z = \begin{bmatrix} I_{11} & I_{12} & 0 \\ I_{21} & I_{22} & 0 \\ 0 & 0 & I_{33} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = I_{33} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \lambda_3 = I_{33}$$

Let the z -axis be n rotational symmetry (Fig: 3.8), then (r, θ, z) and $(r, \theta + \frac{2\pi}{n}q, z)$ $q = 1, 2, \dots, n-1$ appear in pairs. Since the n mass points are center of mass coordinate and the center of mass is at z -axis, we have $\sum_q m x_q z = m z \sum_q x_q = 0$ and $\sum_q m y_q z = 0$, therefore, same as before $\square$

- The normal of the symmetry surface of homogeneous rigid body is the principal axis of inertia

Proof. If $\sum_\alpha m_\alpha x_\alpha z_\alpha = 0$ and $\sum_\alpha m_\alpha y_\alpha z_\alpha = 0$, then z -axis is principal axis of inertia, if z -axis across the center of mass, it is the principal axis of the center of inertia. And the coordinates appear in pairs. $\square$

3. Non-uniqueness of principal axis of inertia: Take a parallel coordinate system at any point on the principal axis of the center of inertia, then the new coordinate remain principal axis of inertia.

Proof. $x' = x, y' = y, z' = z - a$, since principal axis of the center of inertia, then $\sum_\alpha m_\alpha x'_\alpha y'_\alpha = \sum_\alpha m_\alpha x_\alpha y_\alpha = 0$, $\sum_\alpha m_\alpha x'_\alpha z'_\alpha = \sum_\alpha m_\alpha x_\alpha (z_\alpha - a) = \sum_\alpha m_\alpha x_\alpha z_\alpha - a \sum_\alpha m_\alpha x_\alpha = 0$, since $\sum_\alpha m_\alpha x_\alpha$ is the coordinate of the center of mass, which is zero. Same as $\sum_\alpha m_\alpha y'_\alpha z'_\alpha$ $\square$

Ex 24. The eigen vector of the cube of Fig: 3.7.

The eigen function is $(\mathbf{I} - \lambda m a^2 \mathbb{1})\mathbf{x} = 0$,

$$\det \begin{bmatrix} \frac{2}{3} - \lambda & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{2}{3} - \lambda & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{2}{3} - \lambda \end{bmatrix} = 0 \Rightarrow \lambda_1 = \lambda_2 = \frac{11}{12}, \lambda_3 = \frac{1}{6}$$

For $\lambda_3, x_{13} = x_{23} = x_{33} = \frac{1}{\sqrt{3}}$, after normalization, $\mathbf{x}_3 = \frac{1}{\sqrt{3}}[1 \ 1 \ 1]^T$

For $\lambda_1 = \lambda_2, x_{1\lambda} + x_{2\lambda} + x_{3\lambda} = 0$, we can have $\mathbf{x}_1 = \frac{1}{\sqrt{2}}[-1 \ 1 \ 0]^T$, then $\mathbf{x}_2 = \frac{1}{\sqrt{6}}[1 \ 1 \ -2]^T$

3.4.3 Inertia Ellipsoid

Define a vector $\vec{\rho}$ along the instantaneous axis of rotation,

$$\vec{\rho} = \frac{\vec{n}}{\sqrt{I}} = \frac{\vec{\omega}}{\omega \sqrt{I}} = \frac{\vec{\omega}}{\sqrt{2I}} \quad \vec{\rho} \cdot \vec{I} \cdot \vec{\rho} = 1$$

Let $\vec{\rho} = (\rho_1, \rho_2, \rho_3)$, we have the inertia ellipsoid which is a curved surface equation

$$I_{11}\rho_1^2 + I_{22}\rho_2^2 + I_{33}\rho_3^2 + 2I_{12}\rho_1\rho_2 + 2I_{13}\rho_1\rho_3 + 2I_{23}\rho_2\rho_3 \equiv f(\vec{\rho})$$

- The length of the axis of rotation cutting the inertia ellipsoid is

$$R = |\vec{R}| = |\vec{\rho}| = \frac{1}{\sqrt{I}} \Rightarrow I = \frac{1}{R^2}$$

- The normal direction at the intersection of the axis and the inertia ellipsoid is the angular momentum direction.

Proof. When $f(\vec{\rho}) = 1$, the normal direction is

$$\nabla f(\vec{\rho}) = \frac{\partial}{\partial \vec{\rho}} f(\vec{\rho}) = \frac{\partial}{\partial \vec{\rho}} (\vec{\rho} \cdot \vec{I} \cdot \vec{\rho}) = 2\vec{I} \cdot \vec{\rho} = 2\vec{I} \cdot \frac{\vec{\omega}}{\omega\sqrt{I}} = \sqrt{\frac{2}{T}} \vec{I} \cdot \vec{\omega} = \sqrt{\frac{2}{T}} \vec{J} \quad \square$$

- For the principal of the center of inertia,

$$I_1 \rho_1^2 + I_2 \rho_2^2 + I_3 \rho_3^2 = 1$$

- Three half-axes lengths: $1/\sqrt{I_1}, 1/\sqrt{I_2}, 1/\sqrt{I_3}$
- If $I_1 = I_2$: rotation inertia ellipsoid; If $I_1 = I_2 = I_3$: inertia sphere

- In space system, the inertia ellipsoid rotates with rigid body at real time; in body fixed system, they don't.

3.5 Euler's Dynamics Equations

Newtonian Mechanics Method

In space coordinate system, we have $d\vec{J}/dt = \vec{M}$. Now, we would like to decompose it in body-fixed coordinate system with the reference frame of space system. Use the principal axis coordinate system, in the body-fixed coordinate system, let unit direction vector are $\vec{e}_1, \vec{e}_2, \vec{e}_3$,

$$\vec{J} = \vec{I} \cdot \vec{\omega} = I_1 \omega_1 \vec{e}_1 + I_2 \omega_2 \vec{e}_2 + I_3 \omega_3 \vec{e}_3$$

Since the relation of $\frac{d}{dt} = \frac{d'}{dt} + \vec{\omega} \times$, in space system:

$$\begin{aligned} \frac{d\vec{e}_1}{dt} &= \vec{\omega}_1 \times \vec{e}_1 = \omega_3 \vec{e}_2 - \omega_2 \vec{e}_3, & \frac{d\vec{e}_2}{dt} &= \omega_1 \vec{e}_3 - \omega_3 \vec{e}_1, & \frac{d\vec{e}_3}{dt} &= \omega_2 \vec{e}_1 - \omega_1 \vec{e}_2 \Rightarrow \\ \frac{d\vec{J}}{dt} &= I_1 \dot{\omega}_1 \vec{e}_1 + I_2 \dot{\omega}_2 \vec{e}_2 + I_3 \dot{\omega}_3 \vec{e}_3 + I_1 \omega_1 (\omega_3 \vec{e}_2 - \omega_2 \vec{e}_3) + I_2 \omega_2 (\omega_1 \vec{e}_3 - \omega_3 \vec{e}_1) + I_3 \omega_3 (\omega_2 \vec{e}_1 - \omega_1 \vec{e}_2) \\ &= [I_1 \dot{\omega}_1 - \omega_2 \omega_3 (I_2 - I_3)] \vec{e}_1 + [I_2 \dot{\omega}_2 - \omega_3 \omega_1 (I_3 - I_1)] \vec{e}_2 + [I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2)] \vec{e}_3 \\ &= M_1 \vec{e}_1 + M_2 \vec{e}_2 + M_3 \vec{e}_3 \Rightarrow \end{aligned}$$

Euler's Dynamics Equations:

$$I_1 \dot{\omega}_1 - \omega_2 \omega_3 (I_2 - I_3) = M_1$$

$$I_2 \dot{\omega}_2 - \omega_3 \omega_1 (I_3 - I_1) = M_2$$

$$I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2) = M_3$$

Lagrangian mechanics method

Let (ϕ, θ, ψ) as generalized coordinates, in rigid body coordinate system and for principal inertia axis, we have

$$L = T_{\text{rotation}} - V(\phi, \theta, \psi) = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) - V(\phi, \theta, \psi), \quad \omega = \omega_i(\phi, \theta, \psi), i = 1, 2, 3$$

For the three generalized force responding to three Euler angles, only angle of rotation ψ always in z' -direction, therefore

$$\begin{aligned} M_z &= \frac{d}{dt} \frac{\partial T}{\partial \dot{\psi}} - \frac{\partial T}{\partial \psi} = \frac{d}{dt} \sum_i \frac{\partial T}{\partial \omega_i} \frac{\partial \omega_i}{\partial \dot{\psi}} - \sum_i \frac{\partial T}{\partial \omega_i} \frac{\partial \omega_i}{\partial \psi} \quad \text{and} \quad \frac{\partial T}{\partial \omega_i} I_i \omega_i, \frac{\partial \omega_1}{\partial \dot{\psi}} = \frac{\partial \omega_3}{\partial \dot{\psi}} = 0, \frac{\partial \omega_3}{\partial \dot{\psi}} = 1 \\ &\Rightarrow \begin{cases} \partial_{\psi} \omega_1 = \dot{\phi} \sin \theta \cos \phi - \dot{\theta} \sin \psi = \omega_2 \\ \partial_{\psi} \omega_2 = -\dot{\phi} \sin \theta \sin \psi - \dot{\theta} \cos \psi = -\omega_1 \\ \partial_{\psi} \omega_3 = 0 \end{cases} \Rightarrow I_3 \dot{\omega}_3 - \omega_1 \omega_2 (I_1 - I_2) = M_3 \end{aligned}$$

* For two rigid body with same I_1, I_2, I_3 , then their motion are completely same.

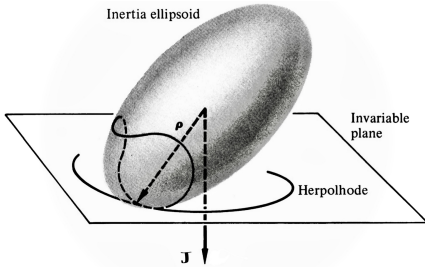


Fig 3.9: The motion of the inertia ellipsoid relative to the invariable plane.

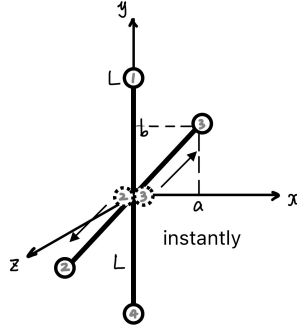


Fig 3.10: Diving problem.

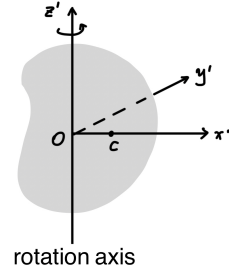


Fig 3.11: Rigid body rotates around a fixed axis.

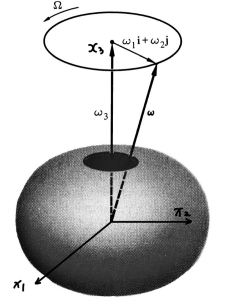


Fig 3.12: d

Ex 25. Diving problems: the space and body-fixed system are same at instant when athletes with open arms, initially $\vec{\omega}_i = \omega \vec{e}_x$, finally $\vec{\omega}_f = ?$ (ignore gravity). (Fig: 3.10)

$$\vec{I}_i \cdot \vec{\omega}_i = \vec{I}_f \cdot \vec{\omega}_f, \quad \mathbf{I} = \mathbf{I}' + \mathbf{I}_c$$

Initially, $\vec{r}_{1,4} = (0, \pm L, 0)$, $\vec{r}_{2,3} = (0, 0, 0)$

$$\mathbf{I}_i = \begin{bmatrix} 4I + 2ML^2 & & \\ & 4I & \\ & & 4I + 2ML^2 \end{bmatrix}, \quad \mathbf{I}_f = \begin{bmatrix} 4I + 2M(L^2 + b^2) & -2Mab & 0 \\ -2Mab & 4I + 2Ma^2 & 0 \\ 0 & 0 & 4I + 2M(L^2 + a^2 + b^2) \end{bmatrix} \Rightarrow$$

$$\vec{\omega}_f = \begin{bmatrix} \omega' \\ \frac{Mab}{2I + MR^2} \\ 0 \end{bmatrix}, \quad \omega' = \frac{(2I + ML^2)(2I + Ma^2)}{(2I + M(L^2 + b^2))(2I + Ma^2) - M^2a^2b^2} \omega$$

Ex 26. Fixed axis rotation lateral constraint force. The applied torque $\vec{M}_F$ is in z -direction, no other applied force. And $\vec{\omega} = \omega \vec{e}_z$, the center of mass is $\vec{r}_c = l \vec{e}_x$. (Fig: 3.11)

The motion of the center of mass is

$$\vec{N} + \vec{N}' = m \vec{a}_c = m \vec{\omega} \times \vec{r}_c + m \vec{\omega} \times (\vec{\omega} \times \vec{r}_c) = -m\omega^2 l \vec{e}_x + m\omega l \vec{e}_y$$

where $\vec{N} + \vec{N}'$ is constraint force, and since the center of mass is not in rotation axis, therefore it need bearing to provide lateral constraint force.

We have $\vec{M}_F + \vec{M}_{N,N'} = d\vec{J}/dt$, and in body-fixed system $\vec{J} = \vec{I} \cdot \vec{\omega} = \omega(I_{13} \vec{e}_x + I_{23} \vec{e}_y + I_{33} \vec{e}_z)$

$$\frac{dJ}{dt} = \frac{d'\vec{J}}{dt} + \vec{\omega} \times \vec{J} = (\dot{\omega}I_{13} - \omega^2 I_{23}) \vec{e}_x + (\dot{\omega}I_{23} + \omega^2 I_{13}) \vec{e}_y + \dot{\omega}I_{33} \vec{e}_z$$

$$\vec{M}_{N,N'} = (-a \vec{e}_z) \times \vec{N} + (a \vec{e}_x \times \vec{N}') = a(N_y - N'_y) \vec{e}_x + a(N'_x - N_x) \vec{e}_y$$

$$\mathbf{M}_F = M_F \vec{e}_z \Rightarrow M_F = \dot{\omega} I_{33}$$

Since the rotation axis is not the principal axis of inertia, so the bearing need to provide torque. When $l = 0$, the rotation don't need lateral constraint force, and if $\vec{M}_F = 0$, then the rigid body will rotate in a constant speed in the fixed axis.

3.6 Torque-Free Motion of a Rigid Body

The motion of a rigid body doesn't subject to any net forces or torque, so the angular momenta. The Euler's dynamics equations reduce to

$$I_1 \dot{\omega}_1 = \omega_2 \omega_3 (I_2 - I_3)$$

$$I_2 \dot{\omega}_2 = \omega_3 \omega_1 (I_3 - I_1)$$

$$I_3 \dot{\omega}_3 = \omega_1 \omega_2 (I_1 - I_2)$$

And it is a free rigid body, the angular momentum is conserved.

3.6.1 Asymmetric Euler Gyroscope

$I_1 \neq I_2 \neq I_3$, the integration of motion is $\vec{J}$ and E , we have

$$(I_1 \omega_1)^2 + (I_2 \omega_2)^2 + (I_3 \omega_3)^2 = J^2 = \text{const}; \quad \frac{1}{2}(I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) = T = \text{const} \Rightarrow \rho \cdot \frac{\vec{J}}{J} = \frac{\vec{\omega} \cdot \vec{I} \cdot \vec{\omega}}{\sqrt{I} \omega J} = \frac{\sqrt{2T}}{J} = \text{const}$$

which means The tangent plane direction of the inertia ellipsoid does not change, the point touch to the plane speed is always 0, the inertia ellipsoid do pure rolling in the plane.

Stability of Rotation

(* Or Tennis racket theorem) If the rotating axis (z -axis) deviates slightly from the main axis of inertia, how does this deviation develop? Consider $\omega_1, \omega_2 \ll \omega_3$, from Euler's dynamics equations, we have

$$I_3 \dot{\omega}_3 \approx 0 \quad (\omega_3 \approx \text{const}) \quad \text{then} \quad \ddot{\omega}_1 + \left[\frac{(I_3 - I_2)(I_3 - I_1)}{I_2 I_1} \omega_3^2 \right] \omega_1 = 0, \quad \ddot{\omega}_2 + \left[\frac{(I_3 - I_1)(I_3 - I_2)}{I_1 I_2} \omega_3^2 \right] \omega_2 = 0$$

- If $I_3 > I_{1,2}$ or $I_3 < I_{1,2}$, then $\omega_{1,2}$ has stable solution of oscillation
- If $I_1 < I_3 < I_2$ or $I_1 > I_2 > I_3$, then $\omega_{1,2}$ has unstable solution in exponential growth

3.6.2 Symmetric Euler Gyroscope

$I_1 = I_2$, then the Euler's equations reduce to

$$\begin{aligned} I_1 \dot{\omega}_1 &= \omega_2 \omega_3 (I_1 - I_3) \\ I_1 \dot{\omega}_2 &= \omega_3 \omega_1 (I_3 - I_1) \\ I_3 \dot{\omega}_3 &= 0 \end{aligned}$$

Therefore, $\omega_3 = \text{const}$, and

$$\ddot{\omega}_1 + \Omega^2 \omega_1 = 0, \quad \ddot{\omega}_2 + \Omega^2 \omega_2 = 0 \quad \Rightarrow \quad \omega_1 = A \cos(\Omega t + \alpha), \quad \omega_2 = A \sin(\Omega t + \alpha)$$

- Precession angular velocity

$$\vec{\Omega} = \frac{I_3 - I_1}{I_1} \vec{\omega}_3$$

- $\omega_1^2 + \omega_2^2 = \text{const}$, $\omega_1^2 + \omega_2^2 + \omega_3^2 = \text{const}$
 - $I_2 > I_1, \Omega > 0$, $\vec{\omega}$ project in in $x_1 - x_2$ plane and rotate with counterclockwise
 - $I_3 < I_1, \Omega < 0$, $\vec{\omega}$ project in in $x_1 - x_2$ plane and rotate with clockwise

Then, the angular momenta are

$$J_1 = I_1 \omega_1 = I_1 A \cos(\Omega t + \alpha), \quad J_2 = I_2 \omega_2 = I_2 A \sin(\Omega t + \alpha), \quad J_3 = I_3 \omega_3 = \text{const}$$

- $J_1^2 + J_2^2 = I_1^2 A^2 = \text{const}$ (circle)

Ex 27. The movement of the Earth's North Pole

x_3 -axis is earth's axis of symmetry (geographical earth axis); $\vec{\omega}$ -axis is rotation axis (astronomical earth axis).

The period of the movement of the earth's pole is $\frac{2\pi}{\Omega} = \frac{2\pi}{\omega_c} \frac{I_1}{I_3 - I_1} \approx 300 \text{ days} = 10 \text{ months}$ (Actually 14 months)

Discussion in Space Reference System

Take them in Euler's motion equations, we can got the Euler's angles

$$\begin{cases} \omega_1 = A \cos(\Omega t + \alpha) = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi \\ \omega_2 = A \sin(\Omega t + \alpha) = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \\ \omega_3 = \text{const} = \dot{\phi} \cos \theta + \dot{\psi} \end{cases} \Rightarrow \begin{cases} \dot{\theta} = A \cos(\Omega t + \alpha + \psi) \\ \dot{\phi} = A \csc \theta \sin(\Omega t + \alpha + \psi) \\ \dot{\psi} = \omega_3 - A \cot \theta \sin(\Omega t + \alpha + \psi) \end{cases}$$

Since $\vec{J}$ is conserved, and let direction is z' -axis, then

$$J_3 = I_3 \omega_3 = \text{const} = J \cos \theta \quad \Rightarrow \quad \theta = \text{const}$$

Take it into, then ⁴

$$\begin{cases} \Omega t + \alpha + \psi = \text{const}, & \text{and let } \alpha = \frac{\pi}{2}, \text{ so } \psi = -\Omega t \\ \dot{\phi} = \frac{A}{\sin \theta}, \\ -\Omega = \omega_3 - \frac{A}{\tan \theta}, \end{cases} \quad \text{then } A = \frac{I_3}{I_1} \omega_3 \tan \theta$$

⁴ $\csc \theta = \frac{1}{\sin \theta}, \cot \theta = \frac{1}{\tan \theta}$

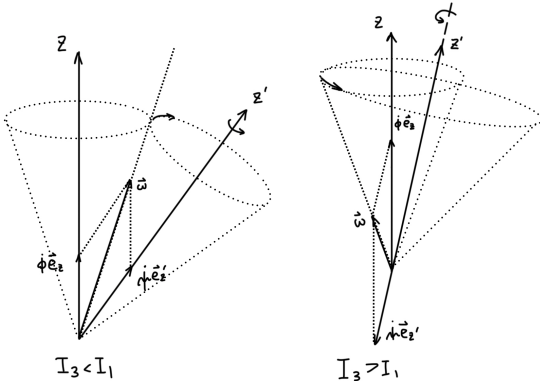


Fig 3.13: The space cone and body cone of symmetric Euler gyroscope.

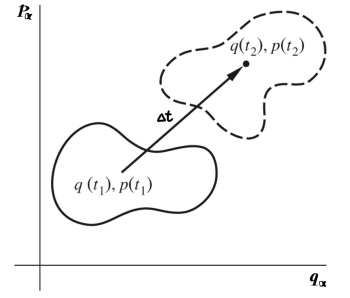


Fig 3.14: Motion of a volume in two-dimensional phase space.

Finally, we got

$$\dot{\theta} = 0, \quad \dot{\psi} = -\frac{I_3 - I_1}{I_1} \omega_3, \quad \dot{\phi} = \frac{I_3}{I_1 \cos \theta} \omega_3$$

- which means the angle of nutation remains, and angle of rotation and precession change with constant speed
- The magnitude of angular velocity:

$$\vec{\omega}_z = \dot{\psi} \vec{e}'_z + \dot{\phi} \vec{e}_z, \quad \text{then} \quad \omega_3 = \dot{\phi} + \dot{\psi} \cos \theta = \text{const} \quad \Rightarrow \quad |\vec{\omega}| = \sqrt{\dot{\psi}^2 + \dot{\phi}^2 + 2\dot{\psi}\dot{\phi} \cos \theta} = \text{const}$$

Two conditions of the Euler gyroscope and how their space cone and body cone look like. Assume $\omega_3 > 0$, and consider θ is an acute angle, so $\dot{\phi} \neq 0$ (Fig: 3.13)

- When $I_3 < I_1$, $\dot{\psi} > 0$, $\vec{\omega}$ rotates around z' -axis in body-fixed system in clockwise
- When $I_3 > I_1$, $\dot{\psi} < 0$, $\vec{\omega}$ rotates around z' -axis in body-fixed system in counterclockwise

Chapter 4

Hamilton Mechanics $\mathfrak{E}$

4.1 The Canonical Equations of Hamilton

4.1.1 Legendre Transformations

Consider $f(x, y) \rightarrow df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = udx + vdy$, Legendre transformations:

$$\begin{aligned} x, y &\rightarrow u, y \quad ; \quad g(u, y) = -f + ux \quad \rightarrow \quad dg = xdu - vdy \\ x, y &\rightarrow x, v \quad ; \quad g'(x, v) = -f + vy \quad \rightarrow \quad dg' = -udx + ydv \\ x, y &\rightarrow u, v \quad ; \quad g''(u, v) = -f + ux + vy \quad \rightarrow \quad dg'' = xdu + ydv \end{aligned}$$

4.1.2 The Hamilton Equations of Motion

Transformation from $(q, \dot{q}, t)$ to (q, p, t) :

$$\begin{aligned} dL &= \frac{\partial L}{\partial q_\alpha} dq_\alpha + \frac{\partial L}{\partial \dot{q}_\alpha} d\dot{q}_\alpha + \frac{\partial L}{\partial t} dt \\ &= \dot{p}_\alpha dq_\alpha + p_\alpha d\dot{q}_\alpha + \frac{\partial L}{\partial t} dt \quad * \quad p_\alpha d\dot{q}_\alpha = d(p_\alpha \dot{q}_\alpha) - \dot{q}_\alpha dp_\alpha \\ \Rightarrow \quad d(L - p_\alpha \dot{q}_\alpha) &= p_\alpha d\dot{q}_\alpha - \dot{q}_\alpha dp_\alpha + \frac{\partial L}{\partial t} dt = -dH(q, p, t) \end{aligned}$$

Hamiltonian:

$$H(q, p, t) = \dot{q}_\alpha p_\alpha - L(q, \dot{q}, t)$$

- Differential: $-dH = -\left(\frac{\partial H}{\partial q_\alpha} dq_\alpha + \frac{\partial H}{\partial p_\alpha} dp_\alpha + \frac{\partial H}{\partial t} dt\right)$
- Generalized energy: $H(q, \dot{q}, t)$

Then, we obtain $2n + 1$ relations

Canonical equations of Hamilton:

$$\left. \begin{aligned} \dot{q}_\alpha &= \frac{\partial H}{\partial p_\alpha} \\ -\dot{p}_\alpha &= \frac{\partial H}{\partial q_\alpha} \end{aligned} \right\} \star \quad ; \quad -\frac{\partial L}{\partial t} = \frac{\partial H}{\partial t}$$

- $\star$ are the canonical equations of Hamilton, constitute the desired set of $2n$ first-order equations of motion replacing the n second-order Lagrange equations
- q_α, p_α : canonical variables
- $\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha} \Leftrightarrow \frac{\partial L}{\partial \dot{q}_\alpha} = p_\alpha$; $-\dot{p}_\alpha = \frac{\partial H}{\partial q_\alpha} \Leftrightarrow$ Lagrange's Equations

- Phase space: $(q_1, \dots, q_s, p_1, \dots, p_s)$ span to a $2s$ -dimensional space
 - Phase point: in the space any point represent a determined a state of motion of the system
 - Phase orbit: trajectories drawn by the phase points (non-intersect since the evolution of a mechanical system is uniquely determined by the initial state and the dynamical equations)
- If a canonical variable (cyclic coordinate) q_α or p_α is one that does not appear explicitly in the Hamiltonian, its conjugate variables (cyclic integral) p_α or q_α is conserved.

Ex 28. One-dimensional harmonic oscillator problem. dof=(x)

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2, \Rightarrow p = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

$$H = p\dot{x} - L = \frac{p^2}{2m} + \frac{1}{2}kx^2, \Rightarrow \dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}; \dot{p} = -\frac{\partial H}{\partial x} = -kx \rightarrow m\ddot{x} + kx = 0$$

The phase orbit is elliptical:

$$x = A \cos(\omega t + \phi); p = -m\omega A \sin(\omega t + \phi) \quad \left(\frac{x}{A}\right)^2 + \left(\frac{p}{m\omega A}\right)^2 = 1 \quad \omega = \sqrt{\frac{k}{m}}$$

Ex 29. Central forced field problem, consider the center of mass in $V(r)$. dof=(r, θ)

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r), \Rightarrow p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r} \rightarrow \dot{r} = \frac{p_r}{m}; p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \rightarrow \dot{\theta} = \frac{p_\theta}{mr^2}$$

$$H = p_r\dot{r} + p_\theta\dot{\theta} - L = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + V(r), \Rightarrow \dot{p}_r = -\frac{\partial H}{\partial r} = -\frac{p_\theta^2}{mr^3} - \frac{\partial V}{\partial r}; \dot{p}_\theta = -\frac{\partial H}{\partial \theta} = 0$$

Therefore, $p_\theta = \text{const} \equiv J \Rightarrow \dot{\theta} = \frac{J}{mr^2}, m\ddot{r} = \dot{p}_r = -\frac{J^2}{mr^3} - \frac{\partial V}{\partial r}$

Ex 30. The motion of charged particles in electromagnetic field problem. dof=($\vec{v}$)

Generalized potential: $U = e\phi - e\vec{A} \cdot \vec{v}, L = T - U = \frac{1}{2}m\vec{v}^2 - e\phi + e\vec{A} \cdot \vec{v}, \Rightarrow \vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} + e\vec{A} \rightarrow \vec{v} = \frac{\vec{p} - e\vec{A}}{m}$

$$H = \vec{p} \cdot \vec{v} - L = \frac{1}{2}m\vec{v}^2 + e\phi = \frac{(\vec{p} - e\vec{A})^2}{2m} + e\phi$$

Consider $L \rightarrow L' = L + df(q, t)/dt$, if the motion of equations don't change. (Hamiltonian Mechanics influence?)

$$L' = L(q, \dot{q}, t) + \frac{\partial f(q, t)}{\partial q_\beta} \dot{q}_\beta + \frac{\partial f(q, t)}{\partial t}$$

New canonical variables are (q_α, p'_α) , and

$$p'_\alpha = \frac{\partial L'}{\partial \dot{q}_\alpha} = \frac{\partial L}{\partial \dot{q}_\alpha} + \frac{\partial f}{\partial q_\alpha} = p_\alpha + \frac{\partial f}{\partial q_\alpha} \Rightarrow p_\alpha = p_\alpha(q_\alpha, p'_\alpha, t) = p'_\alpha - \frac{\partial f(q_\alpha, t)}{\partial q_\alpha}$$

Then, new Hamiltonian is

$$H' = p'_\alpha \dot{q}_\alpha - L' = \left(p_\alpha + \frac{\partial f}{\partial q_\alpha}\right) \dot{q}_\alpha - \left(L + \frac{\partial f}{\partial q_\beta} \dot{q}_\beta + \frac{\partial f}{\partial t}\right) = p_\alpha \dot{q}_\alpha - L \frac{\partial f}{\partial t} = H - \frac{\partial f}{\partial t}$$

The new canonical equations is

$$\dot{q}_\alpha = \frac{\partial H'}{\partial p'_\alpha} = \frac{\partial H}{\partial p_\alpha} \quad \dot{p}'_\alpha = -\frac{\partial H'}{\partial q_\alpha}$$

Proof.

$$\dot{q}_\alpha = \frac{\partial H'}{\partial p'_\alpha} = \frac{\partial}{\partial p'_\alpha} H'(q, p', t) = \frac{\partial}{\partial p'_\alpha} \left[H(q_\alpha, p_\alpha(q, p', t), t) - \frac{\partial f(q_\alpha, t)}{\partial t} \right] \Big|_{q_\alpha, t} = \frac{\partial H}{\partial p_\beta} \frac{\partial p_\beta}{\partial p'_\alpha} = \frac{\partial H}{\partial p_\beta} \frac{\partial}{\partial p'_\alpha} \left(p'_\beta - \frac{\partial f(q_\beta, t)}{\partial q_\beta} \right) = \frac{\partial H}{\partial p_\alpha}$$

$$\dot{p}'_\alpha = \frac{d}{dt} \left(p_\alpha + \frac{\partial f}{\partial q_\alpha} \right) = \dot{p}_\alpha + \frac{d}{dt} \left(\frac{\partial f(q_\alpha, t)}{\partial q_\alpha} \right) = \dot{p}_\alpha + \frac{\partial^2 f}{\partial q_\beta \partial q_\alpha} \dot{q}_\beta + \frac{\partial^2 f}{\partial t \partial q_\alpha}$$

$$-\frac{\partial H'}{\partial q_\alpha} = -\frac{\partial}{\partial q_\alpha} \left[H(q_\alpha, p_\alpha(q, p', t), t) - \frac{\partial f(q_\alpha, t)}{\partial t} \right] \Big|_{p'_\alpha, t} = -\left[\frac{\partial H}{\partial q_\alpha} + \frac{\partial H}{\partial p_\beta} \frac{\partial p_\beta}{\partial q_\alpha} - \frac{\partial}{\partial q_\alpha} \left(\frac{\partial f}{\partial t} \right) \right] \Big|_{p'_\alpha, t}$$

$$= -\frac{\partial H}{\partial q_\alpha} - \frac{\partial H}{\partial p_\beta} \frac{\partial}{\partial q_\alpha} \left(p'_\beta - \frac{\partial f(q_\beta, t)}{\partial q_\beta} \right) \Big|_{p'_\alpha, t} + \frac{\partial^2 f}{\partial q_\alpha \partial t} = -\frac{\partial H}{\partial q_\alpha} + \frac{\partial H}{\partial p_\beta} \frac{\partial^2 f}{\partial q_\alpha \partial q_\beta} + \frac{\partial^2 f}{\partial q_\alpha \partial t} = \dot{p}'_\alpha$$

4.1.3 Hamilton's Principle in Phase Space

Hamilton's Principle: $\delta S = 0 \iff$ Canonical Equations of Hamilton

Proof. Notice: δ means variation in configuration space, $\{\delta q_\alpha\}$ is dependent with each other, but δq_α and δp_α not sure. Consider contemporaneous variation,

$$\begin{aligned} 0 = \delta S &= \delta \int_{t_1}^{t_2} L dt = \delta \int_{t_1}^{t_2} (p_\alpha \dot{q}_\alpha - H) dt = \int_{t_1}^{t_2} \left[(p_\alpha \delta \dot{q}_\alpha + \dot{q}_\alpha \delta p_\alpha) - \left(\frac{\partial H}{\partial q_\alpha} \delta q_\alpha + \frac{\partial H}{\partial p_\alpha} \delta p_\alpha \right) \right] dt \\ &= \cancel{p_\alpha \delta q_\alpha} \Big|_{t_1}^{t_2} + \int_{t_1}^{t_2} \left[(-\dot{p}_\alpha \delta q_\alpha + \dot{q}_\alpha \delta p_\alpha) - \left(\frac{\partial H}{\partial q_\alpha} \delta q_\alpha + \frac{\partial H}{\partial p_\alpha} \delta p_\alpha \right) \right] dt = \int_{t_1}^{t_2} \left[\left(-\dot{p}_\alpha - \frac{\partial H}{\partial q_\alpha} \right) \delta q_\alpha + \left(\dot{q}_\alpha - \frac{\partial H}{\partial p_\alpha} \right) \delta p_\alpha \right] dt \end{aligned}$$

Therefore, from canonical equations $\Rightarrow \delta S = 0$. However, from ΔS cannot get canonical equations. But the following equation is always true,

$$\frac{\partial H}{\partial p_\alpha} = \frac{\partial}{\partial p_\alpha} [p_\beta \dot{q}_\beta(q, p, t) - L(q, \dot{q}(q, p, t), t)] = \dot{q}_\alpha + \cancel{p_\beta \frac{\partial \dot{q}_\beta}{\partial p_\alpha}} - \cancel{\frac{\partial L}{\partial \dot{q}_\beta} \frac{\partial \dot{q}_\beta}{\partial p_\alpha}} = \dot{q}_\alpha$$

so that we can $\delta S = 0 \Rightarrow$ canonical equations. □

We can introduce variation of phase space from variation of configuration space, and from $\{\delta q_\alpha\} \rightarrow \{\delta q_\alpha, \delta p_\alpha\}$ in contemporaneous variation.

Hamilton's principle in phase space: $\iff$ Hamilton's canonical equations

$$0 = \delta S = \delta \int_{t_1}^{t_2} [p_\alpha \dot{q}_\alpha(p, q, t) - H(p, q, t)] dt$$

4.2 Canonical Transformations

4.2.1 The Equations of Canonical Transformation

Let $Q_\alpha = Q_\alpha(q, t)$ as cyclic coordinate. $Q_\alpha = Q_\alpha(q, p, t)$; $P_\alpha = P_\alpha(q, p, t)$ still satisfy

$$\dot{Q}_\alpha = \frac{\partial K}{\partial P_\alpha}; \quad \dot{P}_\alpha = -\frac{\partial K}{\partial Q_\alpha}$$

- K : new Hamiltonian, and

$$K = P_\alpha \dot{Q}_\alpha - L$$

The Condition of Canonical Transformation

From Hamilton's principle in phase space, Q_i and P_i must satisfy

$$0 = \delta S = \delta \int_{t_1}^{t_2} [p_\alpha \dot{q}_\alpha - H(q, p, t)] dt \Rightarrow 0 = \delta S = \delta \int_{t_1}^{t_2} [P_\alpha \dot{Q}_\alpha - K(Q, P, t)] dt$$

For the integrand function F , if $\frac{dF}{dt} \left(\delta F \Big|_{t_1}^{t_2} = 0 \right)$, we have

$$p_\alpha \dot{q}_\alpha - H(q, p, t) = P_\alpha \dot{Q}_\alpha - K(Q, P, t) + \frac{dF}{dt} \Rightarrow$$

The condition of canonical transformation

$$\begin{aligned} p_\alpha \dot{q}_\alpha - P_\alpha dQ_\alpha + (K - H) dt &= dF \\ \text{or } p_\alpha \delta q_\alpha - P_\alpha \delta Q_\alpha &= \delta F \end{aligned}$$

Generating Function

F called the generating function of the transformation acts as a bridge between the two sets of canonical variables.

1. $F = F_1(q, Q, t)$

$$p_\alpha dq_\alpha - P_\alpha dQ_\alpha + (K - H)dt = dF_1(q, Q, t) = \frac{\partial F_1}{\partial q_\alpha} dq_\alpha + \frac{\partial F_1}{\partial Q_\alpha} dQ_\alpha + \frac{\partial F_1}{\partial t} dt \Rightarrow$$

$$p_\alpha = \frac{\partial F_1}{\partial q_\alpha} \quad P_\alpha = -\frac{\partial F_1}{\partial Q_\alpha} \quad K = H + \frac{\partial F_1}{\partial t}$$

2. $F = F_2(q, P, t) - Q_\alpha P_\alpha$

$$-P dQ_\alpha = -d(Q_\alpha P_\alpha) + Q_\alpha dP_\alpha \Rightarrow p_\alpha dq_\alpha + Q_\alpha dP_\alpha + (K - H)dt = d(F + Q_\alpha P_\alpha) = dF_2(q, P, t) \Rightarrow$$

$$p_\alpha = \frac{\partial F_2}{\partial q_\alpha} \quad Q_\alpha = \frac{\partial F_2}{\partial P_\alpha} \quad K = H + \frac{\partial F_2}{\partial t}$$

3. $F = F_3(p, Q, t) + q_\alpha p_\alpha$

$$p_\alpha dq_\alpha = d(p_\alpha q_\alpha) - q_\alpha dp_\alpha \Rightarrow -q_\alpha dp_\alpha - P_\alpha dQ_\alpha + (K - H)dt = d(F - q_\alpha p_\alpha) = dF_3(p, Q, t) \Rightarrow$$

$$q_\alpha = -\frac{\partial F_3}{\partial p_\alpha} \quad P_\alpha = -\frac{\partial F_3}{\partial Q_\alpha} \quad K = H + \frac{\partial F_3}{\partial t}$$

4. $F = F_4(p, P, t) + q_\alpha p_\alpha - Q_\alpha P_\alpha$

$$-q_\alpha dp_\alpha + Q_\alpha dP_\alpha + (K - H)dt = d(F - q_\alpha p_\alpha + Q_\alpha P_\alpha) = dF_4(p, P, t) \Rightarrow$$

$$q_\alpha = -\frac{\partial F_4}{\partial p_\alpha} \quad Q_\alpha = \frac{\partial F_4}{\partial P_\alpha} \quad K = H + \frac{\partial F_4}{\partial t}$$

Examples of Canonical Transformations

1. Identity transformation:

$$F_2(q, P, t) = q_\alpha P_\alpha \Rightarrow p_\alpha = \frac{\partial F_2}{\partial q_\alpha} = P_\alpha, \quad Q_\alpha = \frac{\partial F_2}{\partial P_\alpha} = q_\alpha, \quad K = H$$

$$F_3(p, Q, t) = -p_\alpha Q_\alpha \Rightarrow q_\alpha = -\frac{\partial F_3}{\partial p_\alpha} = Q_\alpha, \quad P_\alpha = -\frac{\partial F_3}{\partial Q_\alpha} = p_\alpha$$

2. Orthogonal transformation: ($\mathbf{A}^\top \mathbf{A} = \mathbf{A} \mathbf{A}^\top = \mathbb{I}$)

$$F_2(q, P, t) = a_{\alpha\beta} P_\alpha q_\beta \Rightarrow p_r = \frac{\partial F_2}{\partial q_r} = a_{\alpha r} P_\alpha, \quad Q_r = \frac{\partial F_2}{\partial P_r} = a_{r\beta} q_\beta \Rightarrow \mathbf{p} = \mathbf{A}^\top \mathbf{P}, \quad \mathbf{Q} = \mathbf{A} \mathbf{q}$$

Ex 31. One-dimensional harmonic oscillator $H = \frac{1}{2}(q^2 + p^2)$,

proof $Q = \frac{1}{2}(q^2 + p^2)$, $P = -\arctan\left(\frac{q}{p}\right)$ is canonical transformation, and write $F_4(p, P, t)$ ^a

Proof. From $p_\alpha \delta q_\alpha - P_\alpha \delta Q_\alpha = \delta F$, we have

$$\delta F = p \delta q - P \delta Q = p \delta q + \arctan\left(\frac{q}{p}\right)(q \delta q + p \delta p) = \left[p + q \arctan\left(\frac{q}{p}\right)\right] \delta q + \arctan\left(\frac{q}{p}\right) \delta p$$

since integrability, there is $\frac{\partial}{\partial p} \left[p + q \arctan\left(\frac{q}{p}\right) \right] = \frac{\partial}{\partial q} \left[\arctan\left(\frac{q}{p}\right) \right] = \frac{p^2}{p^2 + q^2}$ □

$$q = -\frac{\partial F_4}{\partial p} = -p \tan P, \quad Q = \frac{\partial F_4}{\partial P} = \frac{1}{2} p^2 \sec^2 P \Rightarrow$$

$$dF_4(p, P) = \frac{\partial F_4}{\partial p} dp + \frac{\partial F_4}{\partial P} dP = p \tan P dp + \frac{1}{2} p^2 \sec^2 P dP = d\left(\frac{1}{2} p^2 - \tan P\right)$$

Since $K = H = Q$, we have

$$\dot{Q} = \frac{\partial K}{\partial P} = 0 \Rightarrow Q = \text{const} \equiv \frac{1}{2} A_0^2; \text{quad} \dot{P} = -\frac{\partial K}{\partial Q} = -1 \Rightarrow P = -t - t_0$$

Then $q = -\sqrt{2Q} \sin P = A_0 \sin(t + t_0)$; $p = \sqrt{2Q} \cos P = A_0 \cos(t + t_0)$

$$^a (\arctan x)' = \frac{1}{1+x^2}, \quad (\tan x)' = \sec^2 x$$

Table 4.1: Properties of the Four Basic Canonical Transformations

Generating Function	Generating Function Derivatives	Trivial Special Case
$F = F_1(q, Q, t)$	$p_\alpha = \frac{\partial F_1}{\partial q_\alpha} \quad P_\alpha = -\frac{\partial F_1}{\partial Q_\alpha}$	$F_1 = q_i Q_i, \quad Q_\alpha = p_\alpha, \quad P_\alpha = -q_\alpha$
$F = F_2(q, P, t) - Q_\alpha P_\alpha$	$p_\alpha = \frac{\partial F_2}{\partial q_\alpha} \quad Q_\alpha = \frac{\partial F_2}{\partial P_\alpha}$	$F_2 = q_\alpha P_\alpha, \quad Q = q_\alpha, \quad P_\alpha = p_\alpha$
$F = F_3(p, Q, t) + q_\alpha p_\alpha$	$q_\alpha = -\frac{\partial F_3}{\partial p_\alpha} \quad P_\alpha = -\frac{\partial F_3}{\partial Q_\alpha}$	$F_3 = p_\alpha Q_\alpha, \quad Q_\alpha = -q_\alpha, \quad P_\alpha = -p_\alpha$
$F = F_4(p, P, t) + q_\alpha p_\alpha - Q_\alpha P_\alpha$	$q_\alpha = -\frac{\partial F_4}{\partial p_\alpha}, \quad Q_\alpha = \frac{\partial F_4}{\partial P_\alpha}$	$F_4 = p_\alpha P_\alpha, \quad Q_\alpha = p_\alpha, \quad P_\alpha = -q_\alpha$

4.2.2 Poisson Bracket

For physical quantity: $f(q, p, t)$, we have

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial f}{\partial p_\alpha} \dot{p}_\alpha = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial q_\alpha} \frac{\partial H}{\partial p_\alpha} - \frac{\partial f}{\partial p_\alpha} \frac{\partial H}{\partial q_\alpha}$$

Poisson bracket for two functions $u(q, p, t)$, $v(q, p, t)$:

$$[u, v]_{q,p} = \frac{\partial u}{\partial q_\alpha} \frac{\partial v}{\partial p_\alpha} - \frac{\partial u}{\partial p_\alpha} \frac{\partial v}{\partial q_\alpha}$$

Therefore,

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [f, H]_{q,p}$$

Properties

1. $[u, v] = -[v, u]$ (anti-symmetry)
2. $[u, v_1 + v_2] = [u, v_1] + [u, v_2]$ (distribution law)
3. $[u, v_1 v_2] = v_1 [u, v_2] + [u, v_1] v_2$ (association law)
4. $\frac{\partial}{\partial x} [u, v] = \left[\frac{\partial u}{\partial x}, v \right] + \left[u, \frac{\partial v}{\partial x} \right] \quad (x = q_\alpha, p_\alpha, t)$
5. $[u, [v, w]] + [v, [w, u]] + [w, [u, v]] = 0$ (Jacobi's identity)
6. $[u, v]_{q,p} = [u, v]_{Q,P}$ * All Poisson brackets are canonical invariants.

$$[q_\alpha, q_\beta] = [p_\alpha, p_\beta] = 0, \quad [q_\alpha, p_\beta] = \delta_{\alpha\beta}, \quad [q_\alpha, f] = \frac{\partial f}{\partial p_\alpha}, \quad [p_\alpha, f] = -\frac{\partial f}{\partial q_\alpha}$$

Applications

1. $\dot{q}_\alpha = [q_\alpha, H], \quad \dot{p}_\alpha = [p_\alpha, H]$
2. $\frac{df}{dt} = \frac{\partial f}{\partial t} + [f, H] = 0$, where f is an integral of motion
 - if f don't explicit t , then $[f, H] = 0$. i.e. H don't explicit t , then $[H, H] = 0$ and H here is an integral of motion
3. Poisson Theorem: If u, v are integrals of motion, then $[u, v]$ is also integral of motion, there are at most $2s$ integrals of motion in a system.

Proof. It's known that $\frac{\partial u}{\partial t} = [u, H], \quad \frac{\partial v}{\partial t} = -[v, H]$

$$\frac{d}{dt} [u, v] = \frac{\partial}{\partial t} [u, v] + [[u, v], H] = \left[\frac{\partial u}{\partial t}, v \right] + \left[u, \frac{\partial v}{\partial t} \right] + [[u, v], H] = -[[u, H], v] - [u, [v, H]] - [H, [u, v]] = 0$$

Ex 32. A mass point move in a conserved field, if J_x, J_y is conserved, $\vec{J}$ is conserved.

Proof.

$$\begin{aligned} [J_x, J_y] &= [yp_z - zp_y, zp_x - xp_z] = [yp_z, zp_x - xp_z] - [zp_y, zp_x - xp_z] = y[p_z, zp_x - xp_z] - [z, zp_x - xp_z] \\ &= -y \frac{\partial}{\partial z}(zp_x - xp_z) - \left[\frac{\partial}{\partial z}(zp_x - xp_z) \right] p_y = -yp_x + xp_y = J_z \end{aligned} \Rightarrow$$

□

4.2.3 Liouville's Theorem

Consider the ensemble system of N particles with r degrees of freedom for each particle, so the system's degrees of freedom is $s = Nr$. In an infinitesimal volume in phase space, the volume contained by points defining the volume move about in phase space. The dashed curve (phase orbit) in Fig: 3.14 indicates the evolution of the infinitesimal volume dV with time.

- $(q_1, \dots, q_s, p_1, \dots, p_s)$: one of phase points in $2s$ -dimensional phase space, determines the state of motion of the system at a certain moment
- $dV = dq_1 \cdots dq_s dp_1 \cdots dp_s$: infinitesimal volume in phase space
- $\rho(q_1, \dots, q_s, p_1, \dots, p_s, t)$: the density of points in dV vary with time
- $dN = \rho(q_\alpha, p_\alpha) dV$: the number in the dV at certain moment

We have the density at $t + dt$:

$$\begin{aligned} \rho(q_\alpha + \dot{q}_\alpha dt, p_\alpha + \dot{p}_\alpha dt, t + dt) &= \rho(q_\alpha, p_\alpha, t) + \left[\sum_\alpha \left(\frac{\partial \rho}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial \rho}{\partial p_\alpha} \dot{p}_\alpha \right) + \frac{\partial \rho}{\partial t} \right] dt = \rho(q_\alpha, p_\alpha, t) + \frac{d\rho}{dt} dt \\ \text{or } \frac{d\rho}{dt} &= [\rho, H] + \frac{\partial \rho}{\partial t} \end{aligned}$$

Liouville Theorem:

$$\frac{d\rho}{dt} = 0 \quad \text{or} \quad \frac{\partial \rho}{\partial t} = 0, [\rho, H] = 0 \quad \text{when in statistical equilibrium}$$

which states that the density of systems in the neighborhood of some given system in phase space with the motion trace of phase orbit remains constant in time. Proof from three aspects:

1. The phase volume don't change in canonical transformations

Proof.

$$d\Gamma = dq_1 \cdots dq_s dp_1 \cdots dp_s \quad \text{phase volume: } \int_{\Sigma} d\Gamma$$

Consider canonical transformations $(q, p) \rightarrow (Q, P)$, then

$$\begin{aligned} \int \cdots \int dQ_1 \cdots dQ_s dP_1 \cdots dP_s &= \int \cdots \int D dq_1 \cdots dq_s dp_1 \cdots dp_s \\ \text{Jacobi determinant: } D &= \frac{\partial(Q_1 \cdots Q_s, P_1 \cdots P_s)}{\partial(q_1 \cdots q_s, p_1 \cdots p_s)} = \frac{\partial(Q_1 \cdots Q_s, p_1 \cdots p_s)}{\partial(q_1 \cdots q_s, P_1 \cdots P_s)} \frac{\partial(q_1 \cdots q_s, P_1 \cdots P_s)}{\partial(q_1 \cdots q_s, p_1 \cdots p_s)} \\ &= \left\{ \frac{\partial(Q_1 \cdots Q_s)}{\partial(q_1 \cdots q_s)} \right\}_{p=\text{const}} \bigg/ \left\{ \frac{\partial(p_1 \cdots p_s)}{\partial(P_1 \cdots P_s)} \right\}_{r=\text{const}} \end{aligned}$$

$$\text{For } i \text{ row, } j \text{ column, former: } \frac{\partial Q_i}{\partial q_k} = \frac{\partial}{\partial q_k} \left(\frac{\partial F_2}{\partial P_i} \right) = \frac{\partial^2 F_2}{\partial q_k \partial P_i}, \text{ latter: } \frac{\partial p_i}{\partial P_k} = \frac{\partial}{\partial P_k} \left(\frac{\partial F_2}{\partial q_i} \right) = \frac{\partial^2 F_2}{\partial P_k \partial q_i}$$

Then the same as j row, i column, we can get it is transpose symmetry. Therefore, $D = 1$

□

2. The change of q, p of motion can be thought as a series of canonical transformations

Proof. At time $t \rightarrow q_t, p_t$; and at time $t + \Delta t \rightarrow q_{t+\Delta t} \equiv Q = q_{t+\Delta t}(q_t, p_t, t, \Delta t)$; $p_{t+\Delta t} \equiv P = p_{t+\Delta t}(q_t, p_t, t, \Delta t)$

Hamilton's main function: $S = \int L dt$

$$\delta S = \frac{\partial L}{\partial \dot{q}} \delta q \Big|_{t_1}^{t_2} + \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) \delta q dt$$

Consider variation in not-fixed endpoint, then $\delta q_{t_1} \delta q_{t_2} \neq 0$, in physic process therefore,

$$\delta S = p_2 \delta q_2 - p_1 \delta q_1 \equiv p_{t+\Delta t} \delta q_{t+\Delta t} - p_t \delta q_t \Rightarrow \begin{cases} p_{t+\Delta t} = -\frac{\partial(-s)}{\partial q_{t+\Delta t}} \\ p_t = \frac{\partial(-s)}{\partial q_t} \end{cases}$$

We can have $-s = \mp 1$, since $P = \frac{\partial F_1}{\partial Q}$ and $p = \frac{\partial F_1}{\partial q}$

□

3. The number in the neighborhood area is constant

Proof. We have already known that $\int_{\Sigma} d\Gamma = \text{const}$, since the phase orbit cannot intersect, and in canonical ensemble, then $dN = \text{const}$, therefore $\rho = \frac{dN}{d\Gamma} = \text{const} \Rightarrow \frac{\partial \rho}{\partial t} = 0$

□

4.3 Hamilton-Jacobi Theory

4.3.1 The Hamilton-Jacobi Equation For Hamilton's Principal Function

Idea: Find a Hamiltonian $K = 0$, then P, Q is constant, then we can solve q, p , for then the equations of motion are

$$\begin{aligned} \dot{Q}_\alpha &= \frac{\partial K}{\partial P_\alpha} = 0 \rightarrow Q_\alpha = \xi_\alpha (\text{const}) \\ \dot{P}_\alpha &= \frac{\partial K}{\partial Q_\alpha} = 0 \rightarrow P_\alpha = \eta_\alpha (\text{const}) \end{aligned}$$

Consider $F_2(q, P, t)$, and $p_\alpha = \frac{\partial F_2}{\partial q_\alpha}$, $Q_\alpha = \frac{\partial F_2}{\partial P_\alpha}$, $K = H + \frac{\partial F_2}{\partial t}$. Since $K = 0$, we have

$$H\left(q_1, \dots, q_s; \frac{\partial F_2}{\partial q_i}, \dots, \frac{\partial F_2}{\partial q_s}; t\right) + \frac{\partial F_2}{\partial t} = 0$$

- Hamilton-Jacobi equation
- constitutes a partial differential equation in $(s+1)$ variables
- Denote the solution F_2 of H-J equation by S which is called Hamilton's principal function

Hamilton-Jacobi Equation:

$$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0$$

There are $s+1$ independent constant of integration, one is additive constant which has no importance in a generating function, since only partial derivatives of the generating function occur in the transformation equations. Hence, a complete solution can be written in the form

$$S = S(q_1, \dots, q_s; \eta_1, \dots, \eta_s; t)$$

- Here we have already taken the s constants of integration to be the new (constant) momenta:

$$P_\alpha = \eta_\alpha \Rightarrow p_\alpha = \frac{\partial S(q, \eta, t)}{\partial q_\alpha}, Q_\alpha = \xi_\alpha = \frac{\partial S(q, \eta, t)}{\partial \eta_\alpha} \Rightarrow q_\alpha = q_\alpha(\eta, \xi, t)$$

Consider H don't explicit t , then $H = E = \text{const}$ is conserved, and

$$\frac{\partial S}{\partial t} = -E \Rightarrow S(q, P, t) = -Et + W(q, P)$$

Then, we can get H-J equation when $H = \text{const}$, the H-J equation reduces to

$$H\left(q, \frac{\partial W}{\partial q}\right) = E$$

- $W(q, P)$ here is Hamilton's eigenfunction
- W constitutes s constant of integration (one is additive constant) and E depended on. Hence, from $s - 1$ non-additive constant and E , we can take $P_\alpha = \eta_\alpha$ as well

4.3.2 The Harmonic Oscillator Problem as an Example of The H-J Method

$$H = \frac{1}{2m}(p^2 + m^2\omega^2q^2) \equiv E \quad \text{where} \quad \omega = \sqrt{\frac{k}{m}}$$

Let $p = \frac{\partial S}{\partial q}$, here the H-J equation becomes

$$\frac{1}{2m} \left[\left(\frac{dW}{dq} \right)^2 + m^2\omega^2q^2 \right] = E$$

Then it can be integrated to

$$W = \pm \sqrt{2mE} \int dq \sqrt{1 - \frac{m\omega^2q^2}{2E}}$$

$$S = -Et \pm \sqrt{2mE} \int dq \sqrt{1 - \frac{m\omega^2q^2}{2E}}$$

Take E as new generalized momentum P , we have ¹

$$\xi = Q = \frac{\partial S}{\partial P} = \frac{\partial S}{\partial E} = -t \pm \sqrt{\frac{m}{2E}} \int \frac{dq}{\sqrt{1 - \frac{m\omega^2q^2}{2E}}} = -t \pm \frac{1}{\omega} \arcsin q \sqrt{\frac{m\omega^2}{2E}}$$

Then the solution of motion is

$$q = \pm \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \xi)$$

$$p = \frac{\partial S}{\partial q} = \frac{\partial W}{\partial q} = \pm \sqrt{2mE - m^2\omega^2q^2} = \pm \sqrt{2mE} \cos(\omega t + \xi)$$

$$L = \frac{1}{2m}(p^2 - m^2\omega^2q^2) = 2E \left(\cos^2(\omega t + \beta) - \frac{1}{2} \right)$$

* $Q = \xi$: new generalized coordinate

* $P = E$: new generalized moment / total energy

Summary

The procedure of solving by H-J equation:

1. Write $H(q, p, t)$, let p_α by $\frac{\partial S}{\partial q_\alpha}$ or $\frac{\partial W}{\partial q_\alpha}$ and substitute in H-J equation
2. Solve the H-J equation and get
 - $S(q, \zeta, t)$, where s non-additive constant ζ_α are new canonical momenta
 - Or $W(q, \zeta, E)$ where E and $s-1$ non-additive constant ζ_α are new canonical momenta, besides $S = -Et + W$

¹ $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$

3. $S \equiv F_2$, then the canonical transformations are $\xi_\alpha = \frac{\partial S}{\partial \zeta_\alpha}$, $p_\alpha = \frac{\partial S}{\partial q_\alpha}$

4. Get the solution of motion q, p

The physic meaning of Hamilton's principal function $S(q, P, t)$

$$\frac{dS}{dt} = \frac{\partial S}{\partial t} + \frac{\partial S}{\partial q_\alpha} \dot{q}_\alpha = p_\alpha \dot{q}_\alpha - H = L \Rightarrow S = \int L dt$$

- Action of antiderivative of t
- We cannot get S directly from $\int L dt$ without H-J equation, because we don't know q in $L(q, \dot{q}, t)$

Comparison of "Hamilton"

Hamilton's Action Quantity	$S(q, \dot{q}, t) = \int_{t_1}^{t_2} L dt$
Generalized Energy	$H(q, \dot{q}, t)$
Hamiltonion	$H(q, p, t)$
Hamilton's Principal Function	$S(q, P, t) = \int L dt$
Hamilton's Eigenfunction	$W(q, P)$